GKS Characterisation of Hypersurfaces

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Introduction

The existence of a parallel spinor places strong restrictions on the geometry of the ambient manifold, and these restrictions are inherited by its hypersurfaces through the spinorial Gauss formula. Let M^n be an oriented Riemannian spin hypersurface of a spin manifold Z^{n+1} with a parallel spinor. The restriction ψ of this spinor to M satisfies

$$\nabla_X^{\Sigma M}\psi = -\frac{1}{2}W(X)\cdot\psi,$$

where W is the shape operator of the immersion. More generally, given a symmetric endomorphism $W\colon TM \to TM$, a spinor satisfying this equation is called a generalised Killing spinor. This raises a converse question: when do intrinsic generalised Killing spinors come from a genuine hypersurface immersion?

For surfaces, Friedrich's spinorial representation theorem gives a complete answer. A single generalised Killing spinor implies the Gauss and Codazzi equations and therefore determines a local immersion into $\mathbb{R}^3$ [Fri98]. This no longer holds in higher dimensions. The integrability equation for a generalised Killing spinor contains both the Gauss and Codazzi defects, which act through Clifford multiplication in different degrees. On a single spinor these terms may cancel, so one generalised Killing spinor need not imply either classical compatibility equation.

The central question of this thesis is therefore quantitative: how many linearly independent generalised Killing spinors associated with the same endomorphism W are needed before such cancellation becomes impossible? The opening chapters develop the hypersurface spinor conventions and derive the curvature identity that governs the Gauss and Codazzi defects. We then outline how the lifted holonomy representation controls parallel spinors. Using Wang's classification [Wan89], together with the de Rham decomposition and the behaviour of spinors under products, we determine the largest number of parallel spinors that a non-flat Ricci-flat ambient manifold can contribute under hypersurface restriction. When the ambient dimension is even, this means counting parallel spinors in the relevant fixed half-spin module. The extremal examples come from a four-dimensional factor with holonomy $\mathrm{Sp}(1) = \mathrm{SU}(2)$ together with flat directions.

Chapter 4 uses these ambient spaces to produce hypersurfaces showing that the resulting bounds are sharp. If X^4 is a K3 surface with its Ricci-flat metric, then

$$X^4 \times S^{n-4}(r) \subset X^4 \times \mathbb{R}^{n-3}$$

carries

$$2^{\lfloor n/2\rfloor - 1}$$

linearly independent generalised Killing spinors associated with the same shape operator, while the induced data do not satisfy the Euclidean Gauss equation. Separate slice and graph

constructions give the corresponding examples in dimension four, including a real analytic example with non-zero shape operator. In dimension three no analogous multiplicity threshold exists: examples on the Heisenberg group and on hypersurfaces of hyperkähler four-manifolds show that even

$$\dim_{\mathbb{C}} \mathcal{K}_W(M) = \mathrm{rk}_{\mathbb{C}}\, \Sigma M$$

need not force the Euclidean Gauss defect to vanish.

There are two ways to obtain the converse result. In the real analytic category, we use the extension theorem of Ammann–Moroianu–Moroianu [AMM13]. The existence of a non-zero generalised Killing spinor implies the constraint equations for the Ricci-flat Cauchy problem associated with (g, W). We show that all generalised Killing spinors associated with the same W extend to parallel spinors on the same local Ricci-flat extension. The holonomy bounds then imply that, for $n \geq 4$,

$$\dim_{\mathbb{C}} \mathcal{K}_W(M) \geq 2^{\lfloor n/2 \rfloor - 1} + 1$$

forces this extension to be flat. The original data therefore arise locally from a hypersurface immersion into $\mathbb{R}^{n+1}$.

We then remove the real analyticity assumption and prove the same conclusion directly from the integrability equation. After adjoining a formal unit normal, the Gauss and Codazzi defects combine into a single bivector in $\Lambda^2(T_pM \oplus \mathbb{R})$ which annihilates every spinor in $\mathcal{K}_W(M)$ at the point p. Putting this bivector into normal form reduces the problem to estimating the kernel of a two-form acting on the relevant spin representation. In even dimensions a simple weight-pairing argument gives the required half-dimensional bound. In odd dimensions the intrinsic spin representation identifies with one ambient half-spin representation, and the remaining weight problem leads to four exceptional dimensions,

$$n = 3, 7, 11, 15.$$

We treat these cases separately using the curvature symmetries, the first Bianchi identity and the stabilisers of large subspaces of spinors. The dimension-three examples survive, while in dimensions 7, 11 and 15 the additional geometric constraints rule out the exceptional algebraic possibilities. Thus for every smooth spin manifold of dimension $n \geq 4$,

$$\dim_{\mathbb{C}} \mathcal{K}_W(M) \geq 2^{\lfloor n/2 \rfloor - 1} + 1$$

forces the Gauss and Codazzi equations and hence gives a local Euclidean hypersurface immersion. Together with the examples above, this determines the optimal multiplicity threshold in every dimension $n \geq 4$.

1

Hypersurfaces and Generalised Killing Spinors

1.1 Geometric setup and conventions

Let $(Z^{n+1}, \bar{g})$ be an oriented Riemannian spin manifold, let (M^n, g) be an oriented Riemannian manifold, and let

$$F : (M^n, g) \longrightarrow (Z^{n+1}, \bar{g})$$

be an isometric immersion, so that $g = F^*\bar{g}$. Although F need not be injective, $dF_p : T_pM \to T_{F(p)}Z$ is injective at every $p \in M$. We shall therefore identify TM with its image under dF whenever no confusion can arise, and usually write X in place of $dF(X)$.

The ambient tangent bundle along the immersion is the pullback bundle $F^*TZ \to M$, whose fibre at p is $T_{F(p)}Z$. Its sections are the vector fields along F, and we shall occasionally write $TZ|_M$ for F^*TZ.

1.2 The induced orientation and unit normal

Along the immersion there is an orthogonal splitting

$$TZ|_M = TM \oplus NM,$$

where NM is the normal line bundle. The orientations of TM and $TZ|_M$ determine an orientation of NM by requiring that the product orientation on $TM \oplus NM$ agree with the ambient orientation.

Equivalently, if $(e_1, \dots, e_n)$ is a positively oriented orthonormal basis of T_pM, there is a unique unit vector $\nu_p \in N_pM$ such that $(e_1, \dots, e_n, \nu_p)$ is positively oriented in $T_{F(p)}Z$. These vectors depend smoothly on p and define a global unit normal $\nu \in \Gamma(NM)$.

1.3 Ambient and intrinsic Levi-Civita connections

The ambient Levi-Civita connection induces a connection, again denoted ∇^Z, on F^*TZ. Let $(E_1, \dots, E_{n+1})$ be a local frame of TZ defined near the image of F. If V is a vector field along F and

$$V = \sum_{a=1}^{n+1} v_a (E_a \circ F),$$

then

$$\nabla^Z_X V = \sum_{a=1}^{n+1} X(v_a)(E_a \circ F) + \sum_{a=1}^{n+1} v_a \big(\nabla^Z_{dF(X)} E_a\big) \circ F.$$

This definition is independent of the chosen ambient frame.

For tangent vector fields X, Y, decompose $\nabla^Z_X Y$ into tangential and normal components:

$$\nabla^Z_X Y = D_X Y + h(X,Y)\nu.$$

The tangential connection D is metric and torsion-free, hence $D = \nabla^M$. The normal coefficient h is symmetric and is the second fundamental form.

Differentiating $\bar{g}(\nu,\nu) = 1$ shows that $\nabla^Z_X \nu$ is tangent. We therefore define the shape operator $A \in \Gamma(\mathrm{End}(TM))$ by

$$A(X) := -\nabla^Z_X \nu.$$

Differentiating $\bar{g}(Y,\nu) = 0$ then gives

$$h(X,Y) = -g\big(Y, \nabla^Z_X \nu\big) = g(A(X), Y).$$

Thus the symmetry of h is equivalent to the self-adjointness of A. Our Weingarten and Gauss formulae are

$$\nabla^Z_X \nu = -A(X), \qquad \nabla^Z_X Y = \nabla^M_X Y + g(A(X), Y)\nu. \tag{1.1}$$

These are our sign conventions for the Weingarten and Gauss formulae; compare [BGM05, Section 3, Equation (6)].

1.4 Gauss and Codazzi defects

We use the curvature convention

$$R^Z(X,Y)U = \nabla^Z_X \nabla^Z_Y U - \nabla^Z_Y \nabla^Z_X U - \nabla^Z_{[X,Y]} U, \tag{1.2}$$

and the same convention for R^M and the spin curvatures used later.

Proposition 1.1 (Gauss and Codazzi identities). *Let A be the shape operator of the immersion. For tangent vector fields X, Y, U, one has*

$$\big(R^Z(X,Y)U\big)^\top = R^M(X,Y)U - g(A(Y),U)A(X) + g(A(X),U)A(Y), \tag{1.3}$$

$$\bar{g}\big(R^Z(X,Y)U, \nu\big) = g\big((\nabla^M_X A)(Y) - (\nabla^M_Y A)(X), U\big). \tag{1.4}$$

Proof. Substituting the Gauss and Weingarten formulae into the definition of $R^Z(X,Y)U$ and separating tangential and normal components gives the two identities. In the normal component, one uses

$$(\nabla^M_X h)(Y,U) = g\big((\nabla^M_X A)(Y), U\big),$$

which follows from $h(Y,U) = g(A(Y),U)$ and $\nabla^M g = 0$. □

We now formulate these identities for an arbitrary symmetric endomorphism $W \in \Gamma(\mathrm{End}(TM))$.

Definition 1.2. The wedge square of W is defined by

$$(W \wedge W)(X,Y)U := g(W(Y),U)W(X) - g(W(X),U)W(Y). \tag{1.5}$$

The Gauss and Codazzi defects of W are

$$\mathcal{G}_W(X,Y)U := R^M(X,Y)U - (W \wedge W)(X,Y)U, \tag{1.6}$$

$$\mathcal{C}_W(X,Y) := (\nabla^M_X W)(Y) - (\nabla^M_Y W)(X). \tag{1.7}$$

Corollary 1.3. *When $W = A$ is the shape operator of the immersion,*

$$\mathcal{G}_A(X,Y)U = \big(R^Z(X,Y)U\big)^\top, \qquad \mathcal{C}_A(X,Y) = -R^Z(X,Y)\nu. \tag{1.8}$$

Proof. The first identity is Equation (1.3). For the second, Equation (1.4) gives

$$g\big(\mathcal{C}_A(X,Y),U\big) = \bar{g}\big(R^Z(X,Y)U,\nu\big).$$

Since $R^Z(X,Y)$ is skew-adjoint,

$$\bar{g}\big(R^Z(X,Y)U,\nu\big) = -\bar{g}\big(U,R^Z(X,Y)\nu\big).$$

Moreover, $R^Z(X,Y)\nu$ is tangent, because $\bar{g}(R^Z(X,Y)\nu,\nu) = 0$. Both sides of

$$\mathcal{C}_A(X,Y) = -R^Z(X,Y)\nu$$

are therefore tangent vectors, and the equality follows by pairing with arbitrary tangent U. □

Thus $\mathcal{G}_A$ records the tangential component of the ambient curvature, whereas $\mathcal{C}_A$ records its mixed tangent-normal component. In particular,

$$\mathcal{C}_A = 0 \quad \Longleftrightarrow \quad R^Z(X,Y)\nu = 0$$

for all tangent X, Y.

This condition can also be expressed using only tangent vectors. Indeed, for tangent U,

$$\bar{g}\big(R^Z(X,Y)U,\nu\big) = -\bar{g}\big(U,R^Z(X,Y)\nu\big).$$

Hence $R^Z(X,Y)\nu = 0$ implies that $R^Z(X,Y)U$ has no normal component and is therefore tangent. Conversely, if $R^Z(X,Y)U$ is tangent for every tangent U, then $R^Z(X,Y)\nu$ is a tangent vector orthogonal to every tangent vector, and must therefore vanish. Consequently,

$$\mathcal{C}_A = 0 \quad \Longleftrightarrow \quad R^Z(X,Y)U \in TM$$

for all tangent X, Y, U.

If Z is Euclidean, both defects vanish, and one recovers the usual Gauss and Codazzi equations

$$R^M = A \wedge A, \qquad (\nabla^M_X A)(Y) = (\nabla^M_Y A)(X).$$

More generally, suppose that Z has constant sectional curvature κ. Then

$$R^Z(X,Y)V = \kappa\big(\bar{g}(Y,V)X - \bar{g}(X,V)Y\big).$$

Taking $V = \nu$ and using the fact that X and Y are tangent gives

$$R^Z(X,Y)\nu = 0,$$

and therefore $\mathcal{C}_A = 0$. Taking instead $V = U$ tangent gives

$$\mathcal{G}_A(X,Y)U = \kappa\big(g(Y,U)X - g(X,U)Y\big).$$

1.5 The induced spin structure on the hypersurface

We use the standard hypersurface construction of the induced spin structure; see [BGM05, Section 3] and [LM89, Chapter II].

Let

$$\Lambda_Z : P_Z \longrightarrow Q_Z$$

be a fixed spin structure on Z, where Q_Z is the oriented orthonormal frame bundle of TZ, and let $\lambda_k : \mathrm{Spin}(k) \to \mathrm{SO}(k)$ denote the standard double covering. Pulling the ambient spin structure back along F gives principal bundles $F^*P_Z \to M$ and $F^*Q_Z \to M$, whose fibres at $p \in M$ are

$$(F^*P_Z)_p = (P_Z)_{F(p)}, \qquad (F^*Q_Z)_p = (Q_Z)_{F(p)}.$$

Thus $(F^*Q_Z)_p$ consists of the positively oriented orthonormal frames of $T_{F(p)}Z$, while $(F^*P_Z)_p$ is the corresponding fibre of ambient spin frames. The ambient covering induces an equivariant double covering

$$\widetilde{\Lambda}_Z : F^*P_Z \longrightarrow F^*Q_Z, \qquad \widetilde{\Lambda}_Z(p,u) = \big(p, \Lambda_Z(u)\big).$$

Let Q_M be the oriented orthonormal frame bundle of TM. Appending the chosen unit normal defines

$$\iota : Q_M \longrightarrow F^*Q_Z, \qquad \iota(e_1, \dots, e_n) = (e_1, \dots, e_n, \nu).$$

Its image is the bundle of adapted ambient frames.

The relevant inclusions are

$$\jmath_n : \mathrm{SO}(n) \hookrightarrow \mathrm{SO}(n+1), \qquad B \longmapsto \begin{pmatrix} B & 0 \\ 0 & 1 \end{pmatrix},$$

and the inclusion $\widetilde{\jmath}_n : \mathrm{Spin}(n) \hookrightarrow \mathrm{Spin}(n+1)$ induced by $\mathrm{Cl}_n \hookrightarrow \mathrm{Cl}_{n+1}$. The map $\jmath_n(B)$ acts by B on $\mathbb{R}^n$ and fixes e_{n+1}, and

$$\iota(q \cdot B) = \iota(q) \cdot \jmath_n(B)$$

for $q \in Q_M$ and $B \in \mathrm{SO}(n)$.

Lemma 1.4. *The standard inclusions are compatible with the spin coverings:*

$$\lambda_{n+1} \circ \widetilde{\jmath}_n = \jmath_n \circ \lambda_n.$$

Moreover,

$$\lambda_{n+1}^{-1}\big(\jmath_n(\mathrm{SO}(n))\big) = \widetilde{\jmath}_n(\mathrm{Spin}(n)).$$

Proof. An element $u \in \mathrm{Spin}(n)$ is represented by an even product of unit vectors in $\mathbb{R}^n$. When regarded as an element of $\mathrm{Spin}(n+1)$, it induces the same conjugation action on $\mathbb{R}^n$ and commutes with e_{n+1}, since each factor anticommutes with e_{n+1}. Its action therefore fixes e_{n+1}, proving

$$\lambda_{n+1}\big(\widetilde{\jmath}_n(u)\big) = \jmath_n\big(\lambda_n(u)\big).$$

This immediately gives

$$\widetilde{\jmath}_n(\mathrm{Spin}(n)) \subset \lambda_{n+1}^{-1}\big(\jmath_n(\mathrm{SO}(n))\big).$$

Conversely, suppose that $u \in \mathrm{Spin}(n+1)$ satisfies $\lambda_{n+1}(u) = \jmath_n(B)$ for some $B \in \mathrm{SO}(n)$. Choose $v \in \mathrm{Spin}(n)$ with $\lambda_n(v) = B$. Then u and $\widetilde{\jmath}_n(v)$ have the same image under λ_{n+1}, so

$$u\,\widetilde{\jmath}_n(v)^{-1} \in \ker \lambda_{n+1} = \{\pm 1\}.$$

Since $\{\pm 1\} \subset \widetilde{\jmath}_n(\mathrm{Spin}(n))$, it follows that $u \in \widetilde{\jmath}_n(\mathrm{Spin}(n))$. □

Equivalently, the first identity in Lemma 1.4 is expressed by the commutative diagram

$$\begin{array}{ccc} \mathrm{Spin}(n) & \xhookrightarrow{\widetilde{\jmath}_n} & \mathrm{Spin}(n+1) \\ \downarrow{\scriptstyle \lambda_n} & & \downarrow{\scriptstyle \lambda_{n+1}} \\ \mathrm{SO}(n) & \xhookrightarrow[\jmath_n]{} & \mathrm{SO}(n+1). \end{array}$$

We now define the bundle of adapted ambient spin frames by

$$P_M := \widetilde{\Lambda}_Z^{-1}\big(\iota(Q_M)\big) \subset F^* P_Z.$$

Lemma 1.5. *The formula*

$$\widetilde{p} \cdot h := \widetilde{p} \cdot \widetilde{\jmath}_n(h), \qquad \widetilde{p} \in P_M, \quad h \in \mathrm{Spin}(n),$$

defines a free and transitive right $\mathrm{Spin}(n)$*-action on each fibre of* $P_M \to M$*. Hence* $P_M \to M$ *is a principal* $\mathrm{Spin}(n)$*-bundle.*

Proof. Equivariance of $\widetilde{\Lambda}_Z$ and Lemma 1.4 give

$$\widetilde{\Lambda}_Z(\widetilde{p}\cdot h) = \widetilde{\Lambda}_Z(\widetilde{p})\cdot \jmath_n\big(\lambda_n(h)\big).$$

Since the right-hand side is again an adapted frame, the action preserves P_M. It is free because the ambient $\mathrm{Spin}(n+1)$-action is free and $\widetilde{\jmath}_n$ is injective.

To prove transitivity, let $\widetilde{p},\widetilde{q}\in(P_M)_p$. Since F^*P_Z is a principal $\mathrm{Spin}(n+1)$-bundle, there is a unique $u\in\mathrm{Spin}(n+1)$ such that $\widetilde{q}=\widetilde{p}\cdot u$. The frames $\widetilde{\Lambda}_Z(\widetilde{p})$ and $\widetilde{\Lambda}_Z(\widetilde{q})$ are both adapted, so they differ by an element of $\jmath_n(\mathrm{SO}(n))$. Hence

$$\lambda_{n+1}(u)\in \jmath_n(\mathrm{SO}(n)).$$

Lemma 1.4 gives $u=\widetilde{\jmath}_n(h)$ for some $h\in\mathrm{Spin}(n)$, and therefore $\widetilde{q}=\widetilde{p}\cdot h$. □

Proposition 1.6. *The map*

$$\Lambda_M : P_M \longrightarrow Q_M, \qquad \Lambda_M = \iota^{-1}\circ\widetilde{\Lambda}_Z|_{P_M},$$

is a spin structure on M.

Proof. The map is well defined by the definition of P_M. For $\widetilde{p}\in P_M$ and $h\in\mathrm{Spin}(n)$,

$$\begin{aligned}\iota\big(\Lambda_M(\widetilde{p}\cdot h)\big) &= \widetilde{\Lambda}_Z(\widetilde{p}\cdot h)\\ &= \widetilde{\Lambda}_Z(\widetilde{p})\cdot \jmath_n\big(\lambda_n(h)\big)\\ &= \iota\big(\Lambda_M(\widetilde{p})\cdot\lambda_n(h)\big).\end{aligned}$$

Since ι is injective,

$$\Lambda_M(\widetilde{p}\cdot h)=\Lambda_M(\widetilde{p})\cdot\lambda_n(h),$$

so Λ_M is equivariant.

Each adapted frame $\iota(q)$ has exactly two lifts under $\widetilde{\Lambda}_Z$, and both lie in P_M. Hence every $q\in Q_M$ has exactly two preimages under Λ_M, so Λ_M is a double covering. Together with Lemma 1.5, this proves that $\Lambda_M : P_M\to Q_M$ is a spin structure on M. □

The diagram

$$\begin{array}{ccc} P_M & \hookrightarrow & F^*P_Z \\ {\scriptstyle\Lambda_M}\downarrow & & \downarrow{\scriptstyle\widetilde{\Lambda}_Z} \\ Q_M & \xhookrightarrow[\iota]{} & F^*Q_Z \end{array}$$

summarises the construction. We call Λ_M the spin structure on M induced by the ambient spin structure and the chosen unit normal.

1.6 Relating ambient and intrinsic Clifford multiplication

The algebraic background on complex Clifford modules can be found in [LM89, Chapter II]. The hypersurface identification and the restriction of ambient spinors are described in [Bou+15, Section 2.4.1] and [BGM05, Section 3].

All Clifford algebras in this section are complexified. At $x \in M$, define

$$\alpha_x : T_xM \longrightarrow \mathrm{Cl}^0(T_xZ), \qquad \alpha_x(X) = X \cdot \nu.$$

For tangent vectors X, Y, one has

$$\alpha_x(X)\alpha_x(Y) + \alpha_x(Y)\alpha_x(X) = X \cdot Y + Y \cdot X = -2g(X, Y).$$

Thus α_x satisfies the Clifford relations and, by the universal property of the Clifford algebra, extends uniquely to a unital algebra homomorphism

$$\alpha_x : \mathrm{Cl}(T_xM) \longrightarrow \mathrm{Cl}^0(T_xZ).$$

Relative to an adapted frame $(e_1, \ldots, e_n, \nu)$,

$$\alpha_x(e_{i_1} \cdots e_{i_r}) = \begin{cases} e_{i_1} \cdots e_{i_r}, & r \text{ even}, \\ e_{i_1} \cdots e_{i_r}\nu, & r \text{ odd}. \end{cases}$$

Every even ambient Clifford monomial has one of these forms, so α_x is surjective. Since both algebras have complex dimension 2^n, it is an isomorphism. The maps α_x vary smoothly with x and assemble to a bundle isomorphism

$$\alpha : \mathrm{Cl}(TM) \longrightarrow \mathrm{Cl}^0(TZ|_M). \tag{1.9}$$

On the spin group, α is precisely the inclusion $\widetilde{\jmath}_n$: an even product of tangent unit vectors is unchanged when regarded in the larger Clifford algebra.

Let

$$\rho_k : \mathrm{Cl}_k \longrightarrow \mathrm{End}(\Delta_k)$$

be the chosen complex Clifford representation, and let $\kappa_k = \rho_k|_{\mathrm{Spin}(k)}$. Recall that a representation $\rho : A \to \mathrm{End}(V)$ makes V into an A-module by defining $a \cdot v := \rho(a)v$. The module is irreducible if it has no non-zero proper subspace preserved by the action of every $a \in A$. A module isomorphism is a linear isomorphism which intertwines the two algebra actions.

Suppose first that $n = 2m$. Then

$$\mathrm{Cl}_n \cong \mathrm{Mat}_{2^m}(\mathbb{C}),$$

and a full matrix algebra has, up to module isomorphism, a unique irreducible module, namely $\mathbb{C}^{2^m}$ with its standard action. Through α, the ambient spinor space Δ_{n+1} becomes a Cl_n-module

by defining

$$a \cdot \Psi := \rho_{n+1}\big(\alpha(a)\big)\Psi.$$

This module is irreducible. Indeed, the corresponding unital representation

$$\rho_{n+1} \circ \alpha : \mathrm{Cl}_n \longrightarrow \mathrm{End}(\Delta_{n+1})$$

is injective because $\mathrm{Cl}_n \cong \mathrm{Mat}_{2^m}(\mathbb{C})$ is simple: its kernel is a two-sided ideal, and it cannot be the whole algebra since the representation is unital. Moreover,

$$\dim_{\mathbb{C}} \mathrm{Cl}_n = 2^{2m} = \dim_{\mathbb{C}} \mathrm{End}(\Delta_{n+1}).$$

The representation is therefore an isomorphism onto $\mathrm{End}(\Delta_{n+1})$. Hence the Cl_n-action contains every endomorphism of Δ_{n+1}, so no non-zero proper subspace can be invariant under it.

Thus Δ_n and Δ_{n+1} are irreducible Cl_n-modules of the same dimension. By the uniqueness of the irreducible module, we may choose a Cl_n-module isomorphism

$$\Phi : \Delta_n \longrightarrow \Delta_{n+1}.$$

By definition, Φ intertwines the two Clifford actions:

$$\Phi\big(\rho_n(a)\varphi\big) = \rho_{n+1}\big(\alpha(a)\big)\Phi(\varphi)$$

for every $a \in \mathrm{Cl}_n$ and $\varphi \in \Delta_n$. In particular, since $\alpha(X) = X \cdot_{n+1} \nu$,

$$\Phi(X \cdot_n \varphi) = X \cdot_{n+1} \nu \cdot_{n+1} \Phi(\varphi).$$

After multiplying Φ by a constant if necessary, we may assume that it is unitary.

Since $\alpha|_{\mathrm{Spin}(n)} = \widetilde{\jmath}_n$, the map Φ is $\mathrm{Spin}(n)$-equivariant. It therefore induces an isomorphism of associated spinor bundles

$$\mathcal{F} : \Sigma M \longrightarrow \Sigma Z|_M$$

satisfying

$$\mathcal{F}(X \cdot_M \psi) = X \cdot_Z \nu \cdot_Z \mathcal{F}(\psi). \tag{1.10}$$

After making this identification, we shall simply write

$$X \cdot_M \psi = X \cdot_Z \nu \cdot_Z \psi.$$

Now suppose that $n = 2m - 1$. The odd-dimensional algebra

$$\mathrm{Cl}_n \cong \mathrm{Mat}_{2^{m-1}}(\mathbb{C}) \oplus \mathrm{Mat}_{2^{m-1}}(\mathbb{C})$$

has two inequivalent irreducible modules. Their restrictions to $\mathrm{Spin}(n) \subset \mathrm{Cl}_n^0$ are nevertheless

equivalent. The ambient spinor module splits into its half-spin spaces,

$$\Delta_{n+1} = \Delta_{n+1}^{+} \oplus \Delta_{n+1}^{-},$$

and, through α, these two spaces realise the two irreducible Cl_n-modules.

We choose the intrinsic Clifford module so that there is a unitary Cl_n-module isomorphism

$$\Phi_+ : \Delta_n \longrightarrow \Delta_{n+1}^{+}$$

satisfying

$$\Phi_+(X \cdot_n \varphi) = X \cdot_{n+1} \nu \cdot_{n+1} \Phi_+(\varphi).$$

It induces a bundle isomorphism

$$\mathcal{F}_+ : \Sigma M \longrightarrow \Sigma^+ Z|_M$$

with the same intertwining property.

The second irreducible Cl_n-module is obtained by precomposing the Clifford action with the grading automorphism, which sends each vector X to $-X$. Since elements of $\mathrm{Spin}(n)$ are even products of vectors, this automorphism acts trivially on $\mathrm{Spin}(n)$. Hence there is also a $\mathrm{Spin}(n)$-equivariant bundle isomorphism

$$\mathcal{F}_- : \Sigma M \longrightarrow \Sigma^- Z|_M$$

satisfying

$$\mathcal{F}_-(X \cdot_M \psi) = -X \cdot_Z \nu \cdot_Z \mathcal{F}_-(\psi).$$

Consequently,

$$\Sigma Z|_M \cong \Sigma M \oplus \Sigma M,$$

and under this identification,

$$X \cdot_Z \nu \cdot_Z (\psi_+, \psi_-) = \big(X \cdot_M \psi_+, -X \cdot_M \psi_-\big).$$

1.7 Ambient and intrinsic spin connections

Write $\nabla^{\Sigma M}$ and $\nabla^{\Sigma Z}$ for the spin connections induced by ∇^M and ∇^Z. Let $(e_1, \ldots, e_n)$ be a local positively oriented orthonormal frame of TM and set $e_{n+1} = \nu$. If

$$\omega_{ij}^M(X) = g(\nabla_X^M e_i, e_j), \qquad \omega_{kl}^Z(X) = \bar{g}(\nabla_X^Z e_k, e_l),$$

then the Gauss and Weingarten formulae give

$$\omega_{ij}^Z(X) = \omega_{ij}^M(X), \qquad \omega_{i,n+1}^Z(X) = g(A(X), e_i).$$

When n is even, identify ΣM with $\Sigma Z|_M$. When n is odd, identify ΣM with $\Sigma^+ Z|_M$ according to the convention fixed above. In either case, $X \cdot_M \psi = X \cdot_Z \nu \cdot_Z \psi$. The tangent-tangent terms in the local connection formulae agree, while the normal terms give

$$\nabla^{\Sigma Z}_X \psi - \nabla^{\Sigma M}_X \psi = \frac{1}{2} \sum_{i=1}^{n} g(A(X), e_i) e_i \cdot_Z \nu \cdot_Z \psi = \frac{1}{2} A(X) \cdot_M \psi.$$

Hence

$$\nabla^{\Sigma Z}_X \psi = \nabla^{\Sigma M}_X \psi + \frac{1}{2} A(X) \cdot_M \psi. \tag{1.11}$$

This is the spinorial Gauss formula in our sign convention; compare [Bou+15, Section 2.4.1] and [BGM05, Equation (8.1)].

For odd n, retaining the whole restricted ambient bundle gives

$$\nabla^{\Sigma Z}_X (\psi_+, \psi_-) = \left(\nabla^{\Sigma M}_X \psi_+ + \frac{1}{2} A(X) \cdot_M \psi_+, \nabla^{\Sigma M}_X \psi_- - \frac{1}{2} A(X) \cdot_M \psi_- \right).$$

1.8 Restricting parallel spinors

The restriction of ambient parallel spinors to hypersurfaces provides the basic geometric source of generalised Killing spinors; see [AMM13, Equations (1.1)–(1.2)]. Indeed, the spinorial Gauss formula shows directly that the restriction of an ambient parallel spinor satisfies the generalised Killing equation determined by the shape operator.

We shall need one additional fact when n is odd, so that the ambient dimension $n+1$ is even. Let $\omega^Z_{\mathbb{C}}$ denote the complex volume element of Z, normalised so that

$$\left(\omega^Z_{\mathbb{C}}\right)^2 = \mathrm{id}_{\Sigma Z}.$$

Its action on ΣZ is called the chirality operator. The positive and negative half-spinor bundles are its two eigensubbundles:

$$\Sigma^{\pm} Z = \left\{ \Psi \in \Sigma Z : \omega^Z_{\mathbb{C}} \cdot \Psi = \pm \Psi \right\},$$

and hence

$$\Sigma Z = \Sigma^+ Z \oplus \Sigma^- Z.$$

Every ambient spinor therefore decomposes uniquely as

$$\Psi = \Psi^+ + \Psi^-, \qquad \Psi^{\pm} \in \Gamma(\Sigma^{\pm} Z),$$

where

$$\Psi^{\pm} = P_{\pm} \Psi, \qquad P_{\pm} = \frac{1}{2}\left(\mathrm{id}_{\Sigma Z} \pm \omega^Z_{\mathbb{C}}\right).$$

The complex volume element is parallel with respect to the spin connection. Indeed, the Levi-Civita connection preserves the metric and orientation, and therefore preserves the volume

element. Consequently,

$$\nabla^{\Sigma Z}_X\big(\omega^Z_{\mathbb{C}}\cdot\Psi\big)=\omega^Z_{\mathbb{C}}\cdot\nabla^{\Sigma Z}_X\Psi.$$

Thus the projections $P_\pm$ are parallel and the spin connection preserves the two half-spinor bundles:

$$\nabla^{\Sigma Z}_X\Psi^\pm=P_\pm\big(\nabla^{\Sigma Z}_X\Psi\big)\in\Gamma(\Sigma^\pm Z).$$

In particular, if Ψ is parallel, then Ψ^+ and Ψ^- are separately parallel.

Proposition 1.7. *Let $\Psi\in\Gamma(\Sigma Z)$ be parallel.*

If n is even, let $\psi\in\Gamma(\Sigma M)$ correspond to $\Psi|_M$ under $\Sigma M\cong\Sigma Z|_M$. Then

$$\nabla^{\Sigma M}_X\psi=-\frac{1}{2}A(X)\cdot_M\psi.$$

If n is odd, write $\Psi=\Psi^+ + \Psi^-$ and let $\psi_\pm\in\Gamma(\Sigma M)$ correspond to $\Psi^\pm|_M$ under the identifications

$$\Sigma M\cong\Sigma^\pm Z|_M.$$

Then

$$\nabla^{\Sigma M}_X\psi_+=-\frac{1}{2}A(X)\cdot_M\psi_+,\qquad\nabla^{\Sigma M}_X\psi_-=\frac{1}{2}A(X)\cdot_M\psi_-.$$

Proof. When n is odd, the preceding discussion shows that Ψ^+ and Ψ^- are separately parallel. The assertions now follow from (1.11), using the opposite Clifford convention on the negative half-spin summand. □

Definition 1.8. Let $W\in\Gamma(\mathrm{End}(TM))$ be symmetric. A spinor $\psi\in\Gamma(\Sigma M)$ is a *generalised Killing spinor associated with W* if

$$\nabla^{\Sigma M}_X\psi=-\frac{1}{2}W(X)\cdot_M\psi \tag{1.12}$$

for every $X\in TM$.

With our odd-dimensional convention $\Sigma M\cong\Sigma^+Z|_M$, the positive component of an ambient parallel spinor restricts to a generalised Killing spinor associated with A, while the negative component is associated with $-A$.

Every generalised Killing spinor has constant length. Indeed, tangent Clifford multiplication is skew-Hermitian, and therefore

$$X(|\psi|^2)=2\,\mathrm{Re}\langle\nabla^{\Sigma M}_X\psi,\psi\rangle=-\,\mathrm{Re}\langle W(X)\cdot_M\psi,\psi\rangle=0.$$

A non-zero generalised Killing spinor is consequently nowhere vanishing. If $W=0$, it is parallel; if $W=\lambda\,\mathrm{id}_{TM}$ for a constant λ, it is an ordinary Killing spinor with Killing constant $-\lambda/2$. In general, the equation records the full candidate shape operator W.

The next chapter asks when the intrinsic equation (1.12) forces the defects $\mathcal{G}_W$ and $\mathcal{C}_W$ to vanish.

2

Integrability Conditions for Generalised Killing Spinors

2.1 Intrinsic GKS data

Having established the hypersurface origin of the generalised Killing equation, we now study it intrinsically on M, without assuming that M is realised as a hypersurface. Let (M^n, g) be a connected oriented Riemannian spin manifold, and let $W \in \Gamma(\mathrm{End}(TM))$ be symmetric. Suppose that $0 \neq \psi \in \Gamma(\Sigma M)$ satisfies

$$\nabla^{\Sigma M}_X \psi = -\frac{1}{2} W(X) \cdot_M \psi$$

for every $X \in TM$. As shown above, ψ has constant length. Since M is connected and ψ is not identically zero, this constant is positive, so ψ is nowhere vanishing.

When (M, g) is realised as a hypersurface in a spin manifold carrying a parallel spinor, the restriction of that spinor yields a generalised Killing spinor with $W = A$. In odd dimensions, this statement applies to the positive chiral component; the negative component instead yields $W = -A$. The intrinsic question is whether the data (M, g, W, ψ) suffice to recover such a hypersurface description. For a Euclidean hypersurface, the required integrability conditions are precisely $\mathcal{G}_W = 0$ and $\mathcal{C}_W = 0$.

We use the fundamental theorem of hypersurfaces in the formulation of [BGM05, Theorem 7.1].

Theorem 2.1 (Fundamental theorem of hypersurfaces)**.** *Let (M^n, g) be simply connected and let $W \in \Gamma(\mathrm{End}(TM))$ be symmetric. If*

$$\mathcal{G}_W = 0, \qquad \mathcal{C}_W = 0,$$

then there is an oriented isometric immersion $F : M \to \mathbb{R}^{n+1}$ with shape operator $A = W$. It is unique up to an orientation-preserving Euclidean isometry.

Without simple connectedness, the result applies to the universal cover. Thus the vanishing of the two defects is equivalent to the local existence of a Euclidean hypersurface with shape operator W.

In dimension two, Friedrich's spinorial representation theorem gives a complete converse [Fri98, Theorem 1]: a single generalised Killing spinor already forces these conditions. Recall that, for

a local orthonormal frame $(e_1, \dots, e_n)$,

$$D^M\psi = \sum_{j=1}^{n} e_j \cdot \nabla^{\Sigma M}_{e_j}\psi.$$

Clifford contraction of (1.12) gives

$$D^M\psi = -\frac{1}{2}\sum_{j=1}^{n} e_j \cdot W(e_j)\cdot\psi = \frac{1}{2}\operatorname{tr}(W)\psi,$$

since the off-diagonal terms cancel by the symmetry of W.

Theorem 2.2 (Friedrich). *Let (M^2, g) be an oriented Riemannian spin surface and let $H \in C^\infty(M)$. The following data are equivalent:*

(i) *an isometric immersion of the universal cover $\widetilde{M}$ into $\mathbb{R}^3$ with mean curvature H;*

(ii) *a unit spinor $\psi \in \Gamma(\Sigma M)$ satisfying $D^M\psi = H\psi$;*

(iii) *a unit spinor $\psi \in \Gamma(\Sigma M)$ and a symmetric $W \in \Gamma(\operatorname{End}(TM))$ satisfying*

$$\operatorname{tr}(W) = 2H, \qquad \nabla^{\Sigma M}_X\psi = -\frac{1}{2}W(X)\cdot\psi.$$

Under the correspondence, W is the shape operator of the immersion.

Hence, in dimension two, the generalised Killing equation forces $\mathcal{G}_W = \mathcal{C}_W = 0$. In higher dimensions this no longer follows immediately. The natural connection attached to W is

$$\nabla^W_X\varphi := \nabla^{\Sigma M}_X\varphi + \frac{1}{2}W(X)\cdot\varphi. \tag{2.1}$$

A spinor is a generalised Killing spinor associated with W precisely when it is ∇^W-parallel.

2.2 Curvature of the modified connection

This is the standard curvature computation underlying the integrability conditions for generalised Killing spinors; compare [AMM13, Section 3].

The curvature of (2.1) is $R^{\nabla^W}_{X,Y} = \nabla^W_X\nabla^W_Y - \nabla^W_Y\nabla^W_X - \nabla^W_{[X,Y]}$.

Proposition 2.3. *For all $X, Y \in TM$,*

$$R^{\nabla^W}_{X,Y} = R^{\Sigma M}(X,Y) + \frac{1}{2}\mathcal{C}_W(X,Y)\cdot + \frac{1}{4}\big[W(X)\cdot, W(Y)\cdot\big].$$

Consequently, every generalised Killing spinor ψ associated with W satisfies

$$R^{\Sigma M}(X,Y)\psi = -\frac{1}{2}\mathcal{C}_W(X,Y)\cdot\psi - \frac{1}{4}\big[W(X)\cdot, W(Y)\cdot\big]\psi. \tag{2.2}$$

Proof. For $\varphi \in \Gamma(\Sigma M)$, compatibility of the spin connection with Clifford multiplication gives

$$\nabla^{\Sigma M}_X\big(W(Y)\cdot\varphi\big) = \nabla^M_X\big(W(Y)\big)\cdot\varphi + W(Y)\cdot\nabla^{\Sigma M}_X\varphi.$$

Expanding $\nabla^W_X\nabla^W_Y\varphi$, skew-symmetrising in X, Y and subtracting $\nabla^W_{[X,Y]}\varphi$ cancels all first derivatives of φ and yields

$$\begin{aligned} R^{\nabla^W}_{X,Y}\varphi = {} & R^{\Sigma M}(X,Y)\varphi \\ & + \frac{1}{2}\big(\nabla^M_X(W(Y)) - \nabla^M_Y(W(X)) - W([X,Y])\big)\cdot\varphi \\ & + \frac{1}{4}\big(W(X)\cdot W(Y) - W(Y)\cdot W(X)\big)\cdot\varphi. \end{aligned}$$

The torsion-free identity $[X,Y] = \nabla^M_X Y - \nabla^M_Y X$ identifies the middle term with $\frac{1}{2}\mathcal{C}_W(X,Y)\cdot\varphi$. This proves the curvature formula. If ψ is generalised Killing, then $R^{\nabla^W}_{X,Y}\psi = 0$, giving (2.2). □

2.3 The spinorial Gauss–Codazzi identity

Let $(e_1,\dots,e_n)$ be a local orthonormal frame. We use the standard curvature formula for the spin connection [Fri00]. With the convention (1.2), it is

$$R^{\Sigma M}(X,Y)\varphi = \frac{1}{4}\sum_{i,j=1}^{n} g\big(R^M(X,Y)e_i, e_j\big)e_i\cdot e_j\cdot\varphi.$$

Lemma 2.4. *For all $X, Y \in TM$,*

$$\frac{1}{4}\sum_{i,j=1}^{n} g\big((W\wedge W)(X,Y)e_i, e_j\big)e_i\cdot e_j = -\frac{1}{4}\big(W(X)\cdot W(Y) - W(Y)\cdot W(X)\big).$$

Proof. Set $U = W(X)$ and $V = W(Y)$. Since

$$g\big((W\wedge W)(X,Y)e_i, e_j\big) = g(V,e_i)g(U,e_j) - g(U,e_i)g(V,e_j),$$

the left-hand side in the lemma's statement is therefore $\frac{1}{4}(V\cdot U - U\cdot V)$, as required. □

Combining Lemma 2.4 with (2.2) gives the basic integrability identity.

Proposition 2.5. *Let ψ be a generalised Killing spinor associated with W. Then*

$$\frac{1}{4}\sum_{i,j=1}^{n} g\big(\mathcal{G}_W(X,Y)e_i, e_j\big)e_i\cdot e_j\cdot\psi = -\frac{1}{2}\mathcal{C}_W(X,Y)\cdot\psi \tag{2.3}$$

for all $X, Y \in TM$.

Proof. The Clifford expression associated with $\mathcal{G}_W = R^M - W\wedge W$ is

$$R^{\Sigma M}(X,Y) + \frac{1}{4}\big(W(X)\cdot W(Y) - W(Y)\cdot W(X)\big).$$

Applying it to ψ and using (2.2) leaves precisely the right-hand side of (2.3). □

For fixed $X, Y \in T_pM$, the endomorphism $\mathcal{G}_W(X,Y)$ is skew-adjoint. Using the metric to identify skew-adjoint endomorphisms with 2-forms, define

$$\widehat{\mathcal{G}}_W(X,Y) := \frac{1}{4}\sum_{i,j=1}^{n} g\big(\mathcal{G}_W(X,Y)e_i, e_j\big)e_i \cdot e_j \in \mathrm{Cl}^2(T_pM),$$
$$\widehat{\mathcal{C}}_W(X,Y) := \frac{1}{2}\mathcal{C}_W(X,Y) \in \mathrm{Cl}^1(T_pM).$$

The coefficients in the first expression are skew-symmetric, so no scalar term occurs. Equation (2.3) becomes

$$\big(\widehat{\mathcal{G}}_W(X,Y) + \widehat{\mathcal{C}}_W(X,Y)\big)\cdot\psi = 0. \tag{2.4}$$

This is the basic integrability relation for the generalised Killing equation. It turns the geometric problem into a pointwise algebraic one. For fixed $p \in M$ and $X, Y \in T_pM$, the value at p of every spinor in $\mathcal{K}_W(M)$ lies in the kernel of Clifford multiplication by

$$\widehat{\mathcal{G}}_W(X,Y) + \widehat{\mathcal{C}}_W(X,Y).$$

Since a ∇^W-parallel section is determined by its value at one point, the evaluation map

$$\mathcal{K}_W(M) \longrightarrow \Sigma_pM$$

is injective. Thus the dimension of $\mathcal{K}_W(M)$ gives a lower bound for the dimension of this pointwise annihilator. The corresponding algebraic question is how large a subspace of the spinor module can be annihilated by a non-zero Clifford element of this mixed form.

2.4 The possibility of cancellation

The mixed identity (2.4) does not imply by degree considerations alone that both defects vanish. Although its two terms have different Clifford degrees, their actions on a particular spinor may cancel.

There is no such issue for a degree-one term in isolation. If $v \cdot \psi = 0$ for a non-zero spinor ψ, then $v \cdot v \cdot \psi = -|v|^2\psi = 0$, and hence $v = 0$. The difficulty is that a degree-two element may act on ψ in the same way as a vector.

This is particularly transparent in dimension three. Let (e_1, e_2, e_3) be a positively oriented orthonormal basis of T_pM and set

$$\omega := -e_1 \cdot e_2 \cdot e_3.$$

The element ω is central, satisfies $\omega^2 = 1$, and acts on the chosen irreducible spinor module by a sign $\varepsilon \in \{+1, -1\}$. Moreover,

$$e_2 \cdot e_3 = \omega \cdot e_1, \qquad e_3 \cdot e_1 = \omega \cdot e_2, \qquad e_1 \cdot e_2 = \omega \cdot e_3.$$

Every $B \in \mathrm{Cl}^2(T_pM)$ therefore has a unique expression $B = \omega \cdot v$ with $v \in T_pM$, and $B \cdot \psi = \varepsilon v \cdot \psi$. Hence, for $u \in T_pM$,

$$(B + u) \cdot \psi = 0 \quad \Longleftrightarrow \quad u = -\varepsilon v.$$

For example,

$$\big(e_2 \cdot e_3 - \varepsilon e_1\big) \cdot \psi = 0,$$

even though both homogeneous terms are non-zero.

Thus the mixed defect identity alone cannot separate the Gauss and Codazzi defects in dimension three. This does not itself produce a generalised Killing spinor with non-zero defects; it only shows that a pointwise argument using one spinor cannot exclude cancellation. We therefore examine what happens when several generalised Killing spinors are associated with the same endomorphism W, beginning with the extrinsic model supplied by ambient parallel spinors.

3

Parallel Spinors and Ricci-Flat Holonomy

3.1 Parallel spinors and multiplicity

The mixed identity (2.4) need not separate the Gauss and Codazzi defects on a single spinor. This does not show that one generalised Killing spinor is insufficient for immersibility, but it makes the role of multiplicity unavoidable.

For a fixed symmetric endomorphism $W \in \Gamma(\mathrm{End}(TM))$, define

$$\mathcal{K}_W(M) := \{\varphi \in \Gamma(\Sigma M) : \nabla^W \varphi = 0\}. \tag{3.1}$$

By (2.1), this is precisely the vector space of generalised Killing spinors associated with W.

Lemma 3.1 (Constant Gram matrix). *Suppose that M is connected. If $\psi_1, \dots, \psi_k \in \mathcal{K}_W(M)$, then every inner product $\langle \psi_i, \psi_j \rangle$ is constant on M. Consequently, the spinors are linearly independent at one point if and only if they are linearly independent at every point.*

Proof. The connection ∇^W is Hermitian. Indeed, $\nabla^{\Sigma M}$ is Hermitian and Clifford multiplication by the real vector $W(X)$ is skew-Hermitian, so

$$X\langle \varphi, \chi \rangle = \langle \nabla^W_X \varphi, \chi \rangle + \langle \varphi, \nabla^W_X \chi \rangle.$$

For ∇^W-parallel spinors the right-hand side vanishes. Their Gram matrix is therefore constant, and linear independence is equivalent to its non-singularity at any one point. □

Thus a basis of $\mathcal{K}_W(M)$ determines a subspace of the same dimension in every spinor fibre. Two questions now arise. How many elements of $\mathcal{K}_W(M)$ can be produced by restricting parallel spinors from an ambient manifold? Conversely, can a large common space $\mathcal{K}_W(M)$ occur when (g, W) are not locally Euclidean hypersurface data?

The first question is controlled by ambient holonomy. By Proposition 1.7, if n is even, restriction gives an injective linear map

$$\mathcal{P}(Z) \longrightarrow \mathcal{K}_W(M).$$

If n is odd, the chosen hypersurface spinor identification determines a sign $\sigma \in \{+, -\}$ and restriction gives injective linear maps

$$\mathcal{P}^\sigma(Z) \longrightarrow \mathcal{K}_W(M), \qquad \mathcal{P}^{-\sigma}(Z) \longrightarrow \mathcal{K}_{-W}(M).$$

The dimensions of these images are determined by the fixed subspaces of the ambient spin representation. They need not exhaust the corresponding spaces of generalised Killing spinors, so the argument provides lower bounds rather than a classification of all solutions. We therefore turn to the holonomy description of ambient parallel spinors.

3.2 Holonomy and parallel spinors

For parallel transport and the ordinary holonomy group, see [KN96, Chapter II, pp. 68–74]. The fixed-space description for parallel spinors is standard; see [MS00]. For the distinction between full, restricted and local Riemannian holonomy, and for special holonomy, see also [Joy07].

Let $(Z^m, \bar{g})$ be a connected oriented Riemannian spin manifold, and fix a point $z \in Z$. For a piecewise smooth curve γ from z to q, Levi-Civita parallel transport gives an isometry

$$P_\gamma^{TZ} : T_zZ \longrightarrow T_qZ.$$

When γ is a loop based at z, this is an orientation-preserving orthogonal transformation of the single vector space T_zZ. The holonomy group is therefore

$$\mathrm{Hol}_z(Z) := \left\{P_\gamma^{TZ} : \gamma \text{ is a loop based at } z\right\} \subset \mathrm{SO}(T_zZ).$$

Restricting to contractible loops gives the restricted holonomy group $\mathrm{Hol}_z^0(Z)$. For a local application, we instead restrict the metric to a sufficiently small connected simply connected neighbourhood U of the point under consideration. The full and restricted holonomy groups of U then coincide. These should be distinguished from the full and restricted holonomy groups of the original manifold Z.

To describe these transformations relative to a frame, choose an oriented orthonormal frame

$$u = (u_1, \ldots, u_m) \in (Q_Z)_z.$$

Parallel transport around a loop produces the frame

$$\left(P_\gamma^{TZ}u_1, \ldots, P_\gamma^{TZ}u_m\right)$$

in the same fibre $(Q_Z)_z$. Since the right action of $\mathrm{SO}(m)$ on this fibre is free and transitive, there is a unique $h_\gamma \in \mathrm{SO}(m)$ such that

$$\left(P_\gamma^{TZ}u_1, \ldots, P_\gamma^{TZ}u_m\right) = u \cdot h_\gamma.$$

Thus P_γ^{TZ} is the geometric parallel-transport map on T_zZ, while h_γ records its effect in the chosen frame. The corresponding matrix realisation of the ordinary holonomy group is

$$\mathrm{Hol}_z(Z; u) := \{h_\gamma : \gamma \text{ is a loop based at } z\} \subset \mathrm{SO}(m).$$

We now pass to the spin bundle. Recall that the spin structure consists of a principal $\mathrm{Spin}(m)$-bundle $P_Z \to Z$ together with an equivariant double covering

$$\Lambda_Z : P_Z \longrightarrow Q_Z.$$

A *spin frame* at z is simply a point $\widetilde{u} \in (P_Z)_z$ lying above an ordinary frame $u = \Lambda_Z(\widetilde{u})$. The two spin frames above the same ordinary frame are $\widetilde{u}$ and $\widetilde{u} \cdot (-1)$.

Although a spin frame is not itself a basis of the spinor fibre, it provides spinor coordinates through the associated-bundle construction. The spinor bundle is

$$\Sigma Z = P_Z \times_{\kappa_m} \Delta_m,$$

where $\kappa_m : \mathrm{Spin}(m) \to \mathrm{GL}(\Delta_m)$ is the complex spin representation, and the equivalence relation is

$$[\widetilde{v} \cdot g, \varphi] = [\widetilde{v}, \kappa_m(g)\varphi].$$

Once $\widetilde{u} \in (P_Z)_z$ is fixed, the map

$$\Delta_m \longrightarrow \Sigma_z Z, \qquad \varphi \longmapsto [\widetilde{u}, \varphi],$$

is a linear isomorphism. Thus the ordinary frame u gives coordinates on $T_z Z$ relative to $\mathbb{R}^m$, while the spin frame $\widetilde{u}$ gives coordinates on $\Sigma_z Z$ relative to the model spinor space Δ_m.

We briefly recall how the connection acts in these coordinates. Let $\widetilde{\omega}$ be the connection one-form on P_Z, let $\widetilde{s} : U \to P_Z$ be a local spin frame, and write a spinor field on U as

$$\Psi = [\widetilde{s}, \varphi]$$

for a smooth function $\varphi : U \to \Delta_m$. If

$$\widetilde{A} := \widetilde{s}^* \widetilde{\omega}$$

is the corresponding local connection form, then

$$\nabla^{\Sigma Z}_X \Psi = \Big[\widetilde{s},\, X(\varphi) + d\kappa_m\big(\widetilde{A}(X)\big)\varphi\Big]. \tag{3.2}$$

The first term records the change of the coefficients, while the second records the change of the chosen spin frame.

Along a curve γ, write a spinor field as

$$\Psi(t) = [\widetilde{u}(t), \varphi(t)],$$

where $\widetilde{u}(t) \in (P_Z)_{\gamma(t)}$. The preceding formula becomes

$$\nabla^{\Sigma Z}_{\dot\gamma} \Psi = \left[\widetilde{u}(t),\, \frac{d\varphi}{dt} + d\kappa_m\big(\widetilde{\omega}(\dot{\widetilde{u}}(t))\big)\varphi(t)\right].$$

The curve $\widetilde{u}(t)$ is horizontal precisely when

$$\widetilde{\omega}(\dot{\widetilde{u}}(t)) = 0.$$

In a horizontally transported spin frame, therefore,

$$\nabla^{\Sigma Z}_{\dot{\gamma}}\Psi = \left[\widetilde{u}(t), \frac{d\varphi}{dt}\right].$$

Consequently, a spinor field is parallel along γ if and only if its coefficient in the horizontal spin frame is constant. This is the spinorial analogue of the familiar fact that a parallel vector has constant coefficients in a parallel orthonormal frame.

The spin connection therefore gives unitary parallel transport

$$P^{\Sigma}_{\gamma} : \Sigma_z Z \longrightarrow \Sigma_q Z.$$

Starting from

$$\xi = [\widetilde{u}, \varphi] \in \Sigma_z Z,$$

let $\widetilde{u}(t)$ be the horizontal lift of γ satisfying $\widetilde{u}(0) = \widetilde{u}$. Since the coefficient remains constant, one has

$$P^{\Sigma}_{\gamma|_{[0,t]}}\xi = [\widetilde{u}(t), \varphi].$$

Suppose now that γ is a loop based at z. There is a unique $\widetilde{h}_\gamma \in \mathrm{Spin}(m)$ such that

$$\widetilde{u}(1) = \widetilde{u} \cdot \widetilde{h}_\gamma.$$

Applying Λ_Z and using its equivariance gives

$$u \cdot h_\gamma = \Lambda_Z\big(\widetilde{u}(1)\big) = \Lambda_Z(\widetilde{u}) \cdot \lambda_m(\widetilde{h}_\gamma) = u \cdot \lambda_m(\widetilde{h}_\gamma).$$

Freeness of the $\mathrm{SO}(m)$-action yields

$$\lambda_m(\widetilde{h}_\gamma) = h_\gamma.$$

Thus $\widetilde{h}_\gamma$ is the lift of the ordinary holonomy element selected by the spin structure and connection.

Writing the transported spinor once again in the original spin frame gives

$$P^{\Sigma}_{\gamma}[\widetilde{u}, \varphi] = [\widetilde{u}(1), \varphi] = [\widetilde{u} \cdot \widetilde{h}_\gamma, \varphi] = [\widetilde{u}, \kappa_m(\widetilde{h}_\gamma)\varphi].$$

Hence, under the identification

$$\Sigma_z Z \cong \Delta_m$$

determined by $\widetilde{u}$, spinor parallel transport around γ is represented by $\kappa_m(\widetilde{h}_\gamma)$.

The lifted holonomy group determined by $\widetilde{u}$ is

$$\widetilde{\mathrm{Hol}}_z(Z) := \left\{ \widetilde{h}_\gamma : \gamma \text{ is a loop based at } z \right\} \subset \mathrm{Spin}(m).$$

Changing the spin frame conjugates this subgroup. Indeed, if

$$\widetilde{u}' = \widetilde{u} \cdot g$$

for some $g \in \mathrm{Spin}(m)$, then the horizontal lift beginning at $\widetilde{u}'$ is $\widetilde{u}(t) \cdot g$. Hence, if

$$\widetilde{u}(1) = \widetilde{u} \cdot \widetilde{h}_\gamma,$$

then

$$\widetilde{u}'(1) = \widetilde{u}(1) \cdot g = \widetilde{u}' \cdot \big(g^{-1} \widetilde{h}_\gamma g\big),$$

so the lifted holonomy group determined by $\widetilde{u}'$ is

$$g^{-1} \widetilde{\mathrm{Hol}}_z(Z) g.$$

Moreover, $\kappa_m(g^{-1})$ identifies the corresponding fixed subspaces in Δ_m, so their dimensions agree.

The lifted holonomy group should not be confused with the full inverse image of $\mathrm{Hol}_z(Z; u)$ under λ_m. The latter contains both algebraic lifts of every ordinary holonomy element, whereas $\widetilde{\mathrm{Hol}}_z(Z)$ contains only those lifts actually produced by parallel transport of the chosen spin frame around loops. For example, if the ordinary holonomy is trivial, then its full inverse image is $\{1, -1\}$, while the lifted holonomy may be only $\{1\}$.

Alternatively, one may record directly the action of parallel transport on the spinor fibre:

$$\mathrm{Hol}_z^\Sigma(Z) := \left\{ P_\gamma^\Sigma : \gamma \text{ is a loop based at } z \right\} \subset U(\Sigma_z Z).$$

Under the identification $\Sigma_z Z \cong \Delta_m$ determined by $\widetilde{u}$, the map P_γ^Σ is represented by $\kappa_m(\widetilde{h}_\gamma)$. Hence

$$\mathrm{Hol}_z^\Sigma(Z) = \kappa_m\big(\widetilde{\mathrm{Hol}}_z(Z)\big).$$

Let

$$\mathcal{P}(Z) := \left\{ \Psi \in \Gamma(\Sigma Z) : \nabla^{\Sigma Z} \Psi = 0 \right\}$$

denote the space of parallel spinors.

Proposition 3.2. *Evaluation at z identifies parallel spinors with the spinors fixed by holonomy:*

$$\mathcal{P}(Z) \cong (\Sigma_z Z)^{\mathrm{Hol}_z^\Sigma(Z)} \cong (\Delta_m)^{\widetilde{\mathrm{Hol}}_z(Z)}. \tag{3.3}$$

In particular,

$$\dim_{\mathbb{C}} \mathcal{P}(Z) = \dim_{\mathbb{C}} \big((\Delta_m)^{\widetilde{\mathrm{Hol}}_z(Z)}\big).$$

Proof. A parallel spinor is determined by its value at one point, so evaluation at z is injective. Moreover, parallel transport around every loop based at z fixes the value of a parallel spinor at z. Thus evaluation takes values in

$$(\Sigma_z Z)^{\mathrm{Hol}_z^\Sigma(Z)}.$$

Conversely, let $\xi \in \Sigma_z Z$ be fixed by spinor holonomy. For $q \in Z$, choose a curve γ from z to q and define

$$\Psi_q := P_\gamma^\Sigma \xi.$$

If γ_1 and γ_2 are two such curves, then

$$(P_{\gamma_2}^\Sigma)^{-1} P_{\gamma_1}^\Sigma$$

is parallel transport around the loop obtained by following γ_1 and then γ_2 in reverse. Since ξ is fixed by spinor holonomy,

$$P_{\gamma_1}^\Sigma \xi = P_{\gamma_2}^\Sigma \xi.$$

Hence Ψ_q is independent of the chosen curve.

To see that Ψ is smooth, fix $q_0 \in Z$ and choose a geodesically convex neighbourhood V of q_0. Choose a path from z to q_0, and for each $q \in V$ concatenate this path with the unique geodesic in V from q_0 to q. These paths depend smoothly on q, and hence so does parallel transport along them. By path independence, this local expression agrees with Ψ, so Ψ is smooth. By construction, it is parallel.

For the second identification, let

$$I_{\widetilde{u}} : \Delta_m \longrightarrow \Sigma_z Z, \qquad \varphi \longmapsto [\widetilde{u}, \varphi],$$

be the identification determined by the spin frame $\widetilde{u}$. The preceding calculation shows that, for every loop γ based at z,

$$P_\gamma^\Sigma \circ I_{\widetilde{u}} = I_{\widetilde{u}} \circ \kappa_m(\widetilde{h}_\gamma).$$

Thus $I_{\widetilde{u}}$ intertwines the action of spinor holonomy on $\Sigma_z Z$ with the action of $\widetilde{\mathrm{Hol}}_z(Z)$ on Δ_m through κ_m. Consequently, $I_{\widetilde{u}}(\varphi)$ is fixed by every P_γ^Σ if and only if φ is fixed by every $\kappa_m(\widetilde{h}_\gamma)$, and hence

$$(\Sigma_z Z)^{\mathrm{Hol}_z^\Sigma(Z)} \cong (\Delta_m)^{\widetilde{\mathrm{Hol}}_z(Z)}.$$

Here the notation on the right denotes the fixed subspace for the action of $\widetilde{\mathrm{Hol}}_z(Z)$ through κ_m. □

Applying the proposition to a connected simply connected neighbourhood $U \subset Z$ gives the corresponding local statement:

$$\mathcal{P}(U) \cong (\Sigma_z U)^{\mathrm{Hol}_z^\Sigma(U)} \cong (\Delta_m)^{\widetilde{\mathrm{Hol}}_z(U)} = (\Delta_m)^{\widetilde{\mathrm{Hol}}_z^0(U)}.$$

Thus the space of parallel spinors defined on the germ near z is governed by the holonomy of

the restricted metric on U. By contrast, the fixed subspace of the restricted holonomy group of the whole manifold Z describes parallel spinors on its universal cover. Together with the injective restriction of ambient parallel spinors to a hypersurface, this reduces the lower-bound problem for generalised Killing spinors obtained by restriction to a representation-theoretic fixed-space calculation.

3.3 Wang's classification and Ricci-flat holonomy

By (3.3), counting parallel spinors is equivalent to computing the fixed subspace of the lifted holonomy group in the spin representation. We use Wang's classification of the irreducible holonomy groups fixing a non-zero spinor [Wan89], together with the de Rham decomposition theorem. For a modern spinorial account of parallel spinors and special holonomy, see [Bou+15, Sections 8.1–8.2].

We first record that a non-zero parallel spinor forces the metric to be Ricci-flat. If $0 \neq \Psi \in \mathcal{P}(Z)$, then

$$R^{\Sigma Z}(X, Y)\Psi = 0.$$

With our curvature and Clifford conventions,

$$\sum_{j=1}^{m} e_j \cdot R^{\Sigma Z}(X, e_j)\Psi = -\frac{1}{2}\operatorname{Ric}^{\bar{g}}(X) \cdot \Psi. \tag{3.4}$$

The left-hand side vanishes. Since Clifford multiplication by a non-zero vector is invertible in Riemannian signature, it follows that $\operatorname{Ric}^{\bar{g}}(X) = 0$ for every X, and hence

$$\operatorname{Ric}^{\bar{g}} = 0.$$

The same argument applies to a non-zero spinor fixed by the restricted lifted holonomy, since such a spinor determines a local parallel spinor.

The locally symmetric branch of the holonomy classification introduces no additional non-flat case. Indeed, a locally symmetric Riemannian manifold carrying a non-zero parallel spinor is Ricci-flat, while a Ricci-flat locally symmetric Riemannian metric is locally flat.

Recall that the restricted holonomy representation is the action

$$\operatorname{Hol}_z^0(Z) \curvearrowright T_z Z$$

defined by parallel transport around contractible loops based at z. It is *reducible* if there is a non-zero proper subspace $V \subset T_z Z$ such that

$$h(V) = V \qquad \text{for every } h \in \operatorname{Hol}_z^0(Z),$$

and *irreducible* otherwise.

Because the holonomy action is orthogonal, the orthogonal complement of an invariant subspace

is also invariant. Indeed, if $w \in V^{\perp}$, $v \in V$ and $h \in \mathrm{Hol}_z^0(Z)$, then

$$\langle h(w), v\rangle = \langle w, h^{-1}(v)\rangle = 0,$$

since $h^{-1}(v) \in V$. Hence

$$T_z Z = V \oplus V^{\perp}$$

is an orthogonal decomposition into holonomy-invariant subspaces. Repeating this gives an orthogonal decomposition

$$T_z Z = V_0 \oplus V_1 \oplus \cdots \oplus V_k,$$

where

$$V_0 = \{v \in T_z Z : h(v) = v \text{ for every } h \in \mathrm{Hol}_z^0(Z)\}$$

is the fixed subspace and each V_i, $i \geq 1$, is non-trivial and irreducible.

If Z is complete and simply connected, the de Rham decomposition theorem [Joy07, Theorem 3.2.7, p. 47]; see also [KN96, Chapter IV, pp. 179–192], identifies this decomposition with a global Riemannian product

$$Z \cong \mathbb{R}^{m_0} \times Z_1 \times \cdots \times Z_k, \qquad m_0 = \dim V_0,$$

where each Z_i is complete, simply connected, irreducible and non-flat, and

$$T_{z_i} Z_i \cong V_i.$$

Thus the general complete simply connected case reduces to a flat factor and irreducible non-flat factors.

When m is even, set

$$N^{\pm} := \dim_{\mathbb{C}} \left((\Delta_m^{\pm})^{\widetilde{\mathrm{Hol}}_z^0(Z)} \right), \qquad N := N^+ + N^-.$$

When m is odd, set

$$N := \dim_{\mathbb{C}} \left((\Delta_m)^{\widetilde{\mathrm{Hol}}_z^0(Z)} \right).$$

In dimension four, $\mathrm{SU}(2) \cong \mathrm{Sp}(1)$ and their standard representations determine the same subgroup of $\mathrm{SO}(4)$, so they give the same holonomy case.

Theorem 3.3 (Wang's classification and the flat case)**.** *Let Z^m be a connected Riemannian spin manifold. If $\mathrm{Hol}_z^0(Z) = \{1\}$, the fixed-spinor counts are given by the first two rows below. If instead the restricted holonomy representation is irreducible and Z is non-flat, then $\widetilde{\mathrm{Hol}}_z^0(Z)$ fixes a non-zero spinor only for one of the four groups in the remaining rows, with the indicated counts. Conversely, each of these*

four groups fixes a non-zero spinor.

$\mathrm{Hol}^0(Z)$	m	N^+	N^-	N
$\{1\}$	$2s,\ s \geq 1$	2^{s-1}	2^{s-1}	2^s
$\{1\}$	$2s+1,\ s \geq 0$	$-$	$-$	2^s
$\mathrm{SU}(r),\ r \geq 2$	$2r$	$\begin{cases} 2, & r \text{ even}, \\ 1, & r \text{ odd}, \end{cases}$	$\begin{cases} 0, & r \text{ even}, \\ 1, & r \text{ odd}, \end{cases}$	2
$\mathrm{Sp}(r),\ r \geq 1$	$4r$	$r+1$	0	$r+1$
G_2	7	$-$	$-$	1
$\mathrm{Spin}(7)$	8	1	0	1

The flat rows follow from

$$\dim_{\mathbb{C}} \Delta_{2s}^{\pm} = 2^{s-1}$$

and

$$\dim_{\mathbb{C}} \Delta_{2s} = \dim_{\mathbb{C}} \Delta_{2s+1} = 2^s.$$

For the non-flat entries, use

$$\Delta_{2r} \cong \Lambda^{\bullet}\mathbb{C}^r,$$

with chirality given by degree parity. For $\mathrm{SU}(r) \subset \mathrm{SO}(2r)$, the invariant spinors are the scalar 1 and the complex volume form. They have the same chirality when r is even and opposite chiralities when r is odd.

For $\mathrm{Sp}(r) \subset \mathrm{SU}(2r)$, choose a parallel complex structure I and let $\omega_I, \omega_J, \omega_K$ be the corresponding Kähler forms. Then

$$\Omega := \omega_J + i\omega_K$$

is a parallel holomorphic symplectic form with respect to I, and the invariant spinors are

$$1, \Omega, \ldots, \Omega^r,$$

all of even degree. Finally, G_2 and $\mathrm{Spin}(7)$ each fix a one-dimensional spinor space, with the $\mathrm{Spin}(7)$-invariant spinor lying in one half-spin representation. Reversing the orientation exchanges the two half-spin representations.

3.4 Parallel spinors on Riemannian products

We now apply Wang's table to the de Rham decomposition

$$Z \cong \mathbb{R}^{m_0} \times Z_1 \times \cdots \times Z_k, \qquad z = (z_0, z_1, \ldots, z_k).$$

With respect to

$$T_z Z \cong \mathbb{R}^{m_0} \oplus T_{z_1} Z_1 \oplus \cdots \oplus T_{z_k} Z_k,$$

the restricted holonomy group is

$$\mathrm{Hol}^0_z(Z) = \{\mathrm{id}_{\mathbb{R}^{m_0}}\} \times \mathrm{Hol}^0_{z_1}(Z_1) \times \cdots \times \mathrm{Hol}^0_{z_k}(Z_k). \tag{3.5}$$

On each tensor-product summand of the product spinor bundle, the spin connection is the tensor-product connection [LM89, Chapter II]. Thus

$$\nabla^{\Sigma Z}_{(X_0,\ldots,X_k)}(\Psi_0 \otimes \cdots \otimes \Psi_k) = \sum_{i=0}^{k} \Psi_0 \otimes \cdots \otimes \nabla^{\Sigma Z_i}_{X_i}\Psi_i \otimes \cdots \otimes \Psi_k. \tag{3.6}$$

If every Ψ_i is parallel, every term on the right-hand side vanishes, so their tensor product is parallel on Z.

Conversely, on each tensor-product summand the restricted holonomy group is a direct product

$$G_0 \times \cdots \times G_k,$$

where G_i acts only on the spinor factor associated with Z_i. Hence

$$(V_0 \otimes \cdots \otimes V_k)^{G_0 \times \cdots \times G_k} = V_0^{G_0} \otimes \cdots \otimes V_k^{G_k}.$$

This follows, for example, by successively taking invariants under the individual compact groups. Thus the parallel spinors on the product are exactly those obtained from the fixed spinors on the individual factors, subject to the standard decomposition of the product spin representation.

It remains to record the chirality rules. They follow from the graded tensor-product isomorphism

$$\mathrm{Cl}^{\mathbb{C}}_{m_1+m_2} \cong \mathrm{Cl}^{\mathbb{C}}_{m_1} \mathbin{\widehat{\otimes}} \mathrm{Cl}^{\mathbb{C}}_{m_2}$$

[LM89, Chapter II]. For example, if m_1 is even, the Clifford action on $\Delta_{m_1} \otimes \Delta_{m_2}$ is

$$\begin{aligned}(X_1, 0) \cdot (\psi_1 \otimes \psi_2) &= (X_1 \cdot \psi_1) \otimes \psi_2,\\ (0, X_2) \cdot (\psi_1 \otimes \psi_2) &= \omega^{\mathbb{C}}_{m_1} \cdot \psi_1 \otimes (X_2 \cdot \psi_2),\end{aligned}$$

where $\omega^{\mathbb{C}}_{m_1}$ is the complex volume element. Since it anticommutes with Clifford multiplication by vectors, the actions of vectors from the two factors anticommute, as required.

If both dimensions are even, then, under the standard product identification,

$$\omega^{\mathbb{C}}_{m_1+m_2} = \omega^{\mathbb{C}}_{m_1} \otimes \omega^{\mathbb{C}}_{m_2}.$$

The positive half-spin representation therefore consists of tensors whose factors have the same chirality, while the negative half-spin representation consists of tensors whose factors have

opposite chirality:

$$\Delta^+_{m_1+m_2} \cong (\Delta^+_{m_1} \otimes \Delta^+_{m_2}) \oplus (\Delta^-_{m_1} \otimes \Delta^-_{m_2}), \tag{3.7}$$

$$\Delta^-_{m_1+m_2} \cong (\Delta^+_{m_1} \otimes \Delta^-_{m_2}) \oplus (\Delta^-_{m_1} \otimes \Delta^+_{m_2}). \tag{3.8}$$

If exactly one factor is even-dimensional, the product is odd-dimensional and has no chirality decomposition; its spin representation is

$$\Delta_{m_1+m_2} \cong \Delta_{m_1} \otimes \Delta_{m_2}.$$

If both factors are odd-dimensional, the product is even-dimensional and each half-spin representation restricts to the same $\mathrm{Spin}(m_1) \times \mathrm{Spin}(m_2)$-representation:

$$\Delta^\pm_{m_1+m_2} \cong \Delta_{m_1} \otimes \Delta_{m_2}.$$

Thus the two half-spin spaces give the same tensor-product representation of the two odd-dimensional factors.

These rules are applied successively to the de Rham factors, with the flat factor contributing its full complex spin representation.

The counts above are dimensions of fixed spaces of the restricted lifted holonomy group. They therefore give the numbers of parallel spinors locally and on the universal cover. If Z is simply connected, the full and restricted holonomy groups coincide, so they also give the dimensions of the spaces of global parallel spinors. If Z is not simply connected, the global spaces may be smaller because the full spinor holonomy group may have a non-trivial discrete part.

Finally, let $M^n \subset Z^{n+1}$ be an oriented hypersurface with shape operator W. Restriction of ambient parallel spinors is injective. If n is even, then

$$\dim_{\mathbb{C}} \mathcal{K}_W(M) \geq \dim_{\mathbb{C}} \mathcal{P}(Z).$$

If n is odd, there is a sign $\sigma \in \{+, -\}$, determined by the chosen hypersurface spinor identification, such that

$$\dim_{\mathbb{C}} \mathcal{K}_W(M) \geq \dim_{\mathbb{C}} \mathcal{P}^\sigma(Z), \qquad \dim_{\mathbb{C}} \mathcal{K}_{-W}(M) \geq \dim_{\mathbb{C}} \mathcal{P}^{-\sigma}(Z).$$

Thus, in odd hypersurface dimension, the relevant lower bounds use the individual half-spinor counts rather than their sum.

3.5 Optimal ambient products

We now determine the largest parallel-spinor count permitted by Wang's table and identify the holonomy configurations in which equality can occur. The next chapter will construct a four-dimensional factor with the required holonomy and use it to realise these bounds.

A flat factor of real dimension d has trivial holonomy and therefore contributes its full complex spin representation:

$$\dim_{\mathbb{C}} \mathcal{P}(\mathbb{R}^d) = \dim_{\mathbb{C}} \Delta_d = 2^{\lfloor d/2 \rfloor}.$$

Thus, for a non-flat irreducible Ricci-flat factor Y^d carrying parallel spinors, we compare its parallel-spinor count

$$N(Y) := \dim_{\mathbb{C}} \mathcal{P}(Y)$$

with the count contributed by d flat directions. Define the *spinor efficiency* of Y by

$$\eta(Y) := \frac{N(Y)}{2^{\lfloor d/2 \rfloor}}.$$

This is the parallel-spinor invariant introduced by Alekseevsky and Cortés [AC09]. They define the invariant using the real spinor bundle and observe that the same ratio is obtained when it is computed using complex spinors. They also prove that it is multiplicative under Riemannian products. In particular, adjoining a flat factor does not change the invariant.

Their Theorem 4(ii) prints the extremal non-flat value as $1/4$ and identifies the corresponding geometry as a product of a flat manifold with a four-dimensional factor of holonomy $\mathrm{SU}(2) = \mathrm{Sp}(1)$. The printed value appears to be a factor-of-two error. Using the equivalent complex formulation, an $\mathrm{SU}(2)$-holonomy four-manifold has two complex parallel spinors in a complex spinor bundle of rank four, and hence has efficiency

$$\frac{2}{4} = \frac{1}{2}.$$

The calculation below recovers this value directly and retains the dimension and chirality information needed later.

Wang's table gives

$\mathrm{Hol}^0(Y)$	d	$N(Y)$	$\eta(Y)$
$\mathrm{SU}(r),\ r \geq 2$	$2r$	2	2^{1-r}
$\mathrm{Sp}(r),\ r \geq 1$	$4r$	$r+1$	$\dfrac{r+1}{2^{2r}}$
G_2	7	1	$\dfrac{1}{8}$
$\mathrm{Spin}(7)$	8	1	$\dfrac{1}{16}$

The $\mathrm{SU}(r)$ row begins at $r = 2$, since $\mathrm{SU}(1)$ is trivial, and the $\mathrm{SU}(2)$ and $\mathrm{Sp}(1)$ entries describe the same four-dimensional holonomy case.

Proposition 3.4. *Among the non-flat irreducible Ricci-flat factors in Wang's list, the largest spinor efficiency is $\frac{1}{2}$, attained by the four-dimensional holonomy case*

$$\mathrm{Sp}(1) = \mathrm{SU}(2).$$

Consequently, among product models of fixed dimension with at least one non-flat irreducible factor, the total parallel-spinor count is maximised by using exactly one four-dimensional $\mathrm{Sp}(1)$-factor and leaving

all remaining directions flat.

Proof. For $\mathrm{SU}(r)$,

$$\eta = 2^{1-r},$$

which equals $\frac{1}{2}$ for $r = 2$ and is at most $\frac{1}{4}$ for $r \geq 3$. For $\mathrm{Sp}(r)$,

$$\eta = \frac{r+1}{2^{2r}},$$

which equals $\frac{1}{2}$ for $r = 1$ and is at most $\frac{3}{16}$ for $r \geq 2$. The exceptional groups have efficiencies $\frac{1}{8}$ and $\frac{1}{16}$. Hence the largest possible efficiency is $\frac{1}{2}$, attained in dimension four by $\mathrm{Sp}(1) = \mathrm{SU}(2)$.

The efficiency is multiplicative under products. If two factor dimensions are not both odd, then both the number of parallel spinors and the dimension of the model spinor space multiply directly. If both dimensions are odd, the even-dimensional product has two half-spin representations, each containing the tensor product of the two factor representations. This introduces a factor of 2 in both the parallel-spinor count and the flat spinor count, so it cancels in the ratio. Thus

$$\eta(Y_1 \times Y_2) = \eta(Y_1)\eta(Y_2).$$

Iterating over the de Rham factors gives

$$N(Z) = 2^{\lfloor m/2 \rfloor} \prod_i \eta(Y_i), \qquad m = \dim Z,$$

where the flat factor has efficiency 1. Since every non-flat factor has efficiency at most $\frac{1}{2}$, the total count is maximised by using exactly one factor of efficiency $\frac{1}{2}$ and making all remaining directions flat. □

In particular, every such product model satisfies

$$N(Z) \leq 2^{\lfloor m/2 \rfloor - 1},$$

with equality possible only when the de Rham decomposition contains exactly one four-dimensional factor of holonomy $\mathrm{Sp}(1) = \mathrm{SU}(2)$ and all remaining directions are flat.

When the ambient dimension m is even, the two half-spin spaces restrict to generalised Killing spinors with opposite signs of the shape operator. The largest lower bound obtained for one fixed sign is therefore

$$H(Z) := \max\left\{\dim_{\mathbb{C}} \mathcal{P}^+(Z), \dim_{\mathbb{C}} \mathcal{P}^-(Z)\right\}.$$

We must therefore optimise the larger half-spinor count, rather than only the total count

$$N(Z) = \dim_{\mathbb{C}} \mathcal{P}^+(Z) + \dim_{\mathbb{C}} \mathcal{P}^-(Z).$$

For the chirality bookkeeping, organise the factors in the de Rham decomposition into even-

dimensional *blocks*. Each even-dimensional factor is one block. Since m is even, the odd-dimensional factors, including the flat factor if it is odd-dimensional, may be paired, with each pair treated as one even-dimensional block.

For each block B_i, write

$$p_i := \dim_{\mathbb{C}} \mathcal{P}^+(B_i), \qquad q_i := \dim_{\mathbb{C}} \mathcal{P}^-(B_i).$$

A tensor product has positive chirality when an even number of its factors have negative chirality, and negative chirality when an odd number do. Ignoring chirality, each block contributes $p_i + q_i$ choices, so

$$N^+ + N^- = \prod_i (p_i + q_i).$$

To compute the difference, assign the sign $+1$ to a positive-chirality spinor and -1 to a negative-chirality spinor. The sign of a tensor product is then the product of the signs of its factors and therefore agrees with its total chirality. Hence

$$N^+ - N^- = \prod_i (p_i - q_i).$$

Equivalently,

$$N^{\pm} = \frac{1}{2}\left(\prod_i (p_i + q_i) \pm \prod_i (p_i - q_i)\right).$$

We call a block *balanced* if $p_i = q_i$. An even-dimensional flat block is balanced, with

$$p_i = q_i = 2^{\dim B_i/2-1}.$$

An $\mathrm{SU}(r)$-factor with r odd is also balanced, with

$$(p_i, q_i) = (1, 1).$$

Finally, a pair of odd-dimensional factors is balanced: if the two factors carry s_1 and s_2 parallel spinors, then

$$\Delta^{\pm}_{d_1+d_2} \cong \Delta_{d_1} \otimes \Delta_{d_2},$$

so the resulting block has

$$(p_i, q_i) = (s_1 s_2, s_1 s_2).$$

Proposition 3.5. *Let $m \geq 6$ be even. Among Ricci-flat product models of dimension m with at least one non-flat irreducible factor and with non-zero parallel spinors,*

$$H(Z) \leq 2^{m/2-2}.$$

Equality can occur only in the following cases:

1. *the de Rham decomposition contains one four-dimensional $\mathrm{Sp}(1)$-factor and an $(m-4)$-dimensional*

flat factor;

2. $m = 8$ *and the decomposition consists of two four-dimensional* $\mathrm{Sp}(1)$*-factors.*

Proof. The total-efficiency estimate gives

$$N(Z) \leq \frac{1}{2}\, 2^{m/2} = 2^{m/2-1}.$$

Suppose first that the product contains a balanced block. Then one factor in

$$N^+ - N^- = \prod_i (p_i - q_i)$$

vanishes, so

$$N^+ = N^- = \frac{1}{2} N(Z).$$

Therefore

$$H(Z) = \frac{1}{2} N(Z) \leq 2^{m/2-2}.$$

Equality requires equality in the total-efficiency estimate, and hence exactly one non-flat factor of efficiency $\frac{1}{2}$, with all remaining directions flat. This is the first case in the statement.

Suppose now that there is no balanced block. Then every block is an unbalanced even-dimensional non-flat irreducible factor.

If there are at least two blocks, each has total efficiency at most $\frac{1}{2}$, so the product has total efficiency at most $\frac{1}{4}$. Hence

$$H(Z) \leq N(Z) \leq \frac{1}{4}\, 2^{m/2} = 2^{m/2-2}.$$

Equality requires exactly two factors of efficiency $\frac{1}{2}$. The only such irreducible factor is the four-dimensional $\mathrm{Sp}(1)$-factor. Since there can be no flat factor without producing a balanced block, this gives precisely two four-dimensional factors and $m = 8$.

Finally, suppose there is only one block. Then Z is a single unbalanced even-dimensional non-flat irreducible factor. The only such factor with efficiency $\frac{1}{2}$ is the four-dimensional $\mathrm{Sp}(1)$-factor, which is excluded by $m \geq 6$.

The only irreducible factor in Wang's list with efficiency exactly $\frac{1}{4}$ is the six-dimensional $\mathrm{SU}(3)$-case, but this case is balanced, with

$$(N^+, N^-) = (1, 1),$$

and has already been excluded in the present case. Among the remaining unbalanced factors, the largest efficiency is

$$\eta(\mathrm{Sp}(2)) = \frac{3}{16} < \frac{1}{4}.$$

Consequently,

$$H(Z) \leq N(Z) < \frac{1}{4}\, 2^{m/2} = 2^{m/2-2},$$

so equality cannot occur when there is only one unbalanced block. □

The restriction $m \geq 6$ is necessary. In dimension 4, a four-dimensional $\mathrm{Sp}(1)$-factor has

$$H = 2,$$

whereas the formula above would give 1.

For odd ambient dimension there is no half-spin decomposition, so the total efficiency estimate applies directly. Equality in the relevant ambient spinor count can occur only for a product of a four-dimensional $\mathrm{Sp}(1)$-factor with flat directions.

Let U be a sufficiently small simply connected neighbourhood on which the local de Rham decomposition theorem applies [KN96, Chapter IV, pp. 187–192]. The restricted holonomy representation of U then splits into a fixed subspace and irreducible non-flat factors, and Wang's classification applies to each such irreducible factor. Consequently, every Ricci-flat germ carrying parallel spinors is locally a product of factors from Wang's list together with a flat factor. The preceding bounds therefore give the ceiling for multiplicities obtained by restricting ambient parallel spinors.

The next chapter constructs hypersurfaces which attain these ambient lower bounds while failing the Euclidean Gauss equation. It treats the higher-dimensional product family first, followed by the exceptional four- and three-dimensional cases.

4

Hypersurface Examples and Lower Bounds

The preceding chapter determined the largest spaces of parallel spinors available on non-flat Ricci-flat ambient manifolds. We now construct hypersurfaces which realise the resulting lower bounds for generalised Killing spinors while failing the Euclidean Gauss equation. These examples provide the lower-bound side of the multiplicity theorem proved in the next chapter.

The common building block is a non-flat four-dimensional hyperkähler manifold with holonomy $\mathrm{Sp}(1) = \mathrm{SU}(2)$. Products with flat factors yield the examples in dimensions $n \geq 5$. The endpoint in dimension four requires separate slice and graph constructions, while dimension three exhibits a different failure of fixed-sign rigidity.

Throughout this chapter, multiplicity means multiplicity inside one fixed space $\mathcal{K}_W(M)$. Thus all spinors being counted satisfy the same generalised Killing spinor equation for the same symmetric endomorphism W.

4.1 The hyperkähler building block

Let X^4 be a K3 surface equipped with a Ricci-flat Kähler metric, whose existence follows from the Calabi–Yau theorem [Yau78].

The K3 surface is simply connected and spin. Since the metric is Ricci-flat and Kähler of complex dimension two, its holonomy is contained in

$$\mathrm{SU}(2) \subset \mathrm{SO}(4).$$

Because X is simply connected, its full and restricted holonomy groups coincide. If the holonomy were a proper connected subgroup of $\mathrm{SU}(2)$, its real action on $T_xX \cong \mathbb{R}^4$ would be reducible. The de Rham theorem would then split X into lower-dimensional Ricci-flat factors, all of which are flat in dimensions at most three. This would force X itself to be flat, contrary to the choice of metric. Hence

$$\mathrm{Hol}(X) = \mathrm{SU}(2) = \mathrm{Sp}(1).$$

After choosing the orientation appropriately,

$$\dim_{\mathbb{C}} \mathcal{P}^+(X) = 2, \qquad \dim_{\mathbb{C}} \mathcal{P}^-(X) = 0.$$

Thus products of X with flat factors realise the optimal total count from the preceding chapter

and, in even ambient dimension at least six, the optimal half-spinor count. The product $X \times X$ also realises the additional half-spinor equality case in ambient dimension 8.

4.2 Product hypersurfaces in dimension at least 5

Throughout this section let $n \geq 5$ and set

$$Z^{n+1} := X^4 \times \mathbb{R}^{n-3}, \qquad M^n := X^4 \times S^{n-4}(r) \subset X^4 \times \mathbb{R}^{n-3},$$

where $S^{n-4}(r) \subset \mathbb{R}^{n-3}$ is the round sphere of radius $r > 0$. The hypersurface M inherits the product metric $g = g_X + g_{S^{n-4}(r)}$.

Proposition 4.1. *The hypersurface $M = X \times S^{n-4}(r)$ carries at least*

$$2^{\lfloor n/2 \rfloor - 1}$$

complex generalised Killing spinors for the shape operator induced by the embedding $M \subset Z$.

Proof. The product $Z = X \times \mathbb{R}^{n-3}$ is Ricci-flat and spin. The two parallel spinors on X tensor with the constant spinors on the flat factor to produce parallel spinors on Z, and these restrict injectively to generalised Killing spinors on M.

If n is even, then $n+1$ is odd and the relevant quantity is the total parallel-spinor count. Since X has two parallel spinors,

$$\dim_{\mathbb{C}} \mathcal{P}(Z) = 2^{n/2-1},$$

and therefore $\dim_{\mathbb{C}} \mathcal{K}_W(M) \geq 2^{n/2-1}$.

If n is odd, then $n+1$ is even and the relevant quantity is a half-spinor count. The flat factor has positive even dimension $n-3$, so

$$\dim_{\mathbb{C}} \mathcal{P}^{\pm}(\mathbb{R}^{n-3}) = 2^{(n-5)/2}.$$

Since the two parallel spinors on X have the same chirality, the even-even product formula gives

$$\dim_{\mathbb{C}} \mathcal{P}^{+}(Z) = \dim_{\mathbb{C}} \mathcal{P}^{-}(Z) = 2 \cdot 2^{(n-5)/2} = 2^{(n-3)/2}.$$

The two ambient chiralities restrict with opposite signs, but their equal dimensions make the sign choice irrelevant for the numerical bound. Hence $\dim_{\mathbb{C}} \mathcal{K}_W(M) \geq 2^{(n-3)/2}$. Combining the two parity cases gives

$$\dim_{\mathbb{C}} \mathcal{K}_W(M) \geq 2^{\lfloor n/2 \rfloor - 1}.$$

□

We next compute the induced endomorphism W. Here W is the actual shape operator of $M \subset Z$; later, when comparing with Euclidean hypersurface data, the same endomorphism is regarded as part of the prescribed intrinsic data.

The tangent bundle splits orthogonally as $TM = TX \oplus TS^{n-4}(r)$. Let $y \in \mathbb{R}^{n-3}$ denote the Euclidean coordinate and let $\nu_S = y/r$ be the outward unit normal to the sphere. Then the unit normal to $M \subset Z$ is $\nu = (0, \nu_S)$. If $U \in TX$, the normal does not vary in the X-directions, so $\nabla^Z_U \nu = 0$. If $A \in TS^{n-4}(r)$, then

$$\nabla^{\mathbb{R}^{n-3}}_A \nu_S = \nabla^{\mathbb{R}^{n-3}}_A \left(\frac{y}{r}\right) = \frac{1}{r} A.$$

With the convention $W(E) = -\nabla^Z_E \nu$, it follows that

$$W(U) = 0, \qquad W(A) = -\frac{1}{r} A.$$

Changing the normal, or changing the convention for W, reverses the sign. Thus the convention-independent description is

$$W|_{TX} = 0, \qquad W|_{TS^{n-4}(r)} = \varepsilon \, \frac{1}{r} \, \mathrm{id}_{TS^{n-4}(r)}, \qquad \varepsilon \in \{+1, -1\}.$$

The sign will not matter for the obstruction below, since the Gauss term is quadratic in W and vanishes on pure TX-directions.

Proposition 4.2. *The intrinsic data*

$$\big(M,\, g_X + g_{S^{n-4}(r)},\, W\big)$$

are not locally realised as hypersurface data in $\mathbb{R}^{n+1}$ at any point where the curvature of X is non-zero.

Proof. With the convention (1.5), the Euclidean Gauss equation has the form

$$R^M(A, B)C = \langle W(B), C\rangle W(A) - \langle W(A), C\rangle W(B).$$

The Levi-Civita connection of the product $M = X \times S^{n-4}(r)$ splits. If $U, V \in \Gamma(TX)$ and $A, B \in \Gamma(TS^{n-4}(r))$, then

$$\nabla^M_U V = \nabla^X_U V, \qquad \nabla^M_A B = \nabla^{S^{n-4}(r)}_A B, \qquad \nabla^M_U A = \nabla^M_A U = 0.$$

Consequently, for $U, V, T \in TX$,

$$R^M(U, V)T = R^X(U, V)T.$$

On the other hand, $W|_{TX} = 0$, so the right-hand side of the Euclidean Gauss equation vanishes on these vectors. The Euclidean Gauss equation would therefore force $R^X(U, V)T = 0$ for every $U, V, T \in TX$, which is false at any point where $R^X \neq 0$.

The sphere factor is not responsible for the failure: its curvature is exactly of the form produced by $W|_{TS^{n-4}(r)} = \varepsilon r^{-1} \, \mathrm{id}$, and all mixed curvature terms vanish on both sides. The remaining defect is precisely the curvature of the non-flat X-factor. □

Consequently, for every $n \geq 5$, the data

$$\big(X^4 \times S^{n-4}(r),\, g_X + g_{S^{n-4}(r)},\, W\big)$$

carry at least $2^{\lfloor n/2 \rfloor - 1}$ complex generalised Killing spinors, but are not locally realised as hypersurface data in Euclidean space wherever the X-factor has non-zero curvature.

4.3 The four-dimensional slice

The preceding round-sphere construction attains the largest multiplicity produced by the ambient-restriction argument for $n \geq 5$. In dimension four the same bound is attained, but the sphere factor becomes zero-dimensional. We first treat the resulting slice; the next section modifies it to obtain an example whose shape operator is non-zero.

Set

$$Z^5 := X^4 \times \mathbb{R}, \qquad M^4 := X^4 \times \{0\} \subset Z.$$

This is the connected version of the formal S^0 endpoint of the preceding construction: $S^0(r) \subset \mathbb{R}$ consists of two points, and each component of $X \times S^0(r)$ is a slice $X \times \{\text{point}\}$.

Let t denote the coordinate on the $\mathbb{R}$-factor. The unit normal to the slice is ∂_t. Since the product connection splits and ∂_t is parallel, $\nabla^Z_U \partial_t = 0$ for every $U \in TM$. Thus, with the convention $W(U) = -\nabla^Z_U \partial_t$, the shape operator vanishes.

The ambient dimension is odd, so the full ambient spinor bundle restricts to the intrinsic spinor bundle of M. The two parallel spinors on $Z = X \times \mathbb{R}$ consequently restrict to generalised Killing spinors with $W = 0$. Conversely, the generalised Killing spinor equation for $W = 0$ is simply the parallel-spinor equation. Since M is isometric to X,

$$\mathcal{K}_0(M) = \mathcal{P}(M) = \mathcal{P}(X), \qquad \dim_{\mathbb{C}} \mathcal{K}_0(M) = 2 = 2^{\lfloor 4/2 \rfloor - 1}.$$

These data are not locally realised as hypersurface data in $\mathbb{R}^5$ wherever X is curved. Indeed, the Euclidean Gauss equation with $W = 0$ would force $R^M = 0$, whereas the induced metric is g_X and hence $R^M = R^X \neq 0$ at such a point. Thus the slice carries two common generalised Killing spinors while failing the Euclidean Gauss equation, with the special feature that its shape operator vanishes.

4.4 Four-dimensional graph examples

We now modify the slice construction to obtain a real analytic example whose shape operator is non-zero. Since X is non-flat, choose $p \in X$ with

$$R^X_p \neq 0.$$

The Ricci-flat metric g_X is real analytic in harmonic coordinates [DK81]. After shrinking around p, choose a real analytic coordinate neighbourhood $\mathcal{U} \subset X$ with coordinates $x^1, \dots, x^4$ centred at p, and define

$$f : \mathcal{U} \longrightarrow \mathbb{R}, \qquad f = (x^1)^2.$$

Consider the graph

$$M_f := \{(x, f(x)) : x \in \mathcal{U}\} \subset \mathcal{U} \times \mathbb{R}.$$

Its natural parametrisation

$$F : \mathcal{U} \longrightarrow M_f, \qquad F(x) = (x, f(x)),$$

is a diffeomorphism. Let $\iota : M_f \hookrightarrow \mathcal{U} \times \mathbb{R}$ be the inclusion and let $\bar{g} = g_X + dt^2$ be the ambient product metric. The metric induced on the graph is $h_f := \iota^* \bar{g}$, and, using F to identify $\mathcal{U}$ with its graph, we write

$$g_f := F^* h_f = (\iota \circ F)^* \bar{g}.$$

Fix $x \in \mathcal{U}$ and $U \in T_x X$. If γ is a curve in $\mathcal{U}$ with $\gamma(0) = x$ and $\gamma'(0) = U$, then

$$dF_x(U) = \frac{d}{ds}\Big|_{s=0} \big(\gamma(s), f(\gamma(s))\big) = U + df_x(U)\partial_t.$$

Writing $\widetilde{U} := dF(U)$, we therefore have

$$T_{(x,f(x))} M_f = \{U + df_x(U)\partial_t : U \in T_x X\}.$$

For $U, V \in T_x X$,

$$g_f(U, V) = \bar{g}(\widetilde{U}, \widetilde{V}) = g_X(U, V) + df(U)df(V),$$

and hence

$$g_f = g_X + df \otimes df.$$

To determine the normal, regard ∇f as a horizontal vector field on $\mathcal{U} \times \mathbb{R}$ and set

$$N := \partial_t - \nabla f.$$

For $\widetilde{U} = U + df(U)\partial_t$, one has

$$\bar{g}(N, \widetilde{U}) = -g_X(\nabla f, U) + df(U) = 0,$$

so N is normal to the graph. Since $|N|^2 = 1 + |\nabla f|^2$, the upward unit normal is

$$\nu = \phi(\partial_t - \nabla f), \qquad \phi := (1 + |\nabla f|^2)^{-1/2}.$$

Let W_f denote the shape operator of the graph, pulled back to $T\mathcal{U}$ using F. Its second funda-

mental form satisfies

$$g_f(W_f(U), V) = -\bar{g}\left(\nabla^Z_{\widetilde{U}}\nu, \widetilde{V}\right).$$

Differentiating the normal gives

$$\nabla^Z_{\widetilde{U}}\nu = \widetilde{U}(\phi)(\partial_t - \nabla f) + \phi\,\nabla^Z_{\widetilde{U}}(\partial_t - \nabla f).$$

The first term is normal and therefore disappears when paired with $\widetilde{V}$. Moreover, ∂_t is parallel and ∇f is extended independently of the t-coordinate, so

$$\nabla^Z_{\widetilde{U}}(\partial_t - \nabla f) = -\nabla^X_U \nabla f.$$

It follows that

$$g_f(W_f(U), V) = \frac{\operatorname{Hess} f(U, V)}{\sqrt{1 + |\nabla f|^2}}. \tag{4.1}$$

For the chosen function $f = (x^1)^2$, one has

$$df_p = 0, \qquad \operatorname{Hess}_p(f) = 2\,dx^1 \otimes dx^1 \neq 0.$$

The Hessian identity holds even though the analytic coordinates need not be normal: in the coordinate expression for $\operatorname{Hess} f$, the Christoffel-symbol term vanishes at the critical point p. Since $df_p = 0$, the graph is tangent to the horizontal slice at $F(p)$:

$$T_{F(p)}M_f = T_pX \oplus \{0\}.$$

Moreover,

$$(g_f)_p = (g_X)_p, \qquad \phi(p) = 1,$$

and (4.1) reduces to

$$g_X(W_{f,p}(U), V) = \operatorname{Hess}_p(f)(U, V).$$

Thus

$$W_{f,p} = \left(\operatorname{Hess}_p f\right)^{\sharp_{g_X}} \neq 0.$$

The two parallel spinors on X restrict to parallel spinors on $\mathcal{U}$ and tensor with the constant spinor on $\mathbb{R}$ to give two parallel spinors on $\mathcal{U} \times \mathbb{R}$. Their restrictions to the graph are linearly independent generalised Killing spinors associated with W_f. Hence

$$\dim_{\mathbb{C}} \mathcal{K}_{W_f}(M_f) \geq 2.$$

It remains to show that these graph data are not Euclidean hypersurface data near $F(p)$. Since $df_p = 0$, tangent vectors to the graph at $F(p)$ are horizontal. Hence, for $U, V, T \in T_pX$, the product-curvature formula gives

$$\bar{R}(U, V)T = R^X(U, V)T,$$

and this vector is tangent to the graph.

Write

$$Q_W(U,V)T := g_f(W(V),T)W(U) - g_f(W(U),T)W(V).$$

Thus $Q_W = W \wedge W$ in the convention of (1.5). The Gauss equation for the actual embedding $M_f \subset \mathcal{U} \times \mathbb{R}$ gives

$$R^{g_f}(U,V)T = R^X(U,V)T + Q_{W_{f,p}}(U,V)T.$$

If the same abstract data (M_f, g_f, W_f) were locally realised as a hypersurface in $\mathbb{R}^5$, the Euclidean Gauss equation would instead give

$$R^{g_f}(U,V)T = Q_{W_{f,p}}(U,V)T.$$

Subtracting the two equations would force $R^X_p = 0$, contrary to the choice of p. Hence the graph gives a real analytic four-dimensional example for which the shape operator and Euclidean Gauss defect are both non-zero at p.

Thus two generalised Killing spinors in one space $\mathcal{K}_{W_f}(M_f)$ do not suffice to force the Euclidean Gauss defect to vanish in dimension four. Consequently, any universal threshold for this conclusion must be at least three.

4.5 Artacho's Heisenberg example

In dimension three the corresponding fixed-sign threshold statement fails. The putative threshold is $2^{\lfloor 3/2 \rfloor - 1} + 1 = 2$, but two linearly independent spinors in one space $\mathcal{K}_W(M)$ need not force the Euclidean Gauss defect to vanish.

A particularly simple counterexample is contained in Artacho's classification of invariant generalised Killing spinors on three-dimensional Lie groups [Art25]. Consider the simply connected Heisenberg group H_3 with a left-invariant metric admitting an oriented orthonormal frame (e_1, e_2, e_3) such that

$$[e_1, e_2] = ae_3, \qquad [e_1, e_3] = [e_2, e_3] = 0,$$

where $a > 0$.

The left-invariant frame trivialises the oriented orthonormal frame bundle and its spin lift, so $\Sigma H_3 \cong H_3 \times \Delta_3$. In this trivialisation, an invariant spinor is represented by a constant map $H_3 \to \Delta_3$. Since $\dim_{\mathbb{C}} \Delta_3 = 2$, the invariant spinors form a two-dimensional complex vector space.

An invariant spinor need not be parallel: its coefficients are constant in the left-invariant frame, but the frame itself is not parallel. The Koszul formula gives

$$\nabla_{e_1} e_2 = \frac{a}{2} e_3, \qquad \nabla_{e_2} e_1 = -\frac{a}{2} e_3,$$
$$\nabla_{e_1} e_3 = \nabla_{e_3} e_1 = -\frac{a}{2} e_2, \quad \nabla_{e_2} e_3 = \nabla_{e_3} e_2 = \frac{a}{2} e_1,$$

with the remaining derivatives equal to zero.

For an invariant spinor ψ, the ordinary derivatives of its components vanish, so its spinorial covariant derivative is determined entirely by these connection coefficients:

$$\nabla_X^{\Sigma H_3}\psi = \frac{1}{4}\sum_{i,j=1}^{3} g(\nabla_X e_i, e_j) e_i \cdot e_j \cdot \psi.$$

After fixing the action of the complex volume element, bivector multiplication in dimension three can be rewritten as vector multiplication. With one choice of convention, every invariant spinor satisfies

$$\nabla_{e_1}^{\Sigma H_3}\psi = -\frac{a}{4} e_1 \cdot \psi, \qquad \nabla_{e_2}^{\Sigma H_3}\psi = -\frac{a}{4} e_2 \cdot \psi, \qquad \nabla_{e_3}^{\Sigma H_3}\psi = \frac{a}{4} e_3 \cdot \psi.$$

Reversing the convention for the complex volume element reverses all three signs and does not affect the argument.

Thus, in Artacho's convention $\nabla_X \psi = A(X) \cdot \psi$, every invariant spinor is a generalised Killing spinor for the same symmetric endomorphism

$$A(e_1) = -\frac{a}{4} e_1, \qquad A(e_2) = -\frac{a}{4} e_2, \qquad A(e_3) = \frac{a}{4} e_3.$$

More generally, Artacho proves that if one non-zero invariant spinor on a three-dimensional Lie group is a generalised Killing spinor, then every invariant spinor satisfies the same equation with the same endomorphism [Art25].

Our convention is $\nabla_X^{\Sigma M}\psi = -\frac{1}{2} W(X) \cdot \psi$, so $W = -2A$ and therefore

$$W(e_1) = \frac{a}{2} e_1, \qquad W(e_2) = \frac{a}{2} e_2, \qquad W(e_3) = -\frac{a}{2} e_3.$$

Every invariant spinor belongs to $\mathcal{K}_W(H_3)$. Since there are two linearly independent invariant spinors, $\dim_{\mathbb{C}} \mathcal{K}_W(H_3) \geq 2$. The spinor bundle has complex rank two, and evaluation at one point is injective on ∇^W-parallel spinors, so in fact

$$\dim_{\mathbb{C}} \mathcal{K}_W(H_3) = 2.$$

The same connection coefficients give

$$K(e_1, e_2) = -\frac{3a^2}{4}, \qquad K(e_1, e_3) = K(e_2, e_3) = \frac{a^2}{4}.$$

For example,

$$R(e_1, e_2)e_2 = -\nabla_{e_2}\nabla_{e_1} e_2 - \nabla_{[e_1, e_2]} e_2 = -\frac{3a^2}{4} e_1.$$

On the other hand, the frame diagonalises W, with eigenvalues $a/2, a/2, -a/2$. If (g, W) satisfied

the Euclidean Gauss equation, the sectional curvatures would instead be

$$K(e_1, e_2) = \frac{a^2}{4}, \qquad K(e_1, e_3) = K(e_2, e_3) = -\frac{a^2}{4}.$$

These disagree with the actual sectional curvatures. Thus the Euclidean Gauss defect is non-zero despite $\dim_{\mathbb{C}} \mathcal{K}_W(H_3) = 2$, proving that the proposed two-spinor threshold fails in dimension three.

4.6 Hypersurfaces of hyperkähler four-manifolds

We now give a second three-dimensional construction using the same four-dimensional hyperkähler building block as in the preceding examples.

We continue with the K3 surface X^4 introduced above, equipped with its non-flat Ricci-flat Kähler metric. Since $\mathrm{Hol}(X) = \mathrm{SU}(2) = \mathrm{Sp}(1)$, the metric is hyperkähler. Thus X carries three parallel orthogonal complex structures I, J, K satisfying $I^2 = J^2 = K^2 = -\mathrm{id}$ and $IJ = K = -JI$, with parallel Kähler forms $\omega_I(U, V) = g_X(IU, V)$, $\omega_J(U, V) = g_X(JU, V)$ and $\omega_K(U, V) = g_X(KU, V)$.

The hyperkähler structure determines a preferred orientation. At a point $x \in X$, choose a unit vector e_1, set $e_2 = Ie_1$, $e_3 = Je_1$ and $e_4 = Ke_1$, and declare (e_1, e_2, e_3, e_4) to be positively oriented. In the dual coframe,

$$\begin{aligned} \omega_I &= e^1 \wedge e^2 + e^3 \wedge e^4, \\ \omega_J &= e^1 \wedge e^3 - e^2 \wedge e^4, \\ \omega_K &= e^1 \wedge e^4 + e^2 \wedge e^3. \end{aligned}$$

Each form is self-dual. Since $\dim \Lambda^2 T_x^* X = \binom{4}{2} = 6$, the Hodge decomposition $\Lambda^2 T_x^* X = \Lambda^2_+ T_x^* X \oplus \Lambda^2_- T_x^* X$ has two three-dimensional summands, and $\omega_I, \omega_J, \omega_K$ form a basis of $\Lambda^2_+ T_x^* X$.

We recall how this determines the chirality of the parallel spinors. Ordinary holonomy acts on two-forms by $(h \cdot \alpha)(U, V) = \alpha(h^{-1}U, h^{-1}V)$. Since the three Kähler forms are parallel, they are fixed by holonomy, and hence $\Lambda^2_+ T_x^* X$ is fixed pointwise.

The metric identifies $\Lambda^2 T_x^* X$ with $\mathfrak{so}(T_x X)$ by associating to a skew-symmetric endomorphism A the form $\alpha_A(U, V) = g_X(AU, V)$. Under this identification, $h \cdot \alpha_A = \alpha_{hAh^{-1}}$, so the action on two-forms corresponds to the adjoint action on $\mathfrak{so}(4)$.

The double covering $\lambda : \mathrm{Spin}(4) \to \mathrm{SO}(4)$ induces an isomorphism $d\lambda_e : \mathfrak{spin}(4) \to \mathfrak{so}(4)$. In dimension four,

$$\mathrm{Spin}(4) \cong \mathrm{SU}(2)_+ \times \mathrm{SU}(2)_-, \qquad \mathfrak{so}(4) \cong \mathfrak{su}(2)_+ \oplus \mathfrak{su}(2)_-,$$

and this Lie-algebra decomposition agrees with the Hodge decomposition: $\Lambda^2_+ T_x^* X \cong \mathfrak{su}(2)_+$ and $\Lambda^2_- T_x^* X \cong \mathfrak{su}(2)_-$. Conjugation in the direct product is componentwise,

$$\mathrm{Ad}_{(g_+, g_-)}(A_+, A_-) = \big(g_+ A_+ g_+^{-1}, g_- A_- g_-^{-1}\big).$$

Thus $\mathrm{SU}(2)_+$ acts on $\Lambda^2_+ T^*_x X$ by its adjoint representation, while $\mathrm{SU}(2)_-$ acts trivially there; the roles are reversed on $\Lambda^2_- T^*_x X$.

Let $\widetilde{\mathrm{Hol}}_x(X) \subset \mathrm{Spin}(4)$ be the lifted holonomy group. Since holonomy fixes $\Lambda^2_+ T^*_x X$ pointwise, its projection to $\mathrm{SU}(2)_+$ lies in the kernel of the adjoint representation, namely the centre $\{\pm 1\}$. The restricted lifted holonomy group is connected, so this projection is trivial. Since X is simply connected and has full holonomy $\mathrm{SU}(2)$,

$$\widetilde{\mathrm{Hol}}_x(X) = \{1\} \times \mathrm{SU}(2)_-.$$

The half-spin representations have complex dimension two. Under $\mathrm{Spin}(4) \cong \mathrm{SU}(2)_+ \times \mathrm{SU}(2)_-$, the positive half-spin representation is the standard representation of $\mathrm{SU}(2)_+$ and the negative half-spin representation is the standard representation of $\mathrm{SU}(2)_-$. Hence $\widetilde{\mathrm{Hol}}_x(X)$ fixes all of Δ_4^+ and has no non-zero fixed vector in Δ_4^-. Therefore

$$\dim_{\mathbb{C}} \mathcal{P}^+(X) = 2, \qquad \dim_{\mathbb{C}} \mathcal{P}^-(X) = 0.$$

This recovers the chirality count already obtained from Wang's classification and explains why, with the hyperkähler orientation, both parallel spinors have the same chirality. Reversing the orientation exchanges the two chiralities.

Now let $M^3 \subset X^4$ be an oriented hypersurface with shape operator W. Since the ambient dimension is even, $\Sigma X = \Sigma^+ X \oplus \Sigma^- X$. On restriction to M, each half-spinor bundle is identified with ΣM, but the two chiralities give opposite signs in the induced generalised Killing spinor equation. Choosing the hypersurface identification so that positive chirality corresponds to W, restriction gives injective maps

$$\mathcal{P}^+(X) \longrightarrow \mathcal{K}_W(M), \qquad \mathcal{P}^-(X) \longrightarrow \mathcal{K}_{-W}(M).$$

Both ambient parallel spinors therefore lie in $\mathcal{K}_W(M)$ after restriction, so $\dim_{\mathbb{C}} \mathcal{K}_W(M) \geq 2$. Since ΣM has complex rank two and evaluation at one point is injective on ∇^W-parallel spinors, equality holds on each connected component:

$$\dim_{\mathbb{C}} \mathcal{K}_W(M) = 2.$$

It remains to show that the Euclidean Gauss defect can nevertheless be non-zero.

Lemma 4.3. *Let X^4 be a hyperkähler four-manifold with the orientation for which its parallel Kähler forms are self-dual. Let $M^3 \subset X$ be an oriented hypersurface and let $p \in M$. If $R^X_p \neq 0$, then the tangential ambient curvature term in the Gauss equation of $M \subset X$ is non-zero at p. Equivalently, the Euclidean Gauss defect of the induced data (M, g, W) is non-zero at p.*

Proof. Set $V := T_p X$ and $H := T_p M \subset V$, and let ν be a unit normal, so that $V = H \oplus \mathbb{R}\nu$. We

regard the Riemann tensor as the self-adjoint curvature operator

$$\mathcal{R}_p : \Lambda^2 V \longrightarrow \Lambda^2 V, \qquad \langle \mathcal{R}_p(U \wedge V), S \wedge T \rangle = \left\langle R_p^X(U,V)T, S \right\rangle .$$

With respect to the Hodge decomposition $\Lambda^2 V = \Lambda^2_+ V \oplus \Lambda^2_- V$, the standard four-dimensional curvature decomposition is

$$\mathcal{R}_p = \begin{pmatrix} W_p^+ + \frac{s_p}{12}\,\mathrm{id} & \mathring{\mathrm{Ric}}_p^{\,*} \\ \mathring{\mathrm{Ric}}_p & W_p^- + \frac{s_p}{12}\,\mathrm{id} \end{pmatrix},$$

where $W^{\pm}$ are the Weyl-curvature blocks, $\mathring{\mathrm{Ric}}$ is the trace-free Ricci tensor and s is the scalar curvature [Bes87, p. 51]. Since the hyperkähler metric is Ricci-flat, $\mathring{\mathrm{Ric}} = 0$ and $s = \mathrm{tr}_{g_X} \mathrm{Ric} = 0$, so

$$\mathcal{R}_p = \begin{pmatrix} W_p^+ & 0 \\ 0 & W_p^- \end{pmatrix}.$$

The parallel forms $\omega_I, \omega_J, \omega_K$ form a basis of $\Lambda^2_+ V$. The curvature of the induced connection on two-forms annihilates every parallel form, so each curvature endomorphism $R_p^X(U,V) \in \mathfrak{so}(V)$ acts trivially on $\Lambda^2_+ V$. Under

$$\mathfrak{so}(V) \cong \mathfrak{su}(2)_+ \oplus \mathfrak{su}(2)_- \cong \Lambda^2_+ V \oplus \Lambda^2_- V,$$

this means that $R_p^X(U,V)$ lies in the $\mathfrak{su}(2)_- \cong \Lambda^2_- V$ summand. Hence the image of $\mathcal{R}_p$ is contained in $\Lambda^2_- V$. Since $\mathcal{R}_p$ is self-adjoint, it follows that it vanishes on $\Lambda^2_+ V$. Therefore

$$W_p^+ = 0, \qquad \mathcal{R}_p = \begin{pmatrix} 0 & 0 \\ 0 & W_p^- \end{pmatrix}.$$

A form $\omega \in \Lambda^2 H$ is regarded as a two-form on V by declaring that it vanishes whenever one argument is ν. Writing $\pi_\pm(\omega) = \frac{1}{2}(\omega \pm \star\omega)$, we have $\mathcal{R}_p\omega = W_p^- \pi_-(\omega)$, and hence

$$\langle \mathcal{R}_p\omega_1, \omega_2 \rangle = \left\langle W_p^- \pi_-(\omega_1), \pi_-(\omega_2) \right\rangle$$

for all $\omega_1, \omega_2 \in \Lambda^2 H$.

We claim that

$$\pi_- : \Lambda^2 H \longrightarrow \Lambda^2_- V$$

is an isomorphism. If $\pi_-(\omega) = 0$, then ω is self-dual, so $\omega \wedge \omega = \omega \wedge \star\omega = |\omega|^2 \,\mathrm{vol}_V$. A non-zero such form is non-degenerate. On the other hand, $\iota_\nu \omega = 0$ because $\omega \in \Lambda^2 H$, so ω is degenerate. Hence $\omega = 0$, and π_- is injective. Since $\dim \Lambda^2 H = \binom{3}{2} = 3 = \dim \Lambda^2_- V$, it is an isomorphism.

Suppose now that the tangential ambient curvature term vanished at p. Then $(R_p^X(U,V)T)^H = 0$ for all $U, V, T \in H$, and therefore $\langle \mathcal{R}_p\omega_1, \omega_2 \rangle = 0$ for all $\omega_1, \omega_2 \in \Lambda^2 H$. It follows that

$$\left\langle W_p^- \pi_-(\omega_1), \pi_-(\omega_2) \right\rangle = 0$$

for all such pairs. Since π_- is an isomorphism, this forces $W_p^- = 0$. The full curvature operator then vanishes, contradicting $R_p^X \neq 0$. Thus the tangential ambient curvature term is non-zero at p. □

Since the K3 metric is non-flat, choose $p \in X$ with $R_p^X \neq 0$ and a local oriented hypersurface $M^3 \subset X$ through p. Its induced data satisfy $\dim_{\mathbb{C}} \mathcal{K}_W(M) = 2$, while their Euclidean Gauss defect is non-zero at p. Thus even the full complex rank of the three-dimensional spinor bundle does not force the Euclidean Gauss equation. There is therefore no fixed-sign multiplicity threshold in dimension three.

The same holonomy discussion also shows that the ambient-parallel-spinor method cannot produce a one-spinor non-integrable example in this dimension. Let Y^4 be a Ricci-flat spin four-manifold carrying a local parallel spinor, and work on a sufficiently small simply connected neighbourhood. If its restricted holonomy representation is reducible, the local de Rham theorem splits the metric into Ricci-flat factors of dimensions at most three, which are flat. If the representation is irreducible and the metric is non-flat, Wang's classification gives $\mathrm{Hol}^0(Y) = \mathrm{SU}(2) = \mathrm{Sp}(1)$. The parallel spinors then occupy one half-spin representation of complex dimension two. Thus every non-flat local ambient construction of this kind produces two same-sign generalised Killing spinors on a three-dimensional hypersurface, rather than one.

This does not rule out one-spinor non-integrable examples obtained by other methods. It shows only that they cannot arise by restricting parallel spinors from a non-flat Ricci-flat four-dimensional ambient manifold.

4.7 Summary of the dimensional picture

We end this chapter by summarising what the examples show and how they fit with the analytic theorem proved in the next chapter. For every $n \geq 4$, the examples constructed above carry at least

$$2^{\lfloor n/2 \rfloor - 1}$$

common generalised Killing spinors while failing the Euclidean Gauss equation. Thus this multiplicity cannot by itself force local Euclidean immersibility.

All of these examples may be chosen real analytic. The theorem proved in the next chapter shows that one additional common generalised Killing spinor does suffice in the analytic category: the spinors extend to parallel spinors on a common Ricci-flat extension, and the holonomy bounds of Chapter 3 force that extension to be flat. It follows in particular that the examples above have exactly

$$2^{\lfloor n/2 \rfloor - 1}$$

common generalised Killing spinors. Hence

$$2^{\lfloor n/2 \rfloor - 1} + 1$$

is the sharp local analytic threshold for $n \geq 4$.

The low-dimensional cases behave differently. Dimension two is exceptional in a positive direction: one generalised Killing spinor already suffices, by Friedrich's surface correspondence. Dimension three is exceptional in a negative direction: multiplicity inside a single space $\mathcal{K}_W(M)$ cannot force the Euclidean Gauss defect to vanish, since the full complex rank of the spinor bundle may occur while the defect remains non-zero.

Dim.	**Fixed-sign threshold**	**Comments**
2	1	This is Friedrich's surface correspondence. A single generalised Killing spinor gives a local immersion into $\mathbb{R}^3$. Thus the higher-dimensional threshold does not extend to dimension two.
3	No multiplicity threshold	The fixed-sign threshold statement fails. A hypersurface in a non-flat hyperkähler four-manifold may carry two common generalised Killing spinors, the full complex rank of ΣM, while still having non-zero Euclidean Gauss defect.
4	$3 = 2^{\lfloor 4/2 \rfloor - 1} + 1$	The lower-bound examples have two common generalised Killing spinors and fail the Euclidean Gauss equation. The slice $X^4 \subset X^4 \times \mathbb{R}$ has $W = 0$, while the graph construction gives examples with $W \not\equiv 0$. For real-analytic data, the theorem of the next chapter shows that a third common spinor forces a local Euclidean hypersurface immersion.
$n \geq 5$	$2^{\lfloor n/2 \rfloor - 1} + 1$	The examples $$X^4 \times S^{n-4}(r) \subset X^4 \times \mathbb{R}^{n-3},$$ where X^4 has holonomy $\mathrm{Sp}(1) = \mathrm{SU}(2)$, carry $2^{\lfloor n/2 \rfloor - 1}$ common generalised Killing spinors but fail the Euclidean Gauss equation. For real-analytic data, the theorem of the next chapter shows that one additional common spinor forces a local Euclidean hypersurface immersion.

In odd hypersurface dimension, the two ambient chiralities restrict with opposite signs and therefore contribute separately to $\mathcal{K}_W(M)$ and $\mathcal{K}_{-W}(M)$. Their dimensions cannot be added when applying a statement concerning one fixed generalised Killing equation.

Opposite-sign data may nevertheless be decisive when used jointly. Lawn and Roth proved that, on a simply connected three-dimensional spin manifold, the existence of two non-trivial spinors satisfying the two opposite generalised Killing equations associated with the same symmetric

endomorphism forces the compatibility equations and yields an isometric immersion into a four-dimensional space form [LR10, Theorem 1]. In the Euclidean case, and with our sign conventions, this amounts to the existence of one non-zero spinor in each of

$$\mathcal{K}_W(M) \qquad \text{and} \qquad \mathcal{K}_{-W}(M).$$

Thus the failure of a fixed-sign multiplicity threshold in dimension three does not contradict the strength of paired-sign data. The paired-sign result proved later recovers this Euclidean phenomenon directly from the spinorial curvature identities.

This chapter therefore supplies the lower-bound side of the analytic threshold. The next chapter proves the converse: above this lower bound, common real analytic generalised Killing spinors extend to parallel spinors on a single Ricci-flat extension, and the holonomy estimates of Chapter 3 force that extension to be flat.

5

Analytic Extension and the Sharp Bound

5.1 The analytic converse problem

The previous chapter used ambient parallel spinors to produce generalised Killing spinors on hypersurfaces. In the real analytic category, this direction can be reversed in a precise geometric sense: a non-zero generalised Killing spinor for (g, W) gives a local isometric hypersurface realisation of (M, g) in a Ricci-flat manifold, with shape operator W, and the spinor extends to a parallel ambient spinor. This qualitative converse is due to Ammann–Moroianu–Moroianu [AMM13]; see also [Bou+15, Sections 8.5–8.6] for a systematic account of generalised Killing spinors and the Einstein Cauchy problem.

The construction proceeds in three steps. First, the generalised Killing spinor equation forces the Ricci-flat constraint equations for (g, W). Second, the analytic Einstein Cauchy problem produces a Ricci-flat metric

$$\bar{g} = dt^2 + g_t$$

on a neighbourhood of $\{0\} \times M$, with

$$g_0 = g, \qquad \dot{g}_0(X, Y) = -2g(W(X), Y).$$

Consequently, the inclusion $x \mapsto (0, x)$ is an isometric embedding of (M, g) whose shape operator is W. Finally, the generalised Killing spinor extends to a parallel spinor on this Ricci-flat neighbourhood.

The resulting Ricci-flat extension is determined locally by the Cauchy data (g, W), independently of the particular spinor used to derive the constraints. More precisely, any two analytic Ricci-flat extensions with the same initial data are isometric after restricting to sufficiently small neighbourhoods of the hypersurface. Hence all elements of $\mathcal{K}_W(M)$ extend to parallel spinors on the same local Ricci-flat extension.

Combining this fact with the holonomy bounds of Chapter 3 gives the quantitative refinement proved here: once the multiplicity exceeds the half-rank threshold, the Ricci-flat extension is forced to be flat near the hypersurface. The examples of Chapter 4 show that the resulting bound is exact.

The argument of the chapter is therefore

$$\mathcal{K}_W(M) \neq 0 \Longrightarrow (M, g, W) \text{ has a local Ricci-flat hypersurface realisation,}$$

and, when the multiplicity is sufficiently large,

$$\dim_{\mathbb{C}} \mathcal{K}_W(M) \text{ large} \Longrightarrow \begin{cases} \dim_{\mathbb{C}} \mathcal{P}(Z) \text{ large}, & n \text{ even}, \\ \dim_{\mathbb{C}} \mathcal{P}^{\sigma}(Z) \text{ large}, & n \text{ odd}, \end{cases} \Longrightarrow Z \text{ is flat near } M.$$

Here Z denotes a sufficiently small Ricci-flat neighbourhood of the embedded hypersurface. In odd hypersurface dimension, σ is the fixed ambient chirality determined by the hypersurface spinor identification. The local flatness of Z then shows that (g, W) are locally Euclidean hypersurface data.

Throughout the chapter, (M^n, g) is connected, oriented, spin and real analytic, while $W \in \Gamma(\mathrm{End}(TM))$ is symmetric and real analytic. We retain the notation (3.1). Every non-zero element of $\mathcal{K}_W(M)$ is nowhere vanishing, and evaluation at any point is injective because these spinors are ∇^W-parallel.

5.2 Gaussian coordinates and the initial data

The purpose of this section is simply to put the geometry near the hypersurface into a natural form adapted to the extension problem. Since M has codimension one, we may move away from it locally along normal geodesics and use signed normal distance as the additional coordinate. This identifies a neighbourhood of M with a product $(-\varepsilon, \varepsilon) \times M$ and reduces the ambient geometry to the evolution of a one-parameter family of metrics on M.

Suppose that $(M^n, g) \subset (Z^{n+1}, \bar{g})$ is an oriented hypersurface with unit normal ν and shape operator

$$W(X) = -\bar{\nabla}_X \nu.$$

The normal bundle of M is

$$NM = \{(x, v) : x \in M,\ v \perp T_x M\},$$

and the ambient exponential map restricts to the normal exponential map

$$\exp^{\perp} : NM \longrightarrow Z, \qquad (x, v) \longmapsto \exp_x(v).$$

The zero section of NM is the embedded copy of M given by $x \mapsto (x, 0)$. At a point $(x, 0)$ of this section, one may move either the base point x tangentially in M or the normal vector vertically in the fibre $N_x M$. Thus there is a natural identification

$$T_{(x,0)} NM \cong T_x M \oplus N_x M.$$

Under this identification,

$$d(\exp^{\perp})_{(x,0)}(X, \eta) = X + \eta.$$

Since $T_x Z = T_x M \oplus N_x M$, this differential is an isomorphism. The tubular neighbourhood theo-

rem consequently gives a neighbourhood of the zero section on which $\exp^{\perp}$ is a diffeomorphism onto a neighbourhood of M in Z.

Since M is a hypersurface, each normal space N_xM is one-dimensional. The chosen unit normal ν gives

$$N_xM = \mathbb{R}\nu_x,$$

so every normal vector has a unique expression $t\nu_x$. The normal bundle is therefore trivialised by

$$M \times \mathbb{R} \longrightarrow NM, \qquad (x,t) \longmapsto (x, t\nu_x).$$

After restricting to a sufficiently small neighbourhood, we obtain Gaussian coordinates

$$F : (-\varepsilon, \varepsilon) \times M \longrightarrow Z, \qquad F(t,x) = \exp_x(t\nu_x).$$

The curves $t \mapsto F(t,x)$ are unit-speed geodesics normal to M. By the Gauss lemma they remain orthogonal to the parallel hypersurfaces

$$M_t := F(\{t\} \times M),$$

so, after pulling back by F, the ambient metric takes the form

$$\bar{g} = dt^2 + g_t, \qquad g_0 = g,$$

where g_t is a smooth one-parameter family of metrics on the fixed manifold M. For the tubular neighbourhood construction and related concepts see [Lee18, Theorem 5.25, Proposition 5.26, Proposition 6.41, Corollary 6.42 and Example 6.43].

Writing the metric in this form is called placing it in *Gaussian gauge* relative to M. Equivalently, t is signed normal distance from M, the vector field ∂_t is unit and geodesic, and it is orthogonal to every slice M_t. We refer to [BGM05] for the generalised cylinder formalism and the corresponding spinorial identifications.

5.3 Constraint equations from one generalised Killing spinor

The first normal derivative of g_t is determined by the shape operator. To see this, let $X, Y \in \Gamma(TM)$ and regard them as vector fields on $(-\varepsilon,\varepsilon) \times M$ by keeping them independent of t:

$$X_{(t,x)} = X_x, \qquad Y_{(t,x)} = Y_x.$$

We then push these vector fields forwards by F to the corresponding neighbourhood of M in Z. Since their coefficients in the M-directions do not depend on t, this choice of extension satisfies

$$[\partial_t, X] = [\partial_t, Y] = 0.$$

Thus the vanishing of these brackets is not a property of an arbitrary extension of X and Y, but follows from the particular extension chosen in the product coordinates.

Since the Levi-Civita connection is torsion-free,

$$\bar{\nabla}_{\partial_t} X - \bar{\nabla}_X \partial_t = [\partial_t, X] = 0,$$

and similarly

$$\bar{\nabla}_{\partial_t} Y = \bar{\nabla}_Y \partial_t.$$

It follows that

$$\begin{aligned} \partial_t g_t(X,Y) &= \partial_t \bar{g}(X,Y) \\ &= \bar{g}(\bar{\nabla}_{\partial_t} X, Y) + \bar{g}(X, \bar{\nabla}_{\partial_t} Y) \\ &= \bar{g}(\bar{\nabla}_X \partial_t, Y) + \bar{g}(X, \bar{\nabla}_Y \partial_t). \end{aligned}$$

Writing

$$W_t(X) = -\bar{\nabla}_X \partial_t$$

for the shape operator of M_t, and using its symmetry, we obtain

$$\partial_t g_t(X,Y) = -g_t(W_t(X),Y) - g_t(X, W_t(Y)) = -2g_t(W_t(X),Y).$$

Thus the hypersurface data (g, W) determine the Cauchy data

$$g_0 = g, \qquad \dot{g}_0(X,Y) = -2g(W(X),Y). \tag{5.1}$$

Equivalently,

$$g_t = g - 2t\, g(W(\,\cdot\,),\,\cdot\,) + O(t^2).$$

Constructing an ambient metric with prescribed induced metric and shape operator is therefore an evolution problem for the family g_t.

Proposition 5.1 (Einstein constraint equations). *Let (M^n, g) be an oriented hypersurface in an Einstein manifold $(Z^{n+1}, \bar{g})$ satisfying*

$$\mathrm{Ric}^{\bar{g}} = \lambda \bar{g},$$

and let $W(X) = -\bar{\nabla}_X \nu$ be its shape operator. Then the induced data (g, W) satisfy

$$\mathrm{Scal}^g + \mathrm{tr}(W^2) - (\mathrm{tr}\, W)^2 = (n-1)\lambda$$

and

$$d\,\mathrm{tr}(W) + \delta W = 0,$$

where

$$(\delta W)(X) := -\sum_{i=1}^{n} g\big((\nabla^M_{e_i} W)(e_i), X\big)$$

for any local orthonormal frame (e_i) of TM.

Proof. These identities are obtained by contracting the Gauss and Codazzi equations. Indeed, tracing the Gauss equation over an orthonormal tangent frame gives

$$\mathrm{Scal}^{g} = \mathrm{Scal}^{\bar{g}} - 2\,\mathrm{Ric}^{\bar{g}}(\nu,\nu) + (\operatorname{tr} W)^2 - \operatorname{tr}(W^2).$$

Since $\bar{g}$ is Einstein,

$$\mathrm{Scal}^{\bar{g}} = (n+1)\lambda, \qquad \mathrm{Ric}^{\bar{g}}(\nu,\nu) = \lambda,$$

and hence

$$\mathrm{Scal}^{g} + \operatorname{tr}(W^2) - (\operatorname{tr} W)^2 = (n-1)\lambda.$$

Similarly, contracting the Codazzi equation over a tangent orthonormal frame gives

$$\mathrm{Ric}^{\bar{g}}(\nu, X) = \big(d\operatorname{tr}(W) + \delta W\big)(X).$$

The Einstein condition implies

$$\mathrm{Ric}^{\bar{g}}(\nu, X) = \lambda \bar{g}(\nu, X) = 0$$

for every $X \in TM$, and therefore

$$d\operatorname{tr}(W) + \delta W = 0.$$

□

In particular, if the ambient metric is Ricci-flat, so that $\lambda = 0$, the induced data must satisfy the following constraint equations:

$$d\operatorname{tr}(W) + \delta W = 0, \qquad \mathrm{Scal}^{g} + \operatorname{tr}(W^2) - (\operatorname{tr} W)^2 = 0. \tag{5.2}$$

These are the necessary Cauchy constraints for a Ricci-flat extension of (g, W) in Gaussian gauge. The following identities are proved in [AMM13, Corollary 3.1]; we include the derivation to fix conventions.

Proposition 5.2. *If $\mathcal{K}_W(M)$ contains a non-zero spinor, then (g, W) satisfies the Ricci-flat constraints (5.2).*

Proof. Let $0 \neq \varphi \in \mathcal{K}_W(M)$, so that

$$\nabla^{\Sigma M}_X \varphi = -\frac{1}{2} W(X) \cdot \varphi.$$

Its norm is constant, and hence φ is nowhere vanishing. Clifford contraction of the generalised Killing equation gives

$$D\varphi = \frac{1}{2}\operatorname{tr}(W)\varphi.$$

Indeed, at a point where $W(e_i) = \lambda_i e_i$,

$$-\frac{1}{2}\sum_i e_i \cdot W(e_i) \cdot \varphi = \frac{1}{2}\sum_i \lambda_i \varphi.$$

Consequently,

$$D^2\varphi = \frac{1}{2}\big(d\,\mathrm{tr}(W)\big)^\sharp \cdot \varphi + \frac{1}{4}(\mathrm{tr}\,W)^2\varphi.$$

At the same point, choose the orthonormal frame to be geodesic. The generalised Killing equation then gives

$$\begin{aligned}\nabla^*\nabla\varphi &= \frac{1}{2}\sum_i (\nabla^M_{e_i} W)(e_i)\cdot\varphi + \frac{1}{4}\sum_i |W(e_i)|^2\varphi \\ &= -\frac{1}{2}(\delta W)^\sharp\cdot\varphi + \frac{1}{4}\,\mathrm{tr}(W^2)\varphi.\end{aligned}$$

Applying the Schrödinger–Lichnerowicz formula [Fri00],

$$D^2 = \nabla^*\nabla + \frac{1}{4}\,\mathrm{Scal}^g,$$

we obtain

$$\big(\mathrm{Scal}^g + \mathrm{tr}(W^2) - (\mathrm{tr}\,W)^2\big)\varphi = 2\big(d\,\mathrm{tr}(W) + \delta W\big)^\sharp\cdot\varphi. \tag{5.3}$$

The left-hand side is multiplication by a real function and is Hermitian, whereas Clifford multiplication by a real vector is skew-Hermitian. Taking the real part of the Hermitian inner product with φ therefore forces

$$\mathrm{Scal}^g + \mathrm{tr}(W^2) - (\mathrm{tr}\,W)^2 = 0.$$

Equation (5.3) then implies

$$\big(d\,\mathrm{tr}(W) + \delta W\big)^\sharp\cdot\varphi = 0.$$

Clifford multiplication by a non-zero vector is invertible, and φ is nowhere vanishing, so

$$d\,\mathrm{tr}(W) + \delta W = 0.$$

These are precisely the constraints (5.2). □

No multiplicity is needed at this stage: one generalised Killing spinor already places the analytic Cauchy data inside the Ricci-flat constraint set.

5.4 Analytic extension of the Cauchy data

We now use Koiso's analytic Einstein Cauchy theorem [Koi81], in the form recalled in [AMM13, Theorem 2.1].

We shall use the word *germ* in the following elementary sense. A metric germ along $\{0\}\times M$

means a metric defined on some neighbourhood of $\{0\} \times M$ in $\mathbb{R} \times M$, where two such metrics are regarded as the same if they agree after possibly shrinking the neighbourhood. Thus the word records that the construction is local near the initial hypersurface; no global ambient manifold is being asserted.

Theorem 5.3 (Koiso). *Let (M^n, g) be a real analytic Riemannian manifold, and let $W \in \Gamma(\mathrm{End}(TM))$ be a real analytic symmetric endomorphism. Suppose that the Ricci-flat constraint equations*

$$d\,\mathrm{tr}(W) + \delta W = 0, \qquad \mathrm{Scal}^g + \mathrm{tr}(W^2) - (\mathrm{tr}\, W)^2 = 0$$

hold. Then there exists a real analytic metric germ $\bar{g} = dt^2 + g_t$ along $\{0\} \times M$ in $\mathbb{R} \times M$ satisfying $\mathrm{Ric}^{\bar{g}} = 0$, with $g_0 = g$ and shape operator W along the initial slice. Equivalently,

$$\dot{g}_0(X, Y) = -2g(W(X), Y).$$

The metric germ is unique in Gaussian gauge.

Corollary 5.4. *Let (M^n, g) be a real analytic spin manifold, and let W be a real analytic symmetric endomorphism. If $\mathcal{K}_W(M)$ contains a non-zero spinor, then there is a unique real analytic Ricci-flat germ $\bar{g} = dt^2 + g_t$ near $\{0\} \times M$, with $g_0 = g$ and shape operator W at $t = 0$.*

Proof. By Proposition 5.2, the existence of a non-zero spinor in $\mathcal{K}_W(M)$ implies the Ricci-flat constraint equations. Koiso's theorem then gives the required germ. □

The germ is constructed entirely from the Cauchy data

$$g_0 = g, \qquad \dot{g}_0(X, Y) = -2g(W(X), Y),$$

and hence depends only on the pair (g, W). In particular, every spinor in the same space $\mathcal{K}_W(M)$ is associated with this one Ricci-flat ambient germ.

5.5 Extension of the spinor fields

Let $\bar{g} = dt^2 + g_t$ be the Ricci-flat analytic extension of (g, W). Using the hypersurface spinor identification fixed in Chapter 1, a spinor $\varphi \in \mathcal{K}_W(M)$ may be viewed as a spinor along $\{0\} \times M \subset Z$. By the spinorial Gauss formula (1.11),

$$\nabla^{\Sigma Z}_X \varphi = \nabla^{\Sigma M}_X \varphi + \frac{1}{2} W(X) \cdot_M \varphi,$$

where $\cdot_M$ denotes intrinsic Clifford multiplication under this fixed identification. Thus the generalised Killing equation is exactly the statement that

$$\nabla^{\Sigma Z}_X \varphi = 0 \qquad \text{for all } X \in TM.$$

Ammann–Moroianu–Moroianu prove the following analytic extension theorem [AMM13, The-

orem 3.2].

Theorem 5.5 (Ammann–Moroianu–Moroianu). *Let $(Z, \bar{g})$ be a Ricci-flat real analytic spin manifold, and let $M \subset Z$ be an oriented real analytic hypersurface. Suppose that a real analytic spinor field along M is parallel in all tangential ambient directions:*

$$\nabla^{\Sigma Z}_X \psi = 0 \qquad \text{for all } X \in TM.$$

Then, locally near every point of M, ψ extends to a parallel spinor $\Psi \in \Gamma(\Sigma Z)$.

The local proof first extends ψ in the normal direction by solving

$$\nabla^{\Sigma Z}_{\partial_t} \Psi = 0, \qquad \Psi|_{t=0} = \psi.$$

One must then prove that the resulting spinor is also parallel in the directions tangent to the slices M_t. Ricci-flatness, the Bianchi identity and analyticity give a homogeneous analytic system for the failure of tangential parallelness. Since this failure vanishes at $t = 0$, the Cauchy–Kovalevskaya theorem forces it to vanish identically near M. theorem forces it to vanish identically near M.

There is one parity point to record. If n is even, then the ambient dimension $n + 1$ is odd and there is no chirality splitting. If n is odd, then

$$\Sigma Z = \Sigma^+ Z \oplus \Sigma^- Z.$$

By Proposition 1.7, the hypersurface identification fixed in Chapter 1 determines one chirality $\sigma \in \{+, -\}$ for which

$$\Sigma^\sigma Z|_M \cong \Sigma M$$

produces the sign convention defining $\mathcal{K}_W(M)$. The opposite chirality produces the equation defining $\mathcal{K}_{-W}(M)$. Since the ambient complex volume element is parallel, spin parallel transport preserves this splitting.

Corollary 5.6. *Let (M, g, W) be real analytic, and suppose that $\mathcal{K}_W(M)$ contains a non-zero spinor. Let $(Z, \bar{g})$ be the Ricci-flat analytic extension constructed above. After shrinking to a connected germ, every element of $\mathcal{K}_W(M)$ extends uniquely to a parallel spinor on $(Z, \bar{g})$.*

If n is even, extension gives a linear injective map

$$\mathcal{K}_W(M) \longrightarrow \mathcal{P}(Z).$$

If n is odd, extension gives linear injective maps

$$\mathcal{K}_W(M) \longrightarrow \mathcal{P}^\sigma(Z), \qquad \mathcal{K}_{-W}(M) \longrightarrow \mathcal{P}^{-\sigma}(Z),$$

where σ is the chirality determined by the fixed hypersurface identification.

Proof. Let $\varphi \in \mathcal{K}_W(M)$. Since g and W are analytic, φ is analytic as a parallel section of the

analytic connection ∇^W. The spinorial Gauss formula shows that it is tangentially parallel as a spinor along $M \subset Z$, so the extension theorem produces an ambient parallel spinor.

A parallel spinor on a connected germ is determined by its value at one point. The extension is therefore unique, and linearity follows from linearity of the parallel equation. It is injective because restriction to M recovers the original spinor.

If n is odd, Proposition 1.7 identifies $\mathcal{K}_W(M)$ with tangentially parallel data in $\Sigma^\sigma Z|_M$ and $\mathcal{K}_{-W}(M)$ with tangentially parallel data in $\Sigma^{-\sigma} Z|_M$. Parallel transport preserves chirality, so the two extensions lie in the stated half-spin bundles. □

This is the analytic reversal of the restriction argument from Chapter 1. There, ambient parallel spinors gave generalised Killing spinors. Here, analytic generalised Killing spinors with common W extend to ambient parallel spinors on the same Ricci-flat germ.

5.6 The analytic theorem

We now combine the analytic extension result with the total and half-spin holonomy bounds proved in Propositions 3.4 and 3.5.

Theorem 5.7 (Analytic sharp bound). *Let $n \geq 4$, and let (M^n, g) be a connected oriented real analytic spin manifold. Let $W \in \Gamma(\mathrm{End}(TM))$ be a real analytic symmetric endomorphism. Suppose that*

$$\dim_{\mathbb{C}} \mathcal{K}_W(M) \geq 2^{\lfloor n/2 \rfloor - 1} + 1.$$

Then (M, g, W) is locally realised as hypersurface data in Euclidean space $\mathbb{R}^{n+1}$. Equivalently, (g, W) satisfies the Euclidean Gauss and Codazzi equations. In particular, every element of $\mathcal{K}_W(M)$ is locally the restriction of a parallel spinor on $\mathbb{R}^{n+1}$.

Proof. Set

$$k := 2^{\lfloor n/2 \rfloor - 1} + 1$$

and choose linearly independent spinors $\varphi_1, \dots, \varphi_k \in \mathcal{K}_W(M)$.

Suppose, for contradiction, that at least one of the Euclidean Gauss or Codazzi defects is non-zero at some point $p_0 \in M$. Choose a simply connected coordinate ball $B \subset M$ containing p_0. By injectivity of evaluation at p_0, the restrictions $\varphi_1|_B, \dots, \varphi_k|_B$ remain linearly independent.

The constraint proposition and Koiso's theorem give a real analytic Ricci-flat germ $(Z^{n+1}, \bar{g})$ along B, with

$$\bar{g} = dt^2 + g_t, \qquad g_0 = g|_B,$$

and shape operator $W|_B$ along the initial slice. The Gauss and Codazzi equations for $B \subset Z$ identify the Euclidean defects with tangential and mixed-normal components of $\bar{R}$ along B. Since one defect is non-zero at p_0, the germ is non-flat at p_0.

By Corollary 5.6, the k spinors extend to ambient parallel spinors. They remain linearly in-

dependent, since their values at p_0 are the independent values of the original spinors. After shrinking, the germ is diffeomorphic to $(-\varepsilon, \varepsilon) \times B$ and is simply connected, so its full and restricted holonomy groups coincide. The local de Rham and Wang analysis of Chapter 3 therefore applies directly.

Suppose first that n is even. Then $n+1$ is odd, and

$$\dim_{\mathbb{C}} \mathcal{P}(Z) \geq k.$$

Proposition 3.4 gives, for a non-flat Ricci-flat germ,

$$\dim_{\mathbb{C}} \mathcal{P}(Z) \leq 2^{\lfloor n/2 \rfloor - 1}.$$

This contradicts the definition of k.

Now suppose that n is odd. Since $n \geq 4$, we have $n \geq 5$, so the ambient dimension $n+1$ is even and at least six, as required in Proposition 3.5. All elements of $\mathcal{K}_W(B)$ extend into the same half-spin bundle, and hence

$$\dim_{\mathbb{C}} \mathcal{P}^{\sigma}(Z) \geq k.$$

Proposition 3.5 gives

$$H(Z) := \max\left\{\dim_{\mathbb{C}} \mathcal{P}^{+}(Z), \dim_{\mathbb{C}} \mathcal{P}^{-}(Z)\right\} \leq 2^{(n+1)/2-2} = 2^{\lfloor n/2 \rfloor - 1}.$$

Thus

$$k \leq \dim_{\mathbb{C}} \mathcal{P}^{\sigma}(Z) \leq H(Z) \leq 2^{\lfloor n/2 \rfloor - 1},$$

again a contradiction.

The Gauss and Codazzi defects therefore vanish everywhere. Fix a point $p \in M$ and shrink to a simply connected neighbourhood U of p. By the classical fundamental theorem of hypersurfaces, there is a local oriented isometric immersion

$$F : U \longrightarrow \mathbb{R}^{n+1}$$

with shape operator $W|_U$. Since g and W are analytic, the immersion may be chosen analytic.

Place the Euclidean metric in Gaussian gauge about $F(U)$. Its pullback to a germ along $\{0\} \times U$ is an analytic Ricci-flat metric with the same Cauchy data

$$g_0 = g|_U, \qquad \dot{g}_0(X, Y) = -2g(W(X), Y)$$

as the germ constructed by Koiso. Uniqueness of the analytic Ricci-flat Cauchy problem in Gaussian gauge therefore identifies the two germs. Hence the analytic extension is locally flat and (g, W) is locally realised as Euclidean hypersurface data.

Finally, Corollary 5.6 extends every element of $\mathcal{K}_W(U)$ to a parallel spinor on this flat germ. Under the local identification with $\mathbb{R}^{n+1}$, these are parallel Euclidean spinors, which proves the

final assertion. □

Corollary 5.8 (Local analytic quantisation)**.** *Let U be a connected simply connected real analytic spin manifold of dimension $n \geq 4$, and let W be real analytic and symmetric. Set*

$$N := \operatorname{rank}_{\mathbb{C}} \Sigma U = 2^{\lfloor n/2 \rfloor}.$$

Then

$$\dim_{\mathbb{C}} \mathcal{K}_W(U) \in \left\{0, 1, \ldots, \frac{N}{2}\right\} \cup \{N\}.$$

Equivalently, no local analytic multiplicity lies strictly between half rank and full rank.

Proof. If $\dim_{\mathbb{C}} \mathcal{K}_W(U) \leq N/2$, there is nothing to prove. If it is larger than $N/2$, Theorem 5.7 gives a local Euclidean hypersurface realisation. On a simply connected neighbourhood, the restrictions of the appropriate ambient parallel spinors fill the entire rank-N intrinsic spinor module, so $\dim_{\mathbb{C}} \mathcal{K}_W(U) = N$. □

Remark 5.9 (The two signs in odd dimension)**.** Let $n \geq 5$ be odd. For the analytic Ricci-flat extension of (g, W), Corollary 5.6 places $\mathcal{K}_W$ and $\mathcal{K}_{-W}$ in the two opposite ambient half-spin bundles. Thus, if the data are not locally Euclidean,

$$\max\{\dim_{\mathbb{C}} \mathcal{K}_W, \dim_{\mathbb{C}} \mathcal{K}_{-W}\} \leq 2^{\lfloor n/2 \rfloor - 1}.$$

This is the bivariate form of the half-spin obstruction and will be useful when the two signs are considered simultaneously.

The lower-bound examples of Chapter 4 show that the numerical threshold is sharp for $n \geq 4$. These examples may be taken analytic. Ricci-flat Einstein metrics are real analytic in harmonic coordinates [DK81]; the round sphere

$$S^{n-4}(r) \subset \mathbb{R}^{n-3}$$

appearing in the hypersurface

$$X^4 \times S^{n-4}(r) \subset X^4 \times \mathbb{R}^{n-3}$$

is analytic; and products preserve analyticity. In Section 4.4, the graph construction was also carried out locally using a real analytic defining function. Thus there exist analytic data with

$$\dim_{\mathbb{C}} \mathcal{K}_W(M) = 2^{\lfloor n/2 \rfloor - 1}$$

which are not locally Euclidean hypersurface data. The analytic theorem shows that one more common generalised Killing spinor forces Euclidean immersibility.

The exclusion of $n = 3$ is essential. Chapter 4 gives two real analytic counterexamples. Artacho's left-invariant Heisenberg metric carries two common invariant generalised Killing spinors while failing the Euclidean Gauss equation [Art25]; left-invariant metrics and tensors are real analytic.

Likewise, hypersurfaces in non-flat hyperkähler four-manifolds carry two common generalised Killing spinors, the full complex rank of the three-dimensional spinor bundle, while their Euclidean Gauss defect can remain non-zero. Thus the two-spinor threshold fails even in the analytic category.

In dimension $n = 2$, a Ricci-flat three-dimensional germ is flat. Hence a single analytic generalised Killing spinor already forces the Ricci-flat extension to be Euclidean. This agrees with Friedrich's spinorial representation theorem for surfaces [Fri98] and with Theorem 2.2, and explains why the sharp threshold formula is stated only for $n \geq 4$.

6

The Algebraic Approach

6.1 The pointwise annihilator problem

The analytic proof passed through a Ricci-flat ambient extension. We now seek a direct argument from the curvature identity and the algebra of the spin representation.

Fix $p \in M$ and write $E = T_pM$ and $\Delta = \Sigma_pM$. Evaluation is injective on the space of ∇^W-parallel spinors, so

$$V_p := \mathrm{ev}_p\big(\mathcal{K}_W(M)\big) \subset \Delta, \qquad \dim_{\mathbb{C}} V_p = \dim_{\mathbb{C}} \mathcal{K}_W(M).$$

The mixed defect identity (2.4) says that, for every $X, Y \in E$,

$$\big(\widehat{\mathcal{G}}_W(X,Y) + \widehat{\mathcal{C}}_W(X,Y)\big) \cdot V_p = 0,$$

where $\widehat{\mathcal{G}}_W(X,Y) \in \mathrm{Cl}^2(E)$ and $\widehat{\mathcal{C}}_W(X,Y) \in E$. The precise signs of the two terms are irrelevant for the dimension estimates below; only their Clifford degrees matter.

Let E now be any n-dimensional Euclidean vector space with complex spin module Δ, and let $V \subset \Delta$ be a complex subspace. Define the mixed annihilator

$$\mathfrak{A}_V := \{(B,u) \in \mathrm{Cl}^2(E) \oplus E : (B+u) \cdot V = 0\}. \tag{6.1}$$

Thus the geometric defect pair $(\widehat{\mathcal{G}}_W(X,Y), \widehat{\mathcal{C}}_W(X,Y))$ belongs to $\mathfrak{A}_{V_p}$ for every X, Y.

Two projections of $\mathfrak{A}_V$ will be useful. Set

$$\begin{aligned}
\mathfrak{B}_V &:= \{B \in \mathrm{Cl}^2(E) : \text{there is a fixed } u \in E \text{ with } B \cdot \psi = -u \cdot \psi \text{ for all } \psi \in V\},\\
\mathfrak{K}_V &:= \{B \in \mathrm{Cl}^2(E) : B \cdot V = 0\}.
\end{aligned}$$

The word "fixed" in the first definition is essential: the same vector u must work for every spinor in V. The space $\mathfrak{B}_V$ is the projection of $\mathfrak{A}_V$ to $\mathrm{Cl}^2(E)$, while $\mathfrak{K}_V$ is its pure two-form kernel.

The pointwise problem is now precise: how large can V be if $\mathfrak{A}_V$ contains a non-zero pair? If $\mathfrak{A}_V = 0$ above a given dimension threshold, the geometric consequence is immediate: $\widehat{\mathcal{G}}_W = \widehat{\mathcal{C}}_W = 0$, hence $\mathcal{G}_W = \mathcal{C}_W = 0$.

The difficulty is the possible cancellation already seen in dimension three. A two-form need not annihilate V outright; it may imitate the action of a single vector on V. We first control this representation-theoretically by putting the two-form in normal form and studying the

weights of the spin module. In even dimensions chirality separates the two Clifford degrees. In odd dimensions an auxiliary even-dimensional space performs the same role, but exceptional extremal configurations remain. The final part of the argument uses the curvature symmetries of the Gauss defect, including the Bianchi identity, to eliminate those configurations.

Thus the algebraic route is

$$\text{many common spinors} \Longrightarrow \dim V_p \text{ large} \Longrightarrow \mathfrak{A}_{V_p} \text{ constrained} \Longrightarrow \mathcal{G}_W = \mathcal{C}_W = 0.$$

6.2 Normal form for two-forms

We begin with the degree-two part of the problem. Let E be an n-dimensional Euclidean vector space, let Δ be its complex spin module, and write $r = \lfloor n/2 \rfloor$. Strictly speaking, the Clifford algebra is only $\mathbb{Z}_2$-graded. Throughout this chapter, $\mathrm{Cl}^2(E)$ denotes the image of $\Lambda^2 E$ under the standard antisymmetrisation map $\Lambda^* E \to \mathrm{Cl}(E)$; it is a vector subspace, not a subalgebra. The normal form used here is not a special fact about spinors. It is the ordinary orthogonal normal form for a real skew-adjoint endomorphism; see, for example, [HJ13]. We recall the argument because the passage from an exterior two-form to a degree-two Clifford element is important below.

Let $\omega \in \Lambda^2 E^*$, and associate to it the skew-adjoint endomorphism $K \in \mathfrak{so}(E)$ defined by

$$\omega(X, Y) = \langle KX, Y \rangle.$$

The endomorphism $-K^2$ is self-adjoint and non-negative. If e_1 is a unit eigenvector with eigenvalue $\lambda_1^2 > 0$, set $e_2 = \lambda_1^{-1} K e_1$. Then e_1, e_2 are orthonormal and

$$Ke_1 = \lambda_1 e_2, \qquad Ke_2 = -\lambda_1 e_1.$$

Moreover, the plane $E_1 = \mathrm{span}\{e_1, e_2\}$ and its orthogonal complement are both K-invariant. Repeating the construction on $E_1^\perp$ and finally choosing an orthonormal basis of $\ker K$ gives an orthogonal decomposition

$$E = E_1 \oplus \cdots \oplus E_s \oplus \ker K,$$

where each E_α is an oriented two-plane on which K has matrix

$$\begin{pmatrix} 0 & -\lambda_\alpha \\ \lambda_\alpha & 0 \end{pmatrix}.$$

After adjoining zero coefficients if necessary, and choosing an oriented orthonormal basis $e_1, \ldots, e_n$, this says

$$\omega = \sum_{\alpha=1}^{r} \lambda_\alpha e^{2\alpha-1} \wedge e^{2\alpha}.$$

The coefficients may be chosen non-negative before the orientation of the full basis is prescribed. We shall instead allow arbitrary signs: if the basis obtained above has the wrong

orientation, changing the sign of one basis vector restores the orientation and changes only the corresponding λ_α.

Under the metric identification $E \cong E^*$ and the standard inclusion $\Lambda^2 E \subset \mathrm{Cl}(E)$, orthogonal vectors satisfy $e_i \wedge e_j = e_i \cdot e_j$. Hence the degree-two Clifford element corresponding to ω has the normal form

$$B = \lambda_1 e_1 \cdot e_2 + \lambda_2 e_3 \cdot e_4 + \cdots + \lambda_r e_{2r-1} \cdot e_{2r}. \tag{6.2}$$

The subsequent weight calculation is standard spin representation theory; see [LM89, Chapter II].

Set

$$J_\alpha := e_{2\alpha-1} \cdot e_{2\alpha}, \qquad 1 \le \alpha \le r.$$

Then

$$J_\alpha^2 = -1, \qquad J_\alpha J_\beta = J_\beta J_\alpha \quad \text{for all } \alpha, \beta.$$

Thus the operators $J_1, \ldots, J_r$ are commuting complex structures on Δ. Since we are working over $\mathbb{C}$, they can be diagonalised simultaneously.

In representation theory, the simultaneous eigenvalue data of a commuting family of operators is called a weight. More precisely, a weight records the eigenvalue of each operator in the family, and the corresponding common eigenspace is called a weight space. This terminology is useful because the operators in a representation generally do not all commute, and therefore cannot all be diagonalised simultaneously. One instead chooses a sufficiently large commuting family and uses its common eigenspaces to organise the representation.

In the present case, each J_α satisfies $J_\alpha^2 = -1$, so its eigenvalues are $+i$ and $-i$. The simultaneous eigenvalues are therefore recorded by a sign vector $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_r) \in \{\pm 1\}^r$, where ε_α indicates that J_α acts by $i\varepsilon_\alpha$.

For each such sign vector, define

$$\Delta_\varepsilon := \{\psi \in \Delta : J_\alpha \psi = i\varepsilon_\alpha \psi \text{ for every } \alpha\}.$$

Then

$$\Delta = \bigoplus_{\varepsilon \in \{\pm 1\}^r} \Delta_\varepsilon,$$

and each Δ_ε is one-dimensional. When $n = 2r$, this is the usual weight decomposition of the full spin representation. When $n = 2r + 1$, the same decomposition holds for the irreducible odd-dimensional spin representation; the remaining basis vector e_{2r+1} does not enter the normal form of B.

In this decomposition, the action of B is diagonal. If $\psi \in \Delta_\varepsilon$, then

$$B \cdot \psi = i \left(\sum_{\alpha=1}^{r} \varepsilon_\alpha \lambda_\alpha \right) \psi.$$

Hence

$$\ker(B : \Delta \to \Delta) = \bigoplus_{\substack{\varepsilon \in \{\pm 1\}^r \\ \sum_\alpha \varepsilon_\alpha \lambda_\alpha = 0}} \Delta_\varepsilon.$$

Thus the spinorial kernel of a two-form is governed by a finite signed-sum problem. The weight space Δ_ε lies in the kernel precisely when

$$\varepsilon_1 \lambda_1 + \cdots + \varepsilon_r \lambda_r = 0.$$

The following estimate will be used repeatedly.

Lemma 6.1. *Let $0 \neq B \in \mathrm{Cl}^2(E)$. Then*

$$\dim_{\mathbb{C}} \ker(B : \Delta \to \Delta) \leq 2^{r-1}.$$

Proof. Put B into the normal form above, with coefficients $\lambda_1, \ldots, \lambda_r$, not all zero. Choose an index β with $\lambda_\beta \neq 0$. Pair sign vectors by flipping the βth sign:

$$\varepsilon \longleftrightarrow \varepsilon', \qquad \varepsilon'_\beta = -\varepsilon_\beta, \qquad \varepsilon'_\alpha = \varepsilon_\alpha \text{ for } \alpha \neq \beta.$$

The corresponding signed sums differ by $2\varepsilon_\beta \lambda_\beta$, which is non-zero. Hence at most one sign vector in each such pair can give a zero signed sum. Since there are 2^{r-1} pairs, at most 2^{r-1} weight spaces lie in the kernel. Therefore

$$\dim_{\mathbb{C}} \ker B \leq 2^{r-1}.$$

□

For example, a simple non-zero two-form has no spinorial kernel. In normal form this means that only one coefficient λ_α is non-zero, and then no signed sum can vanish. Non-zero kernels arise from cancellations between at least two normal-form coefficients. Thus

$$B = \lambda(e_1 \cdot e_2 + e_3 \cdot e_4)$$

annihilates exactly the weight spaces with $\varepsilon_1 + \varepsilon_2 = 0$. This is the first appearance of the Kähler-type behaviour that will become important later: sums of commuting two-plane rotations can have unusually large kernels on the spin representation.

We finish by recording the parity distinction. In even dimension $n = 2r$, the spin representation splits into half-spinors,

$$\Delta = \Delta^+ \oplus \Delta^-.$$

The operators J_α preserve chirality, and the half-spinor spaces are obtained by separating the weight spaces according to the parity of the sign vector. After fixing the orientation convention,

$$\Delta^+ = \bigoplus_{\varepsilon_1 \cdots \varepsilon_r = 1} \Delta_\varepsilon, \qquad \Delta^- = \bigoplus_{\varepsilon_1 \cdots \varepsilon_r = -1} \Delta_\varepsilon,$$

possibly after interchanging the labels $+$ and $-$. Thus, in even dimension, a degree-two Clifford element preserves chirality, while a vector reverses chirality.

In odd dimension $n = 2r + 1$, there is no chirality splitting. The remaining basis vector e_{2r+1} commutes with every J_α, and hence preserves each weight line Δ_ε. The other basis vectors $e_{2\alpha-1}$ and $e_{2\alpha}$ flip the αth sign. Thus vector multiplication can interact with the same weight decomposition used for two-forms, but there is no chirality separation. This is the main reason the odd-dimensional mixed annihilator problem is harder.

6.3 Even dimensions: chirality separates the mixed terms

We first treat the even-dimensional case. Here the spin representation has a chirality splitting, and this prevents a degree-two term from cancelling a degree-one term on a sufficiently large subspace.

Let $n = 2r$, and let E be an oriented Euclidean vector space of dimension n. The complex spin module decomposes as

$$\Delta = \Delta^+ \oplus \Delta^-, \qquad \dim_{\mathbb{C}} \Delta^+ = \dim_{\mathbb{C}} \Delta^- = 2^{r-1}.$$

Clifford multiplication by a vector reverses chirality, while multiplication by a two-form preserves it:

$$u \cdot \Delta^\pm \subset \Delta^\mp, \qquad B \cdot \Delta^\pm \subset \Delta^\pm.$$

This elementary parity distinction is the main reason the even-dimensional case is clean.

Lemma 6.2. *Let $n = 2r$, and let $V \subset \Delta$ be a complex subspace satisfying*

$$\dim_{\mathbb{C}} V > 2^{r-1}.$$

If $(B, u) \in \mathrm{Cl}^2(E) \oplus E$ satisfies $(B + u) \cdot V = 0$, then $u = 0$.

Proof. Since $\Delta = \Delta^+ \oplus \Delta^-$ and both half-spinor spaces have dimension 2^{r-1}, the assumption $\dim_{\mathbb{C}} V > 2^{r-1}$ implies that V contains a non-zero spinor of pure chirality. Indeed, the projection $V \to \Delta^-$ has target of dimension 2^{r-1}, so its kernel is non-zero. Hence $V \cap \Delta^+ \neq 0$.

Choose $0 \neq \psi \in V \cap \Delta^+$. The equation $(B + u) \cdot \psi = 0$ becomes

$$B \cdot \psi + u \cdot \psi = 0.$$

But $B \cdot \psi \in \Delta^+$, while $u \cdot \psi \in \Delta^-$. Since $\Delta = \Delta^+ \oplus \Delta^-$ is direct, both terms vanish:

$$B \cdot \psi = 0, \qquad u \cdot \psi = 0.$$

If $u \neq 0$, then Clifford multiplication by u is invertible, because $u \cdot u = -|u|^2$. Thus $u \cdot \psi = 0$ would force $\psi = 0$, a contradiction. Hence $u = 0$.

The equation $B \cdot \psi = 0$ for this single spinor is not used to conclude that $B = 0$. A non-zero two-form may kill individual spinors. The two-form will be eliminated only after applying the kernel estimate to all of V. □

Combining this with the kernel estimate from the previous section gives the even-dimensional annihilator theorem.

Proposition 6.3. *Let $n = 2r$, and let $V \subset \Delta$ satisfy*

$$\dim_{\mathbb{C}} V > 2^{r-1}.$$

Then $\mathfrak{A}_V = 0$. Equivalently, if $(B, u) \in \mathrm{Cl}^2(E) \oplus E$ satisfies $(B + u) \cdot V = 0$, then

$$B = 0, \qquad u = 0.$$

Proof. By the preceding lemma, $u = 0$. Hence $B \cdot V = 0$, so $V \subset \ker B$.

If $B \neq 0$, then the kernel estimate for a non-zero two-form gives

$$\dim_{\mathbb{C}} V \leq \dim_{\mathbb{C}} \ker B \leq 2^{r-1},$$

contradicting the hypothesis. Therefore $B = 0$ as well. □

We now apply this pointwise to the spinorial Gauss–Codazzi identity. Suppose that M^n has even dimension $n = 2r$, that M is connected, and that

$$\dim_{\mathbb{C}} \mathcal{K}_W(M) > 2^{r-1}.$$

At each point $p \in M$, the evaluation subspace $V_p \subset \Sigma_p M$ has the same dimension as $\mathcal{K}_W(M)$, hence $\dim_{\mathbb{C}} V_p > 2^{r-1}$. For every $X, Y \in T_pM$, the mixed defect element

$$\left(\widehat{\mathcal{G}}_W(X, Y), \widehat{\mathcal{C}}_W(X, Y)\right)$$

lies in $\mathfrak{A}_{V_p}$. The proposition therefore gives

$$\widehat{\mathcal{G}}_W(X, Y) = 0, \qquad \widehat{\mathcal{C}}_W(X, Y) = 0$$

for all $X, Y \in T_pM$, equivalently

$$\mathcal{G}_W = 0, \qquad \mathcal{C}_W = 0$$

at p.

Since p was arbitrary, we obtain the following smooth even-dimensional result.

Theorem 6.4 (Even-dimensional algebraic rigidity)**.** *Let $n = 2r$, and let (M^n, g) be a connected smooth spin manifold. Let W be a smooth symmetric endomorphism of TM. Suppose that*

$$\dim_{\mathbb{C}} \mathcal{K}_W(M) \geq 2^{\lfloor n/2 \rfloor - 1} + 1.$$

Then

$$\mathcal{G}_W = 0, \qquad \mathcal{C}_W = 0.$$

Consequently, (g, W) satisfies the Euclidean Gauss and Codazzi equations. By the fundamental theorem of hypersurfaces, (M, g, W) is locally realised as hypersurface data in $\mathbb{R}^{n+1}$.

Proof. Since $n = 2r$, the hypothesis is exactly $\dim_{\mathbb{C}} \mathcal{K}_W(M) > 2^{r-1}$. The pointwise argument above gives $\mathcal{G}_W = 0$ and $\mathcal{C}_W = 0$ at every point. These are precisely the Euclidean Gauss and Codazzi equations for the pair (g, W). The smooth fundamental theorem of hypersurfaces then gives, locally near each point, an isometric immersion into $\mathbb{R}^{n+1}$ whose shape operator is W. □

This proves the desired sharp threshold in even dimensions without using analyticity or an ambient Ricci-flat extension. The argument is entirely pointwise: chirality separates the degree-two and degree-one parts, and the normal form of a two-form shows that a non-zero degree-two element cannot annihilate more than half of the spin representation.

It is worth noting how little geometry was used in the even-dimensional argument. Once the spinorial Gauss–Codazzi identity is known, the proof does not use the curvature symmetries of $\mathcal{G}_W$, the Bianchi identity, or relations between the defect elements for different choices of X and Y. Each mixed Clifford element is killed independently. In odd dimensions this brutal separation fails, and the unused geometric structure will become essential.

6.4 Fixing the vector term in odd dimensions

We continue with the odd-dimensional case. Let $n = 2r + 1$, and let E be an oriented Euclidean vector space of dimension n, with complex spin module Δ.

Suppose that $V \subset \Delta$ satisfies $\dim_{\mathbb{C}} V > 2^{r-1}$, and suppose that $(B, u) \in \mathfrak{A}_V$. If $u = 0$, then the pure kernel estimate from the previous section already forces $B = 0$. Thus the genuinely new case is $u \neq 0$. Write

$$u = c\nu, \qquad c = |u| > 0, \qquad |\nu| = 1,$$

and let $F := \nu^{\perp} \subset E$, which has even dimension $2r$. Every two-form $B \in \Lambda^2 E$ decomposes uniquely as

$$B = \beta + \nu \cdot a, \qquad \beta \in \Lambda^2 F, \quad a \in F.$$

Here $\nu \cdot a$ denotes the Clifford product, which agrees with $\nu \wedge a$ since ν and a are orthogonal.

The unit vector ν gives a splitting of the spin module into its $\pm i$-eigenspaces:

$$\Delta = \Delta_\nu^+ \oplus \Delta_\nu^-,$$

where

$$\Delta_\nu^+ = \{\psi \in \Delta : \nu \cdot \psi = i\psi\}, \qquad \Delta_\nu^- = \{\psi \in \Delta : \nu \cdot \psi = -i\psi\}.$$

Both eigenspaces have complex dimension 2^{r-1}. Clifford multiplication by vectors in F interchanges these two spaces, because every vector in F anticommutes with ν. By contrast,

multiplication by a two-form in $\Lambda^2 F$ preserves them.

Thus β preserves the splitting, while $\nu \cdot a$ interchanges the two summands. If $\psi = \psi_+ + \psi_-$ with $\psi_\pm \in \Delta_\nu^\pm$, then $(B + u) \cdot \psi = 0$ becomes

$$(\beta + ic) \cdot \psi_+ + \nu \cdot a \cdot \psi_- = 0,$$
$$\nu \cdot a \cdot \psi_+ + (\beta - ic) \cdot \psi_- = 0.$$

This system is the odd-dimensional replacement for the chirality separation used in even dimensions.

The first useful observation is that the mixed part $\nu \cdot a$ cannot occur above the half-rank threshold.

Lemma 6.5. *Let $n = 2r + 1$, and let $V \subset \Delta$ satisfy*

$$\dim_{\mathbb{C}} V > 2^{r-1}.$$

Suppose $(B, u) \in \mathfrak{A}_V$ with $u \neq 0$. Write $u = c\nu$ and $B = \beta + \nu \cdot a$ as above. Then $a = 0$.

Proof. Assume, for contradiction, that $a \neq 0$. Then Clifford multiplication by $\nu \cdot a$ is invertible, since

$$(\nu \cdot a)^2 = -|a|^2.$$

Consider the kernel of $B + u$ on Δ. We shall show that it has dimension at most 2^{r-1}.

Let $\psi = \psi_+ + \psi_- \in \ker(B + u)$, with $\psi_\pm \in \Delta_\nu^\pm$. The first equation in the coupled system is

$$(\beta + ic) \cdot \psi_+ + \nu \cdot a \cdot \psi_- = 0.$$

Since $\nu \cdot a : \Delta_\nu^- \to \Delta_\nu^+$ is invertible, this equation determines ψ_- uniquely from ψ_+. Hence the projection

$$\ker(B + u) \longrightarrow \Delta_\nu^+, \qquad \psi_+ + \psi_- \longmapsto \psi_+$$

is injective. Therefore

$$\dim_{\mathbb{C}} \ker(B + u) \leq \dim_{\mathbb{C}} \Delta_\nu^+ = 2^{r-1}.$$

But $V \subset \ker(B + u)$, since $(B, u) \in \mathfrak{A}_V$. This contradicts $\dim_{\mathbb{C}} V > 2^{r-1}$. Therefore $a = 0$. □

Consequently, any non-zero mixed annihilator above the half-rank threshold has a very special form. The vector term singles out a unit vector ν, and the two-form term must lie entirely in $F = \nu^\perp$:

$$u = c\nu, \qquad B = \beta \in \Lambda^2 F.$$

For such a pair, the annihilation equation decouples. If $\psi = \psi_+ + \psi_-$ with $\psi_\pm \in \Delta_\nu^\pm$, then $(\beta + c\nu) \cdot \psi = 0$ is equivalent to

$$(\beta + ic) \cdot \psi_+ = 0, \qquad (\beta - ic) \cdot \psi_- = 0.$$

Therefore any subspace $V \subset \ker(\beta + c\nu)$ is contained in

$$\ker(\beta + ic : \Delta_\nu^+ \to \Delta_\nu^+) \oplus \ker(\beta - ic : \Delta_\nu^- \to \Delta_\nu^-).$$

This reduces the odd-dimensional mixed annihilator problem to a spectral problem for a two-form β on the even-dimensional space $F = \nu^\perp$. The question is now:

How large can the eigenspaces of a real two-form on half-spinors be?

More explicitly, put β into normal form on F:

$$\beta = \lambda_1 e_1 \cdot e_2 + \cdots + \lambda_r e_{2r-1} \cdot e_{2r}.$$

The weight spaces of the spin representation of F are indexed by $\varepsilon \in \{\pm 1\}^r$, and β acts on the corresponding line by

$$i \sum_{\alpha=1}^{r} \varepsilon_\alpha \lambda_\alpha.$$

The two eigenspace conditions above therefore become

$$\sum_{\alpha=1}^{r} \varepsilon_\alpha \lambda_\alpha = -c$$

on one half-spinor component and

$$\sum_{\alpha=1}^{r} \varepsilon_\alpha \lambda_\alpha = c$$

on the other.

Thus, above the half-rank threshold, a non-zero mixed annihilator can exist only if a single real two-form β has unusually large eigenspaces at the two opposite non-zero eigenvalues $\pm ic$. This is already a strong restriction. The extremal cases are precisely where Kähler-type forms begin to appear.

6.5 Passage to a half-spin kernel problem

The preceding analysis isolates the way in which the vector term can interact with the two-form term. There is, however, a more invariant formulation which applies to every mixed Clifford element without first placing its vector part into normal form. This formulation will be essential when we return to the curvature identities.

Let $n = 2r + 1$, let E be an n-dimensional Euclidean vector space, and let Δ be the chosen complex spin module of E. Introduce a unit vector e_0 orthogonal to E, and set

$$\widehat{E} := \mathbb{R}e_0 \oplus E, \qquad \dim \widehat{E} = 2r + 2.$$

The map

$$E \longrightarrow \mathrm{Cl}^0(\widehat{E}), \qquad v \longmapsto e_0 \cdot_{\widehat{E}} v,$$

satisfies the Clifford relations of E. Indeed, for $x, y \in E$,

$$\begin{aligned}(e_0 \cdot_{\widehat{E}} x) \cdot_{\widehat{E}} (e_0 \cdot_{\widehat{E}} y) + (e_0 \cdot_{\widehat{E}} y) \cdot_{\widehat{E}} (e_0 \cdot_{\widehat{E}} x) \\ = x \cdot_{\widehat{E}} y + y \cdot_{\widehat{E}} x = -2g(x, y).\end{aligned}$$

It therefore extends to an algebra isomorphism

$$\mathrm{Cl}(E) \cong \mathrm{Cl}^0(\widehat{E}).$$

Each half-spin module of $\widehat{E}$ is a module for $\mathrm{Cl}^0(\widehat{E})$. After choosing the half-spin label compatibly with the chosen odd-dimensional spin representation, there is an intertwining isomorphism

$$\iota : \Delta \longrightarrow \widehat{\Delta}^\tau, \qquad \tau \in \{+, -\},$$

such that

$$\iota(v \cdot_E \psi) = e_0 \cdot_{\widehat{E}} v \cdot_{\widehat{E}} \iota(\psi)$$

for every $v \in E$ and $\psi \in \Delta$.

Products of two vectors act unchanged under this identification. Indeed, for $x, y \in E$,

$$\begin{aligned}\iota(x \cdot_E y \cdot_E \psi) &= e_0 \cdot_{\widehat{E}} x \cdot_{\widehat{E}} e_0 \cdot_{\widehat{E}} y \cdot_{\widehat{E}} \iota(\psi) \\ &= x \cdot_{\widehat{E}} y \cdot_{\widehat{E}} \iota(\psi).\end{aligned}$$

Consequently, for every $B \in \Lambda^2 E$,

$$\iota(B \cdot_E \psi) = B \cdot_{\widehat{E}} \iota(\psi).$$

Combining the degree-one and degree-two terms gives the uniform identity

$$\iota\big((B + u) \cdot_E \psi\big) = (B + e_0 \wedge u) \cdot_{\widehat{E}} \iota(\psi)$$

for every $B \in \Lambda^2 E$ and $u \in E$. Here $e_0 \wedge u = e_0 \cdot_{\widehat{E}} u$, since e_0 and u are orthogonal.

Thus every mixed Clifford element $B + u \in \Lambda^2 E \oplus E$ determines a genuine two-form

$$\Omega := B + e_0 \wedge u \in \Lambda^2 \widehat{E},$$

and

$$(B + u) \cdot_E \psi = 0 \quad \Longleftrightarrow \quad \Omega \cdot_{\widehat{E}} \iota(\psi) = 0.$$

In particular,

$$\ker(B + u : \Delta \to \Delta) \cong \ker\left(\Omega : \widehat{\Delta}^\tau \to \widehat{\Delta}^\tau\right).$$

The odd-dimensional mixed annihilator problem has therefore been reduced to a pure two-form kernel problem on one half-spin module in even dimension $2r + 2$.

Set $q := r + 1$, so that $\dim \widehat{E} = 2q$. Put Ω into normal form. After choosing an oriented orthonormal basis $f_1, \dots, f_{2q}$ of $\widehat{E}$, we may write

$$\Omega = \mu_1 f_1 \cdot f_2 + \cdots + \mu_q f_{2q-1} \cdot f_{2q}$$

for real coefficients $\mu_1, \dots, \mu_q$.

Define

$$J_\alpha := f_{2\alpha-1} \cdot f_{2\alpha}, \qquad 1 \le \alpha \le q.$$

These operators commute and satisfy $J_\alpha^2 = -1$. The full spin module of $\widehat{E}$ therefore decomposes into simultaneous one-dimensional weight spaces indexed by $\varepsilon = (\varepsilon_1, \dots, \varepsilon_q) \in \{\pm 1\}^q$, where

$$J_\alpha \cdot \psi = i\varepsilon_\alpha \psi.$$

On the corresponding weight line, Ω acts by multiplication by

$$i \sum_{\alpha=1}^{q} \varepsilon_\alpha \mu_\alpha.$$

After fixing the convention for the complex volume element, the two half-spin modules are obtained by separating the sign vectors according to the product $\varepsilon_1 \cdots \varepsilon_q$. Thus, for $\sigma \in \{\pm 1\}$, define

$$Z^\sigma(\mu) := \left\{ \varepsilon \in \{\pm 1\}^q : \varepsilon_1 \cdots \varepsilon_q = \sigma, \quad \sum_{\alpha=1}^{q} \varepsilon_\alpha \mu_\alpha = 0 \right\}.$$

Up to the fixed convention identifying $\widehat{\Delta}^\tau$ with one of the two values of σ, one has

$$\dim_{\mathbb{C}} \ker \left(\Omega|_{\widehat{\Delta}^\tau} \right) = |Z^\sigma(\mu)|.$$

6.6 The parity-restricted zero-weight estimate

Recall that, for $\sigma \in \{\pm 1\}$ and $\mu = (\mu_1, \dots, \mu_q) \in \mathbb{R}^q$, we defined

$$Z^\sigma(\mu) := \left\{ \varepsilon \in \{\pm 1\}^q : \varepsilon_1 \cdots \varepsilon_q = \sigma, \quad \sum_{\alpha=1}^{q} \varepsilon_\alpha \mu_\alpha = 0 \right\}.$$

Up to the fixed convention identifying the two values of σ with the two half-spin modules, one has

$$\dim_{\mathbb{C}} \ker \left(\Omega|_{\widehat{\Delta}^\tau} \right) = |Z^\sigma(\mu)|.$$

The maximal cardinality is independent of the chirality label. Indeed, given $\mu = (\mu_1, \dots, \mu_q)$,

set $\mu' = (-\mu_1, \mu_2, \dots, \mu_q)$ and, for $\varepsilon = (\varepsilon_1, \dots, \varepsilon_q)$, define $\varepsilon' = (-\varepsilon_1, \varepsilon_2, \dots, \varepsilon_q)$. Then

$$\sum_{\alpha=1}^{q} \varepsilon'_\alpha \mu'_\alpha = \sum_{\alpha=1}^{q} \varepsilon_\alpha \mu_\alpha, \qquad \varepsilon'_1 \cdots \varepsilon'_q = -\varepsilon_1 \cdots \varepsilon_q.$$

Hence $\varepsilon \mapsto \varepsilon'$ gives a bijection

$$Z^\sigma(\mu) \longrightarrow Z^{-\sigma}(\mu'),$$

and therefore

$$\max_{\mu \neq 0} |Z^+(\mu)| = \max_{\mu \neq 0} |Z^-(\mu)|.$$

We may consequently define

$$K(q) := \max_{\mu \in \mathbb{R}^q \setminus \{0\}} |Z^\sigma(\mu)|,$$

where either fixed choice of σ gives the same value.

The required combinatorial result is the following parity-restricted form of the Littlewood–Offord estimate.

Theorem 6.6 (Parity-restricted Littlewood–Offord estimate). *Let $q \geq 2$, let $\mu = (\mu_1, \dots, \mu_q) \in \mathbb{R}^q \setminus \{0\}$, and let $\sigma \in \{\pm 1\}$. Then*

$$|Z^\sigma(\mu)| \leq \begin{cases} 2^{q-2}, & q \text{ odd}, \\ \max\left\{2^{q-2}, \binom{q}{q/2}\right\}, & q \text{ even}. \end{cases}$$

The estimate is sharp for every q and either fixed choice of σ.

Moreover, suppose that $q \in \{4, 6, 8\}$ and

$$|Z^\sigma(\mu)| > 2^{q-2}.$$

Then every coefficient μ_α is non-zero and

$$|\mu_1| = \cdots = |\mu_q|.$$

In this case,

$$|Z^\sigma(\mu)| = \binom{q}{q/2}, \qquad Z^{-\sigma}(\mu) = \varnothing.$$

The estimate may be viewed as a parity-restricted refinement of the classical Littlewood–Offord problem [LO43; Erd45]. The final assertion is the rigidity statement needed in the three exceptional cases. Since the proof is entirely combinatorial and is not part of the geometric mechanism, it is deferred to Appendix A.

The theorem gives

$$K(q) = \begin{cases} \binom{q}{q/2}, & q \in \{2,4,6,8\}, \\ 2^{q-2}, & q \text{ odd}, \\ 2^{q-2}, & q \text{ even and } q \geq 10. \end{cases}$$

The two possible values arise from two geometrically different coefficient patterns.

The four-dimensional pattern. Suppose that precisely two normal-form coefficients are non-zero and have equal absolute value. A representative compatible with the splitting $\widehat{E} = \mathbb{R}e_0 \oplus E$ is

$$\Omega = \lambda \left(e_0 \cdot \nu + e_1 \cdot e_2 \right), \qquad \lambda \neq 0.$$

The zero-weight equation is $\varepsilon_1 + \varepsilon_2 = 0$. Thus the first two signs must be opposite, while the remaining signs are free apart from the single chirality condition. For $q > 2$, this gives 2^{q-2} zero-weight lines in either half-spin module.

The form is supported on the oriented four-plane

$$P := \operatorname{span}\{e_0, \nu, e_1, e_2\}.$$

With the orientation determined by (e_0, ν, e_1, e_2), one has

$$*\left(e_0 \wedge \nu\right) = e_1 \wedge e_2.$$

Hence $e_0 \wedge \nu + e_1 \wedge e_2$ is self-dual on P. Reversing the relative sign gives the corresponding anti-self-dual form. This is the four-dimensional algebraic pattern underlying the hyperkähler and K3 examples.

The Kähler pattern. Suppose instead that all q normal-form coefficients have the same absolute value. After choosing the orientations of the rotation planes, one may write

$$\Omega = \lambda \left(e_0 \cdot \nu + \sum_{\alpha=1}^{r} e_{2\alpha-1} \cdot e_{2\alpha} \right).$$

The associated exterior two-form

$$\omega = e_0 \wedge \nu + \sum_{\alpha=1}^{r} e_{2\alpha-1} \wedge e_{2\alpha}$$

is the Kähler form of the orthogonal complex structure defined by

$$Je_0 = \nu, \qquad J\nu = -e_0,$$

and

$$Je_{2\alpha-1} = e_{2\alpha}, \qquad Je_{2\alpha} = -e_{2\alpha-1}.$$

The zero-weight equation is

$$\varepsilon_1 + \cdots + \varepsilon_q = 0.$$

It can hold only when q is even. In that case, the number of positive signs must equal the number of negative signs, so exactly $q/2$ entries are positive and the remaining $q/2$ entries are negative. A sign vector is therefore determined uniquely by choosing which $q/2$ of the q positions carry the positive signs. There are

$$\binom{q}{q/2}$$

such choices. Since each sign vector labels a one-dimensional weight space, the kernel contains $\binom{q}{q/2}$ weight lines.

Every such sign vector has exactly $q/2$ negative entries. Therefore

$$\varepsilon_1 \cdots \varepsilon_q = (-1)^{q/2}.$$

Thus all the zero-weight lines lie in the same half-spin module, while the opposite half-spin module has zero kernel.

The central binomial coefficient exceeds the four-dimensional value precisely when

$$\binom{q}{q/2} > 2^{q-2},$$

which occurs for $q \in \{2, 4, 6, 8\}$. Since $q = r+1$ and $n = 2r+1 = 2q-1$, these values correspond to

$$n \in \{3, 7, 11, 15\}.$$

For every other odd dimension, a non-zero mixed annihilator has kernel dimension at most

$$2^{q-2} = 2^{r-1} = \frac{1}{2}\dim_{\mathbb{C}} \Delta.$$

Consequently, if $\dim_{\mathbb{C}} V > 2^{r-1}$, then the mixed annihilator must vanish.

Dimension 3 is a genuine exception. In that dimension, a non-zero Kähler-type two-form can annihilate the entire relevant half-spin module, and hence the entire odd-dimensional spin module under the identification above. Thus no same-sign multiplicity condition can force the mixed annihilator to vanish.

In dimensions 7, 11, and 15, the single-form estimate leaves the following apparent gaps:

n	$\dim_{\mathbb{C}} \Delta_n$	$\max_{\Omega \neq 0} \dim_{\mathbb{C}} \ker\left(\Omega|_{\widehat{\Delta}^r}\right)$	single-form argument	desired threshold
7	8	$\binom{4}{2} = 6$	$\dim_{\mathbb{C}} V \geq 7$	$\dim_{\mathbb{C}} V \geq 5$
11	32	$\binom{6}{3} = 20$	$\dim_{\mathbb{C}} V \geq 21$	$\dim_{\mathbb{C}} V \geq 17$
15	128	$\binom{8}{4} = 70$	$\dim_{\mathbb{C}} V \geq 71$	$\dim_{\mathbb{C}} V \geq 65$

The rigidity assertion in Theorem 6.6 shows, however, that above half rank any surviving non-zero annihilator must be Kähler-type. The remaining task is therefore geometric: we must use the fact that all curvature values annihilate the same spinor space and that their tangential components satisfy the algebraic curvature identities.

6.7 Eliminating the exceptional dimensions

Fix a point $p \in M$ and write

$$E := T_pM, \qquad \widehat{E} := \mathbb{R}e_0 \oplus E.$$

Let

$$V_p := \{\psi(p) : \psi \in \mathcal{K}_W(M)\} \subseteq \Delta, \qquad \widehat{V}_p := \iota(V_p) \subseteq \widehat{\Delta}^\tau.$$

Every generalised Killing spinor is parallel for the modified connection

$$\nabla^W_X := \nabla^{\Sigma M}_X + \frac{1}{2}W(X)\cdot,$$

so evaluation at p is injective. Hence

$$\dim_{\mathbb{C}} \widehat{V}_p = \dim_{\mathbb{C}} \mathcal{K}_W(M).$$

For each $X, Y \in E$, let $\widehat{\mathcal{G}}_W(X,Y) \in \Lambda^2 E$ and $\widehat{\mathcal{C}}_W(X,Y) \in E$ denote the degree-two and degree-one Clifford representatives appearing in the curvature identity. Thus

$$\left(\widehat{\mathcal{G}}_W(X,Y) + \widehat{\mathcal{C}}_W(X,Y)\right) \cdot \psi = 0$$

for every $\psi \in V_p$.

Define

$$\Omega(X,Y) := \widehat{\mathcal{G}}_W(X,Y) + e_0 \wedge \widehat{\mathcal{C}}_W(X,Y) \in \Lambda^2\widehat{E}.$$

The intertwining identity gives

$$\Omega(X,Y) \cdot \widehat{V}_p = 0.$$

Since

$$\Lambda^2\widehat{E} = \Lambda^2 E \oplus e_0 \wedge E$$

is a direct sum, the identity $\Omega(X,Y) = 0$ is equivalent to the simultaneous vanishing of the Gauss and Codazzi defect values.

Define the common annihilator

$$\mathfrak{a}_p := \left\{A \in \Lambda^2\widehat{E} : A \cdot \widehat{V}_p = 0\right\}.$$

Every curvature value $\Omega(X,Y)$ lies in $\mathfrak{a}_p$.

Lemma 6.7 (The common annihilator). *Suppose that $q \in \{4, 6, 8\}$ and $\dim_{\mathbb{C}} \widehat{V}_p > 2^{q-2}$. Under the standard identification*

$$\Lambda^2 \widehat{E} \cong \mathfrak{so}(\widehat{E}),$$

the common annihilator $\mathfrak{a}_p$ is one of the following:

$$\{0\}, \qquad \mathbb{R}\omega_J, \qquad \operatorname{span}\{\omega_I, \omega_J, \omega_K\},$$

where J is an orthogonal complex structure and, in the last case, I, J, K form a quaternionic triple.

Proof. The space $\mathfrak{a}_p$ is linear. It is also closed under the $\mathfrak{so}(\widehat{E})$-bracket. Indeed, the infinitesimal spin representation satisfies

$$\rho([A, B]) = [\rho(A), \rho(B)].$$

If both $\rho(A)$ and $\rho(B)$ vanish on $\widehat{V}_p$, then their commutator also vanishes on $\widehat{V}_p$. Hence $[A, B] \in \mathfrak{a}_p$.

Let $0 \neq A \in \mathfrak{a}_p$. Since

$$\widehat{V}_p \subseteq \ker\left(A|_{\widehat{\Delta}^\tau}\right)$$

and $\dim_{\mathbb{C}} \widehat{V}_p > 2^{q-2}$, the rigidity assertion in Theorem 6.6 implies that all the normal-form coefficients of A are non-zero and equal in absolute value. Regarded as a skew endomorphism of $\widehat{E}$, it therefore satisfies

$$A^2 = -Q(A)\,\mathrm{id}_{\widehat{E}}, \qquad Q(A) := -\frac{1}{2q}\operatorname{tr}(A^2) > 0.$$

The same identity holds trivially for $A = 0$.

Apply this identity to $A + B$. Expanding $(A + B)^2$ and subtracting the identities for A and B gives

$$AB + BA = -2b(A, B)\,\mathrm{id}_{\widehat{E}},$$

where

$$b(A, B) := \frac{1}{2}\left(Q(A + B) - Q(A) - Q(B)\right)$$

is the positive-definite inner product obtained by polarising Q.

Suppose that $\dim \mathfrak{a}_p \geq 2$, and choose b-orthonormal elements $I, J \in \mathfrak{a}_p$. Then

$$I^2 = J^2 = -\,\mathrm{id},$$

while $b(I, J) = 0$ gives $IJ = -JI$. Since $\mathfrak{a}_p$ is closed under brackets,

$$K := IJ = \frac{1}{2}[I, J]$$

also belongs to $\mathfrak{a}_p$. Moreover,

$$K^2 = IJIJ = -I^2J^2 = -\,\mathrm{id}\,.$$

The three operators satisfy

$$IJ = K, \qquad JK = I, \qquad KI = J,$$

and are mutually orthogonal. Thus I, J, K form a quaternionic triple.

There can be no fourth element $L \in \mathfrak{a}_p$ orthogonal to I, J, K. Orthogonality to I and J would give $LI = -IL$ and $LJ = -JL$, and therefore

$$LK = LIJ = IJL = KL.$$

On the other hand, orthogonality to K would give $LK = -KL$. Hence $KL = 0$, which is impossible because both K and L are invertible.

Therefore $\dim \mathfrak{a}_p \leq 3$. Dimension 2 is also impossible, since two orthonormal elements generate the third element K. The only possibilities are dimensions 0, 1, and 3, with the stated forms. □

We shall use only the standard algebraic curvature symmetries of the Gauss defect. Indeed, if

$$\mathbf{G}_W(X, Y, U, V) := g\left(\mathcal{G}_W(X, Y)U, V\right),$$

then $\mathbf{G}_W$ is an algebraic curvature tensor. This follows immediately from the corresponding symmetries of the Riemann curvature tensor and from the symmetry of W, which gives the same pair symmetry and first Bianchi identity for the quadratic term in the Gauss equation.

We now eliminate the two possible non-zero common annihilators.

Lemma 6.8 (Elimination of a Kähler line)**.** *Suppose that*

$$\mathfrak{a}_p = \mathbb{R}\omega_J,$$

where ω_J is the Kähler form of an orthogonal complex structure J on $\widehat{E}$. Then $\Omega(X, Y) = 0$ for every $X, Y \in E$.

Proof. Since every value of Ω lies in the Kähler line, there is a two-form $\alpha \in \Lambda^2 E^*$ such that

$$\Omega(X, Y) = \alpha(X, Y)\omega_J.$$

Decompose

$$\omega_J = \varphi + e_0 \wedge \eta, \qquad \varphi \in \Lambda^2 E, \quad \eta \in E^*.$$

The tangential component gives

$$\widehat{\mathcal{G}}_W(X, Y) = \alpha(X, Y)\varphi.$$

After absorbing the fixed non-zero normalising constant relating $\widehat{\mathcal{G}}_W$ to $\mathbf{G}_W$ into α, pair symmetry gives

$$\alpha(X, Y)\varphi(U, V) = \alpha(U, V)\varphi(X, Y).$$

Choose $U_0, V_0 \in E$ with $\varphi(U_0, V_0) \neq 0$. Then

$$\alpha(X,Y) = c\,\varphi(X,Y), \qquad c := \frac{\alpha(U_0,V_0)}{\varphi(U_0,V_0)}.$$

Thus

$$\mathbf{G}_W = c\,\varphi \otimes \varphi.$$

The first Bianchi identity gives

$$0 = c\big(\varphi(X,Y)\varphi(U,V) + \varphi(Y,U)\varphi(X,V) + \varphi(U,X)\varphi(Y,V)\big).$$

On the other hand,

$$\begin{aligned}(\varphi\wedge\varphi)(X,Y,U,V) = 2\big(&\varphi(X,Y)\varphi(U,V)\\ &+\varphi(Y,U)\varphi(X,V)\\ &+\varphi(U,X)\varphi(Y,V)\big).\end{aligned}$$

Hence

$$c\,\varphi\wedge\varphi = 0.$$

The form φ is the restriction to $E = e_0^{\perp}$ of the Kähler form of J. Its kernel is precisely $\mathbb{R}\,Je_0$. Indeed, for $X \in E$, the identity $\varphi(X,Y) = g(JX,Y)$ vanishes for every $Y \in E$ precisely when JX is orthogonal to E, hence proportional to e_0. It follows that X is proportional to Je_0.

Therefore

$$\operatorname{rank}\varphi = 2q - 2 \geq 6.$$

In particular, $\varphi\wedge\varphi \neq 0$. Hence $c = 0$, so $\alpha = 0$, and both the tangential and e_0-components of Ω vanish:

$$\Omega = 0.$$

□

Lemma 6.9 (Elimination of a quaternionic triple)**.** *Suppose that*

$$\mathfrak{a}_p = \operatorname{span}\{\omega_I, \omega_J, \omega_K\},$$

where I, J, K form a quaternionic triple. Then $\Omega(X,Y) = 0$ for every $X, Y \in E$.

Proof. Write

$$\Omega(X,Y) = \sum_{a=1}^{3} \alpha_a(X,Y)\omega_a,$$

where $(\omega_1,\omega_2,\omega_3) = (\omega_I,\omega_J,\omega_K)$ and $\alpha_a \in \Lambda^2 E^*$. Decompose

$$\omega_a = \varphi_a + e_0 \wedge \eta_a, \qquad \varphi_a \in \Lambda^2 E.$$

The tangential component is therefore

$$\widehat{\mathcal{G}}_W(X,Y) = \sum_{a=1}^{3} \alpha_a(X,Y)\varphi_a.$$

We first show that $\varphi_1, \varphi_2, \varphi_3$ are linearly independent. Suppose that $\sum_{a=1}^{3} c_a\varphi_a = 0$. Then

$$\sum_{a=1}^{3} c_a\omega_a = e_0 \wedge \left(\sum_{a=1}^{3} c_a\eta_a\right).$$

The right-hand side has rank at most 2. On the other hand, every non-zero linear combination $c_1I + c_2J + c_3K$ satisfies

$$(c_1I + c_2J + c_3K)^2 = -(c_1^2 + c_2^2 + c_3^2)\,\mathrm{id},$$

and therefore has full rank. Hence $c_1 = c_2 = c_3 = 0$.

Set

$$F := \mathrm{span}\{\varphi_1, \varphi_2, \varphi_3\} \subseteq \Lambda^2 E.$$

After absorbing the fixed normalising constant into the forms α_a, regard the Gauss defect as the self-adjoint operator

$$T : \Lambda^2 E \longrightarrow \Lambda^2 E, \qquad T(\xi) := \sum_{a=1}^{3} \alpha_a(\xi)\varphi_a.$$

Its image is contained in F.

Let $\zeta \in F^{\perp}$. Since $T(\xi) \in F$ for every ξ, $\langle T(\xi), \zeta\rangle = 0$. The pair symmetry of the Gauss defect means that T is self-adjoint, so $\langle \xi, T(\zeta)\rangle = 0$ for every ξ, and hence $T(\zeta) = 0$. Since the φ_a are linearly independent, $\alpha_a(\zeta) = 0$ for every a. Thus, after identifying two-forms and two-covectors using the metric, each α_a lies in F.

The forms $\varphi_1, \varphi_2, \varphi_3$ are mutually orthogonal and have equal norm. Indeed, the full Kähler forms $\omega_1, \omega_2, \omega_3$ are mutually orthogonal with equal norm, while the vectors Ie_0, Je_0, and Ke_0 are orthonormal. Removing the corresponding e_0-components therefore preserves orthogonality and equality of norms.

It follows that there is a symmetric matrix

$$S = (S_{ab}) \in \mathrm{Sym}^2(\mathbb{R}^3)$$

such that

$$\alpha_a = \sum_{b=1}^{3} S_{ab}\varphi_b.$$

The symmetry of S is precisely the self-adjointness of T in the orthogonal basis $\varphi_1, \varphi_2, \varphi_3$.

After an oriented orthogonal change of the quaternionic triple, we may diagonalise S. An

oriented orthogonal change of the ordered triple (I, J, K) preserves the quaternionic relations. Thus we may suppose

$$T = s_1\,\varphi_1 \otimes \varphi_1 + s_2\,\varphi_2 \otimes \varphi_2 + s_3\,\varphi_3 \otimes \varphi_3.$$

The first Bianchi identity then gives

$$s_1\,\varphi_1 \wedge \varphi_1 + s_2\,\varphi_2 \wedge \varphi_2 + s_3\,\varphi_3 \wedge \varphi_3 = 0.$$

We now choose a quaternionic orthonormal basis. Set

$$e_1 := Ie_0, \qquad e_2 := Je_0, \qquad e_3 := Ke_0.$$

Choose a unit vector e_4 orthogonal to $\operatorname{span}\{e_0, e_1, e_2, e_3\}$ and define

$$e_5 := Ie_4, \qquad e_6 := Je_4, \qquad e_7 := Ke_4.$$

If $\dim \widehat{E} > 8$, continue in the same manner on the remaining orthogonal complement.

Writing $e^{ij} := e^i \wedge e^j$, and making harmless simultaneous sign choices in the basis, the three Kähler forms may be written as

$$\begin{aligned}
\omega_1 &= e^{01} + e^{23} + e^{45} + e^{67} + \cdots, \\
\omega_2 &= e^{02} - e^{13} + e^{46} - e^{57} + \cdots, \\
\omega_3 &= e^{03} + e^{12} + e^{47} + e^{56} + \cdots.
\end{aligned}$$

Their restrictions to $E = e_0^{\perp}$ therefore begin

$$\begin{aligned}
\varphi_1 &= e^{23} + e^{45} + e^{67} + \cdots, \\
\varphi_2 &= -e^{13} + e^{46} - e^{57} + \cdots, \\
\varphi_3 &= e^{12} + e^{47} + e^{56} + \cdots.
\end{aligned}$$

The four-form e^{2345} occurs with non-zero coefficient in $\varphi_1 \wedge \varphi_1$ and does not occur in either of the other two squares. Similarly, e^{1346} occurs only in $\varphi_2 \wedge \varphi_2$, while e^{1247} occurs only in $\varphi_3 \wedge \varphi_3$. Terms from later quaternionic blocks involve indices at least 8 and cannot contribute to any of these three components.

Taking the coefficients of these three four-forms in the Bianchi identity gives

$$s_1 = 0, \qquad s_2 = 0, \qquad s_3 = 0.$$

Hence $T = 0$. Since the φ_a are linearly independent, $\alpha_1 = \alpha_2 = \alpha_3 = 0$, and therefore

$$\Omega = 0.$$

□

We can now resolve the three exceptional odd dimensions.

Theorem 6.10 (Exceptional odd dimensions). *Let $n \in \{7, 11, 15\}$, let (M^n, g) be a connected smooth spin manifold, and let W be a smooth symmetric endomorphism field. Suppose that*

$$\dim_{\mathbb{C}} \mathcal{K}_W(M) \geq 2^{\lfloor n/2 \rfloor - 1} + 1.$$

Then

$$\mathcal{G}_W = 0 \qquad \text{and} \qquad \mathcal{C}_W = 0.$$

Consequently, (M, g, W) is locally realised as hypersurface data in Euclidean space $\mathbb{R}^{n+1}$.

Proof. Fix $p \in M$ and write $n = 2q - 1$, so that $q \in \{4, 6, 8\}$. The hypothesis gives

$$\dim_{\mathbb{C}} \widehat{V}_p \geq 2^{q-2} + 1 > 2^{q-2}.$$

Every curvature value $\Omega(X, Y)$ lies in the common annihilator $\mathfrak{a}_p$. By Lemma 6.7, this space is either zero, a Kähler line, or a quaternionic triple.

If $\mathfrak{a}_p = \{0\}$, every curvature value vanishes immediately. The Kähler-line and quaternionic-triple cases are eliminated by Lemmas 6.8 and 6.9. Hence

$$\Omega(X, Y) = 0$$

for every $X, Y \in E$.

Since

$$\Lambda^2 \widehat{E} = \Lambda^2 E \oplus e_0 \wedge E$$

is a direct sum, the tangential and normal components vanish separately:

$$\mathcal{G}_W(p) = 0, \qquad \mathcal{C}_W(p) = 0.$$

Since p was arbitrary, both defects vanish globally. The smooth fundamental theorem of hypersurfaces then gives the local Euclidean realisation. □

6.8 The sharp smooth threshold

Combining the even-dimensional argument, the non-exceptional odd-dimensional estimate, and Theorem 6.10 gives the uniform smooth result.

Theorem 6.11 (Sharp smooth forcing threshold). *Let $n \geq 4$, let (M^n, g) be a connected smooth spin manifold, and let W be a smooth symmetric endomorphism field. If*

$$\dim_{\mathbb{C}} \mathcal{K}_W(M) \geq 2^{\lfloor n/2 \rfloor - 1} + 1,$$

then

$$\mathcal{G}_W = 0 \qquad \textit{and} \qquad \mathcal{C}_W = 0.$$

Consequently, (M, g, W) *is locally realised as hypersurface data in* $\mathbb{R}^{n+1}$*. The bound is sharp in every dimension* $n \geq 4$*.*

Proof. The upper bound was proved above in even dimensions. In odd dimensions, the parity-restricted zero-weight estimate proves it outside $\{3, 7, 11, 15\}$, while Theorem 6.10 handles $n \in \{7, 11, 15\}$.

The examples constructed earlier from a non-flat four-dimensional hyperkähler factor and a flat complementary factor carry exactly

$$2^{\lfloor n/2 \rfloor - 1}$$

generalised Killing spinors of the prescribed type without satisfying the Euclidean Gauss–Codazzi equations. Hence no smaller threshold can force local Euclidean realisation. □

Dimension 3 is necessarily excluded. In that dimension, a non-zero mixed Clifford element can annihilate the entire spin module, and the existing non-flat hypersurface examples realise this failure geometrically. Thus no multiplicity condition involving generalised Killing spinors of only one sign can force the Euclidean Gauss and Codazzi equations.

The difference between the two parity cases can now be stated succinctly. In even dimensions, chirality forces each individual defect value to vanish. In odd dimensions, a sufficiently large common kernel first confines every possible defect value to Kähler type. The common annihilator is then restricted to a single Kähler line or a quaternionic triple, and the pair symmetry and first Bianchi identity of the Gauss defect eliminate both possibilities.

A

The Parity-Restricted Littlewood–Offord Estimate

This appendix proves Theorem 6.6. The argument is entirely combinatorial and is separated from the main chapter so that the geometric proof of the exceptional dimensions does not depend on a lengthy discussion of signed sums.

For $\mu = (\mu_1, \dots, \mu_q) \in \mathbb{R}^q$ and $\tau \in \{\pm 1\}$, recall that

$$Z^\tau(\mu) := \left\{ \varepsilon \in \{\pm 1\}^q : \varepsilon_1 \cdots \varepsilon_q = \tau, \quad \sum_{\alpha=1}^{q} \varepsilon_\alpha \mu_\alpha = 0 \right\}.$$

We must bound the number of vanishing signed sums after restricting to one of the two possible values of the product of the signs.

A.1 Removing coefficient signs and zero coefficients

We begin with two elementary reductions.

Lemma A.1. *Let $\mu = (\mu_1, \dots, \mu_q) \in \mathbb{R}^q$. For every non-zero coefficient, write*

$$\mu_\alpha = \sigma_\alpha a_\alpha, \qquad \sigma_\alpha \in \{\pm 1\}, \qquad a_\alpha := |\mu_\alpha| > 0.$$

For a zero coefficient, set $\sigma_\alpha := 1$ and $a_\alpha := 0$. Then, for every $\tau \in \{\pm 1\}$,

$$|Z^\tau(\mu)| = \left| Z^{\tau \prod_{\mu_\alpha \neq 0} \sigma_\alpha}(a_1, \dots, a_q) \right|.$$

Moreover, if at least one coefficient μ_α vanishes, then

$$|Z^\tau(\mu)| \le 2^{q-2}.$$

Proof. Given $\varepsilon = (\varepsilon_1, \dots, \varepsilon_q)$, define $\delta_\alpha := \sigma_\alpha \varepsilon_\alpha$. Then $\varepsilon_\alpha \mu_\alpha = \delta_\alpha a_\alpha$ for every α, and therefore

$$\sum_{\alpha=1}^{q} \varepsilon_\alpha \mu_\alpha = 0 \quad \Longleftrightarrow \quad \sum_{\alpha=1}^{q} \delta_\alpha a_\alpha = 0.$$

The product of the new signs is

$$\delta_1 \cdots \delta_q = \left(\prod_{\mu_\alpha \neq 0} \sigma_\alpha \right) \varepsilon_1 \cdots \varepsilon_q,$$

so $\varepsilon \mapsto \delta$ gives the required bijection.

Now suppose that exactly $s \geq 1$ coefficients vanish and put $p := q - s$. After the preceding sign change, the non-zero coefficients may be assumed positive:

$$a_1, \ldots, a_p > 0.$$

For a choice of signs $\delta_1, \ldots, \delta_p$ on these coordinates, define

$$A_\delta := \{\alpha \in \{1, \ldots, p\} : \delta_\alpha = 1\}, \qquad T := \sum_{\alpha=1}^{p} a_\alpha.$$

Since

$$\sum_{\alpha=1}^{p} \delta_\alpha a_\alpha = \sum_{\alpha \in A_\delta} a_\alpha - \sum_{\alpha \notin A_\delta} a_\alpha,$$

the zero-sum condition is equivalent to

$$\sum_{\alpha \in A_\delta} a_\alpha = \frac{T}{2}.$$

The subsets satisfying this equation form an antichain. Indeed, if $A \subsetneq B$, then positivity gives

$$\sum_{\alpha \in A} a_\alpha < \sum_{\alpha \in B} a_\alpha,$$

so both sets cannot have weighted sum $T/2$.

Sperner's theorem [Spe28] therefore gives at most $\binom{p}{\lfloor p/2 \rfloor}$ admissible sign choices on the non-zero coordinates. For each such choice, the s signs corresponding to the zero coefficients are unrestricted by the signed-sum equation. Exactly half of their 2^s choices give either prescribed value of the total sign product. Hence

$$|Z^\tau(\mu)| \leq 2^{s-1} \binom{p}{\lfloor p/2 \rfloor}.$$

Finally,

$$\binom{p}{\lfloor p/2 \rfloor} \leq 2^{p-1}.$$

When p is odd, the two middle binomial coefficients are equal, so either is at most half the sum of all the coefficients. When $p = 2m$ is even,

$$2\binom{2m}{m-1} = \frac{2m}{m+1}\binom{2m}{m} \geq \binom{2m}{m},$$

so the central coefficient is again at most half the full binomial sum. Therefore

$$|Z^\tau(\mu)| \leq 2^{s-1} 2^{p-1} = 2^{q-2}.$$

□

A.2 The antichain estimate

We shall use the following standard strengthening of Sperner's theorem, the Lubell–Yamamoto–Meshalkin inequality [Yam54; Mes63; Lub66].

Lemma A.2 (Lubell–Yamamoto–Meshalkin inequality)**.** *Let $\mathcal{A}$ be an antichain of subsets of a q-element set. Then*

$$\sum_{A\in\mathcal{A}} \frac{1}{\binom{q}{|A|}} \le 1.$$

Proof. A maximal chain in the Boolean lattice has the form

$$\varnothing \subset A_1 \subset A_2 \subset \cdots \subset A_q, \qquad |A_k| = k.$$

There are $q!$ maximal chains, corresponding to the possible orders in which the q elements are inserted.

A fixed subset A of cardinality k belongs to $k!(q-k)!$ maximal chains: the elements of A may be inserted in any of $k!$ orders, followed by the remaining elements in any of $(q-k)!$ orders.

Since an antichain meets each maximal chain at most once, counting pairs consisting of a set in $\mathcal{A}$ and a maximal chain containing it gives

$$\sum_{A\in\mathcal{A}} |A|!(q-|A|)! \le q!.$$

Dividing by $q!$ yields

$$\sum_{A\in\mathcal{A}} \frac{1}{\binom{q}{|A|}} \le 1.$$

□

A.3 Proof of the zero-weight estimate

We now prove the bound in Theorem 6.6.

Proof of the estimate and its sharpness. By Lemma A.1, the signs of the non-zero coefficients may be removed at the cost of possibly interchanging the two chirality labels. We may therefore suppose that $\mu_\alpha \ge 0$ for every α.

If at least one coefficient vanishes, then Lemma A.1 already gives

$$|Z^\tau(\mu)| \le 2^{q-2}.$$

We may therefore assume for the remainder of the upper-bound argument that $\mu_1, \dots, \mu_q > 0$.

Put

$$T := \sum_{\alpha=1}^{q} \mu_\alpha.$$

To a sign vector $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_q)$, associate

$$A_\varepsilon := \{\alpha : \varepsilon_\alpha = 1\}.$$

The equation

$$\sum_{\alpha=1}^{q} \varepsilon_\alpha \mu_\alpha = 0$$

is equivalent to

$$\sum_{\alpha \in A_\varepsilon} \mu_\alpha = \frac{T}{2}.$$

Hence the zero-weight sign vectors correspond to the family

$$\mathcal{A} := \left\{ A \subseteq \{1, \ldots, q\} : \sum_{\alpha \in A} \mu_\alpha = \frac{T}{2} \right\}.$$

This family is an antichain, and it is closed under complementation.

Suppose first that q is odd. Taking complements changes the parity of the cardinality of a subset, and therefore interchanges the two chirality classes. Since complementation is a bijection of $\mathcal{A}$, the two chirality classes contain the same number of zero-weight subsets. Thus

$$|Z^\tau(\mu)| = \frac{1}{2}|\mathcal{A}|.$$

By Sperner's theorem [Spe28],

$$|\mathcal{A}| \leq \binom{q}{(q-1)/2}.$$

The two middle binomial coefficients are equal when q is odd, so

$$2\binom{q}{(q-1)/2} \leq 2^q.$$

Consequently,

$$|Z^\tau(\mu)| \leq \frac{1}{2}\binom{q}{(q-1)/2} \leq 2^{q-2}.$$

Now suppose that q is even. Since

$$\varepsilon_1 \cdots \varepsilon_q = (-1)^{q-|A_\varepsilon|} = (-1)^{|A_\varepsilon|},$$

fixing the chirality label fixes the parity of $|A_\varepsilon|$. Let $\mathcal{A}^\tau$ denote the subfamily corresponding to

the chosen chirality. It is itself an antichain. By Lemma A.2,

$$\sum_{A\in\mathcal{A}^\tau} \frac{1}{\binom{q}{|A|}} \le 1.$$

Since

$$\binom{q}{|A|} \le \binom{q}{q/2}$$

for every A, it follows that

$$\frac{|\mathcal{A}^\tau|}{\binom{q}{q/2}} \le 1.$$

Therefore

$$|Z^\tau(\mu)| \le \binom{q}{q/2}.$$

Combining the cases with and without zero coefficients gives

$$|Z^\tau(\mu)| \le \begin{cases} 2^{q-2}, & q \text{ odd}, \\ \max\left\{2^{q-2}, \binom{q}{q/2}\right\}, & q \text{ even}. \end{cases}$$

We next show that these bounds are sharp.

Consider first the coefficient vector

$$\mu = (1, 1, 0, \dots, 0).$$

The zero-weight equation is

$$\varepsilon_1 + \varepsilon_2 = 0,$$

so the first two signs must be opposite. When $q > 2$, there are two choices for the first two signs, and for either choice there are 2^{q-3} choices of the remaining signs giving a prescribed total sign product. Hence

$$|Z^\tau(\mu)| = 2^{q-2}.$$

When $q = 2$, choosing the relative sign of the two coefficients appropriately places both zero-weight sign vectors in either prescribed chirality.

Now suppose that q is even and take

$$\mu = (1, \dots, 1).$$

The zero-weight equation becomes

$$\varepsilon_1 + \cdots + \varepsilon_q = 0.$$

It requires exactly $q/2$ positive signs and $q/2$ negative signs. Therefore the number of solutions

is

$$\binom{q}{q/2}.$$

Every such sign vector has product $(-1)^{q/2}$, so all the solutions lie in one chirality. Changing the sign of one coefficient interchanges the two chirality classes, and hence the bound is sharp for either prescribed value of τ.

It remains to compare the two possible values when q is even. Define

$$h(q) := 2^{2-q}\binom{q}{q/2}.$$

For even q,

$$\frac{h(q+2)}{h(q)} = \frac{q+1}{q+2} < 1.$$

Thus $h(q)$ decreases strictly along the even integers. Since

$$h(8) = \frac{70}{64} > 1, \qquad h(10) = \frac{252}{256} < 1,$$

one has

$$\binom{q}{q/2} > 2^{q-2}$$

precisely for $q \in \{2, 4, 6, 8\}$. This proves the asserted formula for the maximal cardinality. □

A.4 Rigidity in the exceptional cases

It remains to prove the final assertion of Theorem 6.6. We first isolate the simple comparison which will force equality of the coefficients.

Lemma A.3 (Middle-layer swap lemma). *Let $q = 2m$, let $a_1, \dots, a_q > 0$, and put $T := a_1 + \cdots + a_q$. Suppose that*

$$\mathcal{A} \subseteq \{A \subseteq \{1, \dots, q\} : |A| = m\}$$

consists of subsets satisfying

$$\sum_{\alpha \in A} a_\alpha = \frac{T}{2}.$$

If fewer than

$$\binom{q-2}{m-1}$$

of the m-element subsets are missing from $\mathcal{A}$, then

$$a_1 = \cdots = a_q.$$

Proof. Fix distinct indices i and j. Consider the m-element subsets containing i but not j. Each has the form

$$A = \{i\} \cup B,$$

where

$$B \subseteq \{1, \dots, q\} \setminus \{i, j\}, \qquad |B| = m - 1.$$

There are $\binom{q-2}{m-1}$ such subsets.

Pair each A with

$$A' := (A \setminus \{i\}) \cup \{j\}.$$

These are disjoint pairs of m-element subsets. If every pair had at least one member missing from $\mathcal{A}$, then at least $\binom{q-2}{m-1}$ subsets would be missing. By hypothesis, this is not the case. Hence some pair A, A' lies entirely in $\mathcal{A}$.

Both members have weighted sum $T/2$. Cancelling their common terms gives $a_i = a_j$. Since i and j were arbitrary, all the coefficients are equal. □

We can now prove the required rigidity statement.

Proposition A.4 (Exceptional rigidity). *Let $q \in \{4, 6, 8\}$, let $\mu = (\mu_1, \dots, \mu_q) \in \mathbb{R}^q \setminus \{0\}$, and let $\tau \in \{\pm 1\}$. If*

$$|Z^\tau(\mu)| > 2^{q-2},$$

then every coefficient μ_α is non-zero and

$$|\mu_1| = \cdots = |\mu_q|.$$

Consequently,

$$|Z^\tau(\mu)| = \binom{q}{q/2}, \qquad Z^{-\tau}(\mu) = \varnothing.$$

Proof. If any coefficient vanishes, then Lemma A.1 gives $|Z^\tau(\mu)| \le 2^{q-2}$, contrary to the hypothesis. Hence every coefficient is non-zero.

By changing the signs of the coefficients as in Lemma A.1, and allowing a possible interchange of the chirality labels, we may suppose that $\mu_\alpha = a_\alpha > 0$ for every α.

Put $T := a_1 + \cdots + a_q$ and let $\mathcal{A}$ be the family of subsets corresponding to the chosen chirality and satisfying

$$\sum_{\alpha \in A} a_\alpha = \frac{T}{2}.$$

Since q is even, taking complements preserves the parity of the cardinality. Thus $\mathcal{A}$ is closed under complementation.

We treat the three values of q separately.

The case $q = 4$. Suppose first that the chosen chirality corresponds to subsets of odd cardinality. The possible sizes are 1 and 3. If a_k denotes the number of members of cardinality k, complementation gives $a_1 = a_3$. The LYM inequality gives

$$\frac{a_1}{\binom{4}{1}} + \frac{a_3}{\binom{4}{3}} \le 1,$$

and hence

$$|\mathcal{A}| = a_1 + a_3 \leq 4.$$

This cannot satisfy $|\mathcal{A}| > 4$.

Therefore the chosen chirality corresponds to subsets of even cardinality. The empty and full sets cannot have weight $T/2$, so every member of $\mathcal{A}$ has cardinality 2. There are six such subsets. Since complementation pairs them without fixed points, $|\mathcal{A}|$ is even. The inequality $|\mathcal{A}| > 4$ therefore forces $|\mathcal{A}| = 6$.

No 2-element subsets are missing. Since

$$0 < \binom{2}{1},$$

Lemma A.3 gives

$$a_1 = a_2 = a_3 = a_4.$$

The case $q = 6$**.** Suppose first that the chosen chirality corresponds to subsets of even cardinality. The possible sizes are 2 and 4. Write a_k for the number of members of size k. Complementation gives $a_2 = a_4$. By the LYM inequality,

$$\frac{a_2}{\binom{6}{2}} + \frac{a_4}{\binom{6}{4}} \leq 1.$$

Hence

$$|\mathcal{A}| = a_2 + a_4 \leq 15.$$

Since $|\mathcal{A}|$ is even, in fact

$$|\mathcal{A}| \leq 14 < 16.$$

Thus the chosen chirality must be the odd-cardinality class.

The possible sizes are then 1, 3, and 5. Suppose that a singleton $\{i\}$ belongs to $\mathcal{A}$. Its complement also belongs to $\mathcal{A}$. Any further member containing i would strictly contain $\{i\}$, while any further member not containing i would be strictly contained in its complement. Both possibilities contradict the antichain property. Hence in this case $|\mathcal{A}| = 2$.

Therefore a family satisfying $|\mathcal{A}| > 16$ consists entirely of 3-element subsets. There are

$$\binom{6}{3} = 20$$

such subsets. Since $\mathcal{A}$ is closed under complementation, its cardinality is even. Thus $|\mathcal{A}| \in \{18, 20\}$, so at most two 3-element subsets are missing.

Since

$$2 < \binom{4}{2} = 6,$$

Lemma A.3 gives

$$a_1 = \cdots = a_6.$$

The case $q = 8$. Suppose first that the chosen chirality corresponds to subsets of odd cardinality. If a singleton belongs to $\mathcal{A}$, the same antichain argument as above shows that it and its complement are the only members.

Otherwise, every member has cardinality 3 or 5. If a_k denotes the number of members of cardinality k, complementation gives $a_3 = a_5$. The LYM inequality gives

$$\frac{a_3}{\binom{8}{3}} + \frac{a_5}{\binom{8}{5}} \leq 1,$$

and therefore

$$|\mathcal{A}| = a_3 + a_5 \leq \binom{8}{3} = 56 < 64.$$

Hence the chosen chirality must correspond to subsets of even cardinality.

The possible sizes are 2, 4, and 6. Let a_k again denote the number of members of size k. Complementation gives $a_2 = a_6$. The LYM inequality gives

$$\frac{a_2}{\binom{8}{2}} + \frac{a_4}{\binom{8}{4}} + \frac{a_6}{\binom{8}{6}} \leq 1,$$

or equivalently

$$\frac{2a_2}{28} + \frac{a_4}{70} \leq 1.$$

Hence

$$a_4 \leq 70 - 5a_2,$$

and therefore

$$|\mathcal{A}| = 2a_2 + a_4 \leq 70 - 3a_2.$$

If $a_2 \geq 2$, this gives $|\mathcal{A}| \leq 64$.

It remains to consider the possibility $a_2 = 1$. Let P be the unique 2-element member of $\mathcal{A}$. Its complement is the unique 6-element member. Any 4-element member must meet P in exactly one point. It cannot contain P, since that would violate the antichain property, and it cannot be disjoint from P, since it would then be strictly contained in the zero-weight complement of P.

There are at most

$$\binom{2}{1}\binom{6}{3} = 40$$

such 4-element subsets. Hence

$$|\mathcal{A}| \leq 1 + 40 + 1 = 42 < 64.$$

Consequently, a family satisfying $|\mathcal{A}| > 64$ has $a_2 = a_6 = 0$ and consists entirely of 4-element

subsets. There are

$$\binom{8}{4} = 70$$

such subsets. Since $\mathcal{A}$ is closed under complementation, its cardinality is even. Therefore $|\mathcal{A}| \geq 66$, so at most four 4-element subsets are missing.

Since

$$4 < \binom{6}{3} = 20,$$

Lemma A.3 gives

$$a_1 = \cdots = a_8.$$

We have proved in every case that

$$a_1 = \cdots = a_q.$$

Returning to the original coefficients gives

$$|\mu_1| = \cdots = |\mu_q|.$$

With equal non-zero absolute values, the zero-sum equation requires exactly $q/2$ positive signs and $q/2$ negative signs. Hence there are

$$\binom{q}{q/2}$$

zero-weight sign vectors. Each has the same sign product, so all belong to one chirality and none belongs to the other. Therefore

$$|Z^{\tau}(\mu)| = \binom{q}{q/2}, \qquad Z^{-\tau}(\mu) = \varnothing.$$

□

The proposition completes the proof of Theorem 6.6.

A.5 The corresponding two-form statement

We finish by recording explicitly the geometric form of the rigidity conclusion used in the main chapter.

Corollary A.5. *Let $q \in \{4, 6, 8\}$, let $\widehat{E}$ be a Euclidean vector space of dimension $2q$, and let*

$$0 \neq \Omega \in \Lambda^2 \widehat{E}.$$

Suppose that, on one half-spin module,

$$\dim_{\mathbb{C}} \ker \left(\Omega|_{\widehat{\Delta}^{\tau}}\right) > 2^{q-2}.$$

Then there is an orthogonal complex structure J on $\widehat{E}$ and a non-zero scalar λ such that

$$\Omega = \lambda\, \omega_J.$$

Moreover,

$$\dim_{\mathbb{C}} \ker \left(\Omega|_{\widehat{\Delta}^\tau}\right) = \binom{q}{q/2},$$

while the kernel on the opposite half-spin module is zero.

Proof. Put Ω into normal form:

$$\Omega = \sum_{\alpha=1}^{q} \mu_\alpha f_{2\alpha-1} \wedge f_{2\alpha}.$$

Proposition A.4 gives

$$|\mu_1| = \cdots = |\mu_q| = |\lambda| \neq 0.$$

By reversing the orientation of individual rotation planes when necessary, the signs may be absorbed into an orthogonal complex structure J. Thus

$$\Omega = \lambda\, \omega_J.$$

The statements about the two half-spin kernels follow from the final part of Proposition A.4. □

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